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de les Illes Balears

FINAL DEGREE REPORT

COMPARISON OF THE SMOOTH AND PIECEWISE LINEAR DYNAMICAL ANALYSIS OF AN ANALOGUE CIRCUITAL IMPLEMENTATION OF THE FITZHUGH-NAGUMO MODEL

Joan Camps Tomas

Double Degree in Mathematics and Telematics Engineering

Escola Politècnica Superior

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Tutors: Antonio E. Teruel, Rodrigo Picos

Perquè tota acció individual s'entén dins un marc col·lectiu, gràcies a totes aquelles
persones, amics, familiars i professors que han arribat a definir el meu.
Gràcies als tutors per la seva ajuda i dedicació durant l'elaboració d'aquest treball.

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ACRONYMS

BEB Boundary Equilibrium Bifurcation

IVP Initial Values Problem

KCL Kirchoff's Current Law

ODE Ordinary Differential Equation

PWL Piecewise Linear

SPICE Simulation Program with Integrated Circuit Emphasis

ABSTRACT

This project adopts a usual paradigm of scientific inquiry: formulating simplified theoretical models to study complex phenomena and ultimately translating those insights back to reality. We introduce two mathematically distinct slow-fast systems—one smooth and one piecewise linear (PWL)—under the FitzHugh-Nagumo framework to reproduce the dynamics of a simple circuit based on a non-linear resistor. Phenomena such as the canard explosion, antimonotonicity, and bursting driven by a slow passage through a bifurcation are formalized to identify the differences and limitations of the models. Notably, while both models generally capture these complex dynamics, the PWL model must be modified to exhibit the delay effect that occurs naturally in the smooth system during specific bursting scenarios. The project concludes by validating these phenomena through computer simulations and an analogue implementation of the circuit. Experimental results show that the circuit simulation using Simulation Program with Integrated Circuit Emphasis (SPICE) software fairly aligns with the models' predictions, while the physical analogue results reveal these phenomena within an unexpected range of parameter configurations.

INTRODUCTION

Dynamical systems theory is one of the most interdisciplinary fields in mathematics, drawing on real analysis, topology, linear algebra, differential geometry, and numerical methods. Furthermore, its wide-ranging applicability to physical, biological, and social phenomena makes it a vibrant branch of applied mathematics. Circuit theory, for instance, is deeply related with differential equations. Even in basic circuits featuring fundamental components like inductors and capacitors, the governing state equations naturally rely on derivatives. More specifically, circuits often involve variables that evolve on vastly different time scales. These are known as slow-fast systems, and they produce complex, widely studied dynamics.

The core of this work focuses on an electronic circuit based on a non-linear resistor, whose state equations fall under the framework of slow-fast systems. Furthermore, under specific parameter configurations, this system acts as a FitzHugh-Nagumo type model. This serves as the foundation for proposing two essentially distinct mathematical models to describe the same object: a smooth model and a PWL model. The usage of PWL models in other problems has demonstrated a dynamical richness comparable to that of their smooth counterparts, despite lacking differentiability across their entire domain. Thus, we expect to observe this likeness between models in our specific scenario. Consequently, specific slow-fast dynamics are analysed to observe how they manifest within the circuit. Finally, circuit experiments are conducted using both SPICE simulation software and a physical analogue implementation.

The remainder of this project is structured as follows. Following this introductory chapter, we introduce fundamental concepts of dynamical systems, focusing specifically on qualitative theory, linear systems, and Hopf bifurcations. This background is essential for comprehending the subsequent material. Next, we present the central circuit of the project and derive its governing state equations. From these equations, the two theoretical models are then formulated.

The main results for the models are detailed in the fourth chapter, which is the most comprehensive section of this work. First, we provide a brief review of slow-fast systems and Fenichel theory. Next, we adapt established results regarding the presence

of a Hopf bifurcation and canard explosion to the specific smooth model under study. The same structural approach is taken for the PWL model; however, all theorems and propositions are developed from scratch, inspired by literature on similar systems.

The presence of a canard explosion acts as the spark for the bursting phenomenon when these systems are subjected to periodic forcing. Up to this point, both the smooth and the PWL systems exhibit analogous phenomena, despite their specific particularities. However, applying a forcing with a sufficiently small frequency leads the smooth model to naturally exhibit a delay during the slow passage through a Hopf bifurcation. Although the PWL system does not initially behave in the same manner, we prove that it can manifest a similar delay when appropriately modified. Finally, period-doubling and antimonotonicity phenomena are reported and directly linked to the previously presented concepts of canard dynamics.

The final chapter evaluates the precision of both models by applying these theoretical scenarios to the electronic circuit to observe the complex dynamics. On the one hand, simulations are carried out using SPICE software to verify the predicted behaviours in a controlled environment. On the other hand, we build an analogue implementation of the circuit in the laboratory to validate the results in a real-world physical setting.

An extensive bibliography was consulted during the development of this project. The general concepts in the introductory chapter on dynamical systems are mostly based on [10] and [18], while the bifurcation theory and PWL results are sourced from [19], [15], and [6]. The discussion on Fenichel theory at the beginning of the fourth chapter relies on [4], [5], and [8], and the article [13] becomes essential for modifying the PWL model to manifest the delay effect. However, the most critical sources for this chapter are two pivotal articles: [9] and [2]. The formalism in [9] is adapted to our specific smooth model to prove the existence of canard dynamics. By contrast, the canard explosion in the PWL framework is backed by results formulated and proven specifically for our PWL system, based on the work in [2].

Some of the software resources used in this project can be found in the following *GitHub* public repository. We have shared *Python* scripts for data representation, *Maxima* files containing symbolic computations and SPICE schematics for circuit simulations, among others.

INTRODUCTION TO DYNAMICAL SYSTEMS

The aim of this chapter is to review some of the most fundamental concepts and results from the field of dynamical systems in order to provide the reader with the tools required to properly understand the chapters to come. Apart from a general introduction to differential equations, we will go deeper into qualitative theory and end up presenting several concepts related to the Hopf bifurcation, both the definition for the smooth case and some variants for the PWL framework. It is fundamental to understand the exposed ideas about linear systems, orbits, equilibria and Hopf bifurcations before presenting the actual core of the project.

The proofs for most of the results in the chapter will be delegated to bibliography, specially to [10] and [18] but also to [19], [15] and [6].

2.1 Ordinary Differential Equations

Let us review some basic definitions regarding the fundamental mathematical tool that will be used in the subsequent work.

Given $U \subseteq \mathbb{R}^m$, $V \subseteq \mathbb{R}^n$ and $k \in \mathbb{N}$, $C^k(U, V)$ denotes the set of functions $U \rightarrow V$ having continuous derivatives up to order k . We will abbreviate $C^k(U) = C^k(U, \mathbb{R})$.

An **Ordinary Differential Equation (ODE)** is a functional relation

$$\mathbf{F}\left(t, \mathbf{x}(t), \frac{d\mathbf{x}(t)}{dt}, \dots, \frac{d^k \mathbf{x}(t)}{dt^k}\right) = \mathbf{0}, \quad (2.1)$$

for the unknown function $\mathbf{x} \in C^k(J, \mathbb{R}^n)$, $J \subseteq \mathbb{R}$, and its derivatives. In this case, $\mathbf{F} \in C(U, \mathbb{R}^n)$ for U an open subset of $\mathbb{R}^{(k+1)n+1}$, where k is the highest derivative appearing in $\mathbf{F}$ and it is called **order** of the differential equation. The variable t is usually called the **independent** variable and $\mathbf{x}$ the **dependent** one. The derivative of a function with respect to time, $\frac{d\mathbf{x}(t)}{dt}$, will also be denoted as $\dot{\mathbf{x}}$.

Given an interval $I \subseteq J$, every function $\mathbf{x} \in C^k(I, \mathbb{R}^n)$, such that $\left(t, \mathbf{x}(t), \frac{d\mathbf{x}(t)}{dt}, \dots, \frac{d^k \mathbf{x}(t)}{dt^k}\right) \in U$, $\forall t \in I$ and satisfying equation (2.1) is called a **solution** of the ODE (2.1) in I .

The solution for an ODE is usually not unique. For instance, for any given $C \in \mathbb{R}$, $x(t) = x + C$ is a solution of the problem $\dot{x} = 1$. However, in many cases one encounters extra conditions that the solution should satisfy, such as those that impose them a value at a given time. Now, when considering the condition $x(t = 2) = 4$, there is only one solution $x(t) = x + 2$ for the previous case. The type of problem that imposes the value of the solution at a given time, $\mathbf{x}(t_0) = \mathbf{x}_0$, is known as **Cauchy Problem** or **Initial Values Problem (IVP)**. To emphasize the dependence of the solutions on the initial conditions $(t_0, \mathbf{x}_0)$, we will denote the solution of the ODE (2.1) passing through $\mathbf{x}_0$ at time $t = t_0$ by $\phi(t; t_0, \mathbf{x}_0)$.

We can recall some of the basic theorems in the field regarding the existence and uniqueness of solutions. First, let us remember a fundamental concept. A function $\mathbf{f} : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is said to be **Lipschitz** if there exists a constant $L \in \mathbb{R}$ such that $\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})\| < L \|\mathbf{x} - \mathbf{y}\|$, $\forall \mathbf{x}, \mathbf{y} \in U$, being U an open subset of $\mathbb{R}^n$. Since all norms are equivalent in $\mathbb{R}^n$, the definition does not depend on the chosen norm. When for every $\mathbf{x}_0 \in U$ there exists a neighbourhood $W_{\mathbf{x}_0}$ such that $\mathbf{f}$ is Lipschitz in $W_{\mathbf{x}_0}$, then the function is said to be **locally Lipschitz**. For instance, every $C^1(U)$ function over an open subset $U \subseteq \mathbb{R}^n$ is locally Lipschitz by the Mean Value Theorem.

Sometimes, we say that a function $\mathbf{f}$ defined in $\mathbb{R}^n$ is Lipschitz in a specific variable $\mathbf{y}$, meaning that we apply the previous definitions to $\mathbf{f} = \mathbf{f}(\mathbf{x}, \mathbf{y})$ as a function of $\mathbf{y}$ and considering $\mathbf{x}$ as parameters.

Theorem 1 (Picard). *Consider the rectangle $R = [t_0 - a, t_0 + a] \times B_b(\mathbf{x}_0)$, where $B_b(\mathbf{x}_0) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_0\| \leq b\}$ and let $\mathbf{f}(t, \mathbf{x}) : R \subset \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous and locally Lipschitz with respect to the second variable, defined in R . Then, there exists a unique solution to the IVP:*

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}), \\ \mathbf{x}(t_0) = \mathbf{x}_0, \end{cases} \quad (2.2)$$

defined in the interval $[t_0 - \alpha, t_0 + \alpha]$ where $\alpha = \min(a, \frac{b}{M})$ and $M = \max\{\|\mathbf{f}(t, \mathbf{x})\| : (t, \mathbf{x}) \in R\}$.

For a proof of the previous theorem see Theorem 2.2 in [18].

We say that $\phi : I \rightarrow U \subseteq \mathbb{R}^n$ with $\phi(t; t_0, \mathbf{x}_0)$ is a **maximal solution** of the IVP (2.2) if for every solution $\psi : J \rightarrow U$ with $\psi(t; t_0, \mathbf{x}_0)$, then $J \subseteq I$. Moreover, the interval of definition of the maximal solution is called **maximal interval of definition** and denoted $I_{(t_0, \mathbf{x}_0)}$.

Even though the Picard's Theorem gives a unique solution at local level, the following theorem (Theorem 2) extends the solution to the global domain.

Theorem 2. *Let $\mathbf{f}(t, \mathbf{x}) : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous and Lipschitz with respect to the second variable, then for the IVP (2.2) it holds that $I_{(t_0, \mathbf{x}_0)} = \mathbb{R}$.*

Proof. Consider the interval $[-m, m]$ for $m \in \mathbb{R}$ such that $m > |t_0|$. For every $\hat{t} \in [-m, m]$ there exists a unique solution defined in an interval $(\hat{t} - \alpha, \hat{t} + \alpha)$ by Picard's Theorem. Notice how one can obtain a solution in the whole interval by applying the same theorem to the new points $\hat{t} + \alpha$ and $\hat{t} - \alpha$. Hence, there is a unique solution $\phi_m(t)$ for the IVP defined in $[-m, m] \times \mathbb{R}^n$. Consider

$$\Phi : \bigcup_{m > |t_0|} [-m, m] \rightarrow \mathbb{R}^n \quad t \rightarrow \phi_m(t), \text{ if } t \in [-m, m] \quad ,$$

which is well defined because if $t \in [-m_1, m_1]$ and $t \in [-m_2, m_2]$ with $m_1 < m_2$ then $[-m_1, m_1] \subset [-m_2, m_2]$ and since there is unicity of solutions in the closed interval, $\varphi_{m_1}(t) = \varphi_{m_2}|_{[-m_1, m_1]}(t)$. Moreover, Φ is continuous and verifies the equation because every φ_m does it. Hence, it is a solution for the IVP defined in $\mathbb{R} \times \mathbb{R}^n$. $\square$

As a consequence, the solutions of linear ODE systems like $\mathbf{f}(t, \mathbf{x}) = A\mathbf{x} + \mathbf{b}$, where A is an $n \times n$ matrix and $\mathbf{b}$ an $n \times 1$ vector, are defined on the whole of $\mathbb{R}$. Indeed, as $\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})\| = \|A\mathbf{x} - A\mathbf{y}\| \leq \|A\|\|\mathbf{x} - \mathbf{y}\|$, then $\mathbf{f}$ is both locally and globally Lipschitz with respect to the second variable.

2.2 Qualitative theory

In most cases, finding an analytic expression for the solution of an ODE is difficult. As an alternative, one may use numerical methods to approximate the solution, or rely on qualitative theory to derive structural properties of the system that provide insight into its behaviour without requiring explicit knowledge of the solutions.

We will usually work with **autonomous** ODEs, that is, equations of the form:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \quad (2.3)$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ and $\mathbf{f}: D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$. The domain D of the ODE is also called **phase space** and $\mathbf{f}$ the **vector field**. From now on we will consider $\mathbf{f}$ to be locally Lipschitz. Notice how having autonomous vector fields is merely a technical issue, as non-autonomous fields $\mathbf{f}(t, \mathbf{x})$ can be rewritten as autonomous when considering a new variable $\mathbf{z} = (t, \mathbf{x})$, with the cost of having an extra state variable.

In the context of autonomous systems, we define the **orbits**.

Definition 1. For every $\mathbf{x}_0$ in the phase space D , we call **orbit** through $\mathbf{x}_0$ the set $\gamma_{\mathbf{x}_0} = \{\varphi(t; t_0, \mathbf{x}_0) : t \in I(t_0, \mathbf{x}_0)\} \subseteq D$.

The following result, which is proven at Section 2.2 in [10], justifies the importance of the orbits for the description of the system.

Proposition 1. Given $\mathbf{x}_0, \mathbf{x}_1 \in D$, if $\gamma_{\mathbf{x}_1} \cap \gamma_{\mathbf{x}_0} \neq \emptyset$ then $\gamma_{\mathbf{x}_0} = \gamma_{\mathbf{x}_1}$.

The previous statement indicates that orbits are independent from the point that refers to it and that they do not intersect each other. As a result, the phase space can be divided into its orbits, that is to say, orbits induce a partition $\{\gamma_{\mathbf{x}}\}_{\mathbf{x} \in D}$ of the phase space, what is called **phase portrait**.

The following result gives a topological description of orbits. See Theorem 2.3.2 in [10] and Appendix Classifying 1-Manifolds in [12].

Theorem 3. Every orbit is a single point, homeomorphic to a circumference $\mathbb{S}^1$ or homeomorphic to $\mathbb{R}$.

Orbits homeomorphic to circumferences will be called **periodic orbits**, while singular orbits will be called **critical points** or **equilibrium points**. A periodic orbit isolated from other periodic orbits will also be referred as a **limit cycle**. Note how critical points

are constant solutions $\phi(t; t_0, \mathbf{x}^*) = \mathbf{x}^*$, $\forall t \in I_{(t_0, \mathbf{x}^*)}$, that is to say, zeros of the vector field. The zeros of the individual vector field components will also be important.

Definition 2. Given an autonomous system (2.3), and denoting $\mathbf{f}: D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ as $\mathbf{f} = (f_1, \dots, f_n)$ with $f_i: D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, a **nullcline** is the set of points $\{\mathbf{x}: f_i(\mathbf{x}) = 0\}$. The set of equilibrium points is the intersection of all the nullclines.

The following concept will be used to classify the behaviour of the system in terms of its orbits.

Definition 3. Given the ODE (2.3), consider $\mathbf{f}: U \rightarrow \mathbb{R}^n$ to be locally Lipschitz on $U \subseteq \mathbb{R}^n$. Supposing that for every $\mathbf{x} \in U$, $I_{(0, \mathbf{x})} = \mathbb{R}$, then the function $\Phi: \mathbb{R} \times U \rightarrow \mathbb{R}^n$ given that $\Phi(t, \mathbf{x}) = \phi(t; 0, \mathbf{x})$ is called the **flow** of the system.

A subset $E \subseteq U$ is said to be **positively invariant** if $\forall \mathbf{q} \in E$ then $\Phi(t, \mathbf{q}) \in E$, $\forall t \geq 0$. Conversely, E is said to be **negatively invariant** if $\forall \mathbf{q} \in E$ then $\Phi(t, \mathbf{q}) \in E$, $\forall t \leq 0$. A set is **invariant** when being both positively and negatively invariant. For example, a singular point is an invariant set.

An invariant set E is called **stable** when for any neighbourhood W of E , there exists a neighbourhood V of E such that $\forall \mathbf{p} \in V$ then $\Phi(t, \mathbf{p}) \in W$, $\forall t > 0$. When not being stable, the invariant set is **unstable**.

Given points $\mathbf{p}, \mathbf{q} \in U$, $\mathbf{q}$ is an **α -limit point** of $\mathbf{p}$ if there exists a sequence of numbers $\{t_n\}_{n=0}^\infty$ approaching $-\infty$ such that $\lim_{n \rightarrow \infty} \Phi(t_n, \mathbf{p}) = \mathbf{q}$ and an **ω -limit point** of $\mathbf{p}$ if there is a sequence of numbers $\{t_n\}_{n=0}^\infty$ approaching $+\infty$ such that $\lim_{n \rightarrow \infty} \Phi(t_n, \mathbf{p}) = \mathbf{q}$. The **α -limit set** ($\alpha(\mathbf{p})$) and **ω -limit set** ($\omega(\mathbf{p})$) of a given point $\mathbf{p}$ are made up of the union of all of its α -limit and ω -limit points, respectively.

Given a set E , we define its **stable manifold** $W^s(E)$ as the union of points whose ω -limit set is contained in E and its **unstable manifold** $W^u(E)$ as the union of points whose α -limit set is contained in E . E is said to be **asymptotically stable** or **asymptotically unstable** if $W^s(E)$ or $W^u(E)$ is a neighbourhood of E , respectively.

As an example of a stable but not asymptotically stable manifold, consider the ODE

$$\dot{x} = f(x) := \begin{cases} 0 & \text{if } x = 0, \\ -x^3 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0, \end{cases}$$

where $f \in C^1$. The equilibria of the system are points $x = 0$ and $x = \frac{1}{k\pi}$, $k \in \mathbb{Z}$. Following the criteria discussed after Theorem 6, points $\frac{1}{(2k-1)\pi}$ with $k > 0$ and $\frac{1}{2k\pi}$ with $k < 0$ are asymptotically unstable because they have a negative derivative.

Given a neighbourhood W of 0 we can find two other critical points $\frac{1}{k\pi}$ both at the right and at the left of $x = 0$ contained in W , such that their stable manifolds are completely contained in the neighbourhood. The neighbourhood V of $x = 0$ whose borders coincide with the leftmost and rightmost borders of the previous stable manifolds is made up of points $p \in V$ such that $\Phi(t, p) \in W$. However, for the same reason the set $\{0\}$ is not asymptotically stable, since solutions get close to other equilibria near to the point.

2.2.1 Linear systems

Studying the phase portrait of linear systems will allow the further local comprehension of portraits around equilibria in non-linear systems, see Theorem 6 at the end of the following section. For this reason, we offer a detailed study of linear systems.

We first consider homogeneous linear systems of the form

$$\dot{\mathbf{x}} = A\mathbf{x}, \quad (2.4)$$

where $A \in L(\mathbb{R}^n)$, being $L(\mathbb{R}^n)$ the vector space of linear maps from $\mathbb{R}^n$ to $\mathbb{R}^n$. The set $GL(\mathbb{R}^n)$ will be used for referring invertible linear maps. We will also denote $d = \det(A)$ and $t = \text{trace}(A)$, so from now on the independent variable of the system will be denoted as s , this is: $\mathbf{x} := \mathbf{x}(s)$.

Let us indicate by $\{\mathbf{v}_i\}_{i=0}^{n_s}$ the generalized eigenvectors of A with negative real part; $\{\mathbf{u}_i\}_{i=0}^{n_u}$ the corresponding ones with positive real part and $\{\mathbf{w}_i\}_{i=0}^{n_c}$ the ones with zero real part. In the context of linear systems one defines the **stable subspace** as $E^s := \text{span}\{\{\mathbf{v}_i\}_{i=0}^{n_s}\}$, the **unstable subspace** as $E^u := \text{span}\{\{\mathbf{u}_i\}_{i=0}^{n_u}\}$ and the **central subspace** as $E^c := \text{span}\{\{\mathbf{w}_i\}_{i=0}^{n_c}\}$. The following theorem, presented as Theorem 1 of Section 1.9 in [14], provides a description of the dynamics of the system.

Theorem 4 (Dynamical behaviour of linear systems). *For a linear differential system of the form $\dot{\mathbf{x}} = A\mathbf{x}$ with $A \in GL(\mathbb{R}^n)$ then:*

- i) $\mathbb{R}^n = E^s \oplus E^u \oplus E^c$,
- ii) $W^s(\mathbf{0}) = E^s$,
- iii) $W^u(\mathbf{0}) = E^u$.

Consider now, non-homogeneous linear systems. Solving a non-homogeneous linear system having solved the homogeneous version is a straightforward computation. Suppose $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{b}$, with $A \in GL(\mathbb{R}^n)$, $\mathbf{b} \in \mathbb{R}^n$, and let $\mathbf{e} = -A^{-1}\mathbf{b}$. The change of variables $\mathbf{z} = \mathbf{x} - \mathbf{e}$ leads to $\dot{\mathbf{z}} = \dot{\mathbf{x}} = A(\mathbf{z} + \mathbf{e}) + \mathbf{b} = A\mathbf{z}$. Thus, the general solution $\boldsymbol{\psi}(s, \mathbf{x})$ of the non-homogeneous system is $\boldsymbol{\psi}(s, \mathbf{x}) = \mathbf{e} + \boldsymbol{\varphi}(s, \mathbf{x} - \mathbf{e})$, where $\boldsymbol{\varphi}$ denotes the solution of the homogeneous system.

Planar linear systems

Any matrix $A \in L(\mathbb{R}^n)$ can be related to an equivalent Jordan's normal matrix J through a relation $J = PAP^{-1}$, with $P \in GL(\mathbb{R}^n)$. In the case of planar systems, the change of coordinates $\mathbf{y} = P\mathbf{x}$ transforms the system (2.4) to

$$\dot{\mathbf{y}} = J\mathbf{y}, \quad (2.5)$$

with J looking like one of the following real matrices:

$$J = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}, \quad \text{or} \quad J = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}.$$

The following theorem (see Theorem 2.5.4 in [10]) provides the form of the Jordan's normal equivalent matrix depending on the value of the trace and the determinant of a matrix.

Theorem 5 (Real Jordan's normal form). *Consider a matrix $A \in L(\mathbb{R}^2)$ with $d = \det(A)$ and $t = \text{trace}(A)$. A is equivalent to one of the following matrices J :*

i) *If $d = 0$ and $t = 0$, then*

$$J = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{or} \quad J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

ii) *If $d = 0$ and $t \neq 0$, then*

$$J = \begin{pmatrix} t & 0 \\ 0 & 0 \end{pmatrix}.$$

iii) *If $d > 0$ and $t = 0$, the eigenvalues of A are complex numbers with zero real part and imaginary part $\beta > 0$, and*

$$J = \begin{pmatrix} 0 & -\beta \\ \beta & 0 \end{pmatrix}.$$

iv) *If $d > 0$ and $t^2 - 4d = 0$, there exists exactly one real eigenvalue of A with multiplicity two, λ_1 , and*

$$J = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_1 \end{pmatrix} \quad \text{or} \quad J = \begin{pmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{pmatrix}.$$

v) *If $d > 0$ and $t^2 - 4d > 0$, there exist two real eigenvalues of A , $\lambda_1 > \lambda_2$, and*

$$J = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

vi) *If $d > 0$, $t \neq 0$ and $t^2 - 4d < 0$, the eigenvalues of A are complex numbers with real part $\alpha \neq 0$ and imaginary part $\beta > 0$, and*

$$J = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}.$$

vii) *If $d < 0$, there exist two real eigenvalues of A , $\lambda_1 > 0 > \lambda_2$, and*

$$J = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

We can limit our study to the latter systems (2.5) in order to obtain the qualitative information of a generic linear planar system of the form (2.4). Indeed, given flows $\Phi(s, \mathbf{x})$, $\Psi(s, \mathbf{y})$ of systems (2.4) and (2.5), then $\Phi(s, \mathbf{x}) = P^{-1}\Psi(s, P\mathbf{x})$. Thus, the phase portrait of the system (2.4) is a linear transformation of the phase portrait of the system (2.5).

The following proposition will be useful to describe the phase portrait of the planar system (2.5). See section 3.2 in [18] for a further development.

Proposition 2. *Consider $J \in L(\mathbb{R}^2)$ and $s \in \mathbb{R}$. Then, the flow of the system (2.5) is defined by $\Psi(s, \mathbf{y}) = e^{sJ}\mathbf{y}$.*

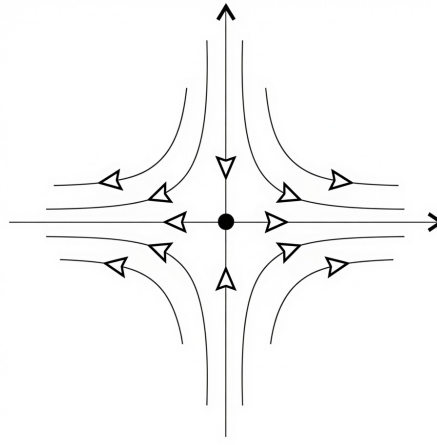


Figure 2.1: Phase portrait of a saddle point. (Image adapted from Chapter 2, Section 2.5.3 in [10].)

$$i) \text{ If } J = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \text{ then } e^{sJ} = \begin{pmatrix} e^{s\lambda_1} & 0 \\ 0 & e^{s\lambda_2} \end{pmatrix}.$$

$$ii) \text{ If } J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}, \text{ then } e^{sJ} = e^{s\lambda} \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}.$$

$$iii) \text{ If } J = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}, \text{ then } e^{sJ} = e^{s\alpha} \begin{pmatrix} \cos(\beta s) & -\sin(\beta s) \\ \sin(\beta s) & \cos(\beta s) \end{pmatrix}.$$

Consider the planar matrix A , we will present the possible scenarios of the phase portrait of (2.4) by taking into account the Proposition 2:

Case $d < 0$:

According to the Jordan's Theorem (Theorem 5) the Jordan's normal matrix has the form $J = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$, with $\lambda_1 > 0 > \lambda_2$. By the Theorem 4, the stable and unstable subspaces of the origin are generated by a single eigenvector. The origin is the only equilibrium point, which is called **saddle point**. The phase portrait can be seen in the Figure 2.1.

Case $d = 0$:

If A is the zero matrix, then $\dim(\ker(A)) = 2$ and every point is a singular point. Otherwise, there is a line of singular points through the origin. If $t = 0$ the straight line is called **non-isolated nilpotent manifold** (see Figure 2.2 (b)); if $t < 0$ (or $t > 0$) the line is called **stable (unstable) normally hyperbolic manifold** (see Figure 2.2 (a) - respectively (c)-). Moreover, the line of equilibria matches the central space E^c , and according to Definition 11 every critical point is normally hyperbolic.

Case $d > 0$:

If $t^2 - 4d < 0$, the matrix A has a pair of conjugate complex eigenvalues and $J = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}$. The phase portrait depends on the sign of $t = 2\alpha$, as can be seen in the Figure

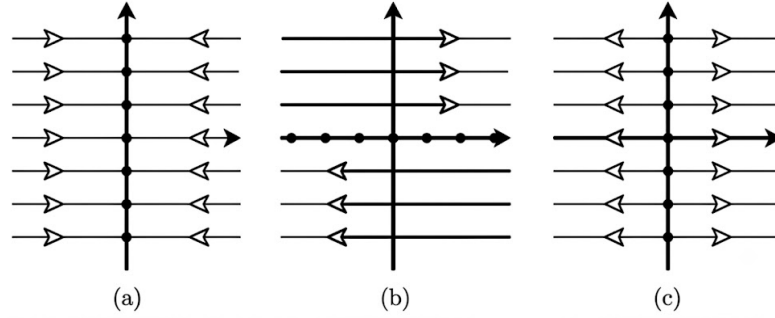


Figure 2.2: Phase portrait of a non-isolated singular points: (a) Stable normally hyperbolic manifold ($t < 0$); (b) Non-isolated nilpotent manifold ($t = 0$); (c) unstable normally hyperbolic manifold $t > 0$. (Image adapted from Chapter 2, Section 2.5.3 in [10].)

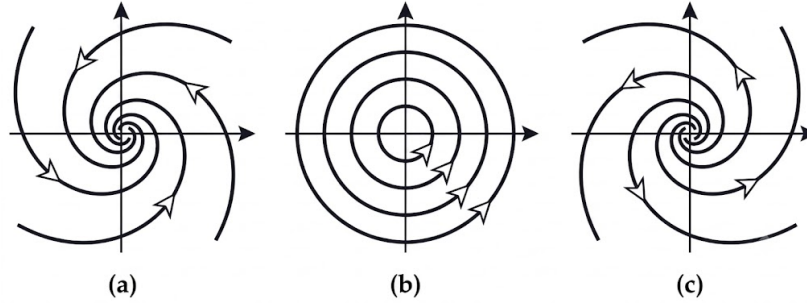


Figure 2.3: (a) Asymptotically unstable focus ($t < 0$); (b) Center ($t = 0$); (c) Asymptotically unstable focus $t > 0$. (Image adapted from Chapter 2, Section 2.5.3 in [10].)

2.3. We will call the origin a **center** when $t = 0$ (see Figure 2.3 (b)), an **asymptotically stable focus** when $t < 0$ (see Figure 2.3 (a)) and an **asymptotically unstable focus** when $t > 0$ (see Figure 2.3 (c)).

If $t^2 - 4d = 0$, the Jordan's form is like $J = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$ or $J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$. In the first case we have **degenerated diagonal nodes** (see Figure 2.4 (a)), both stable and unstable; and **degenerated nodes** (see Figure 2.4 (b)) in the second case.

If $t^2 - 4d > 0$ there are two eigenvalues with the same sign. Consequently, $W^s(\mathbf{0}) = \mathbb{R}^2$ when $t < 0$, what is called an **asymptotically stable node** (see Figure 2.5 (a)); and $W^u(\mathbf{0}) = \mathbb{R}^2$ when $t > 0$, an **asymptotically unstable node** (see Figure 2.5 (b)).

2.3 Non-Linear Systems

The standard procedure to study the local dynamics of non-linear systems near their singular points is to consider simpler linear systems that behave in an analogous way at local level. The theorem presented at the end of the section (Theorem 6) formalises this approach. However, we first need to introduce the notion of conjugacy and equivalence.

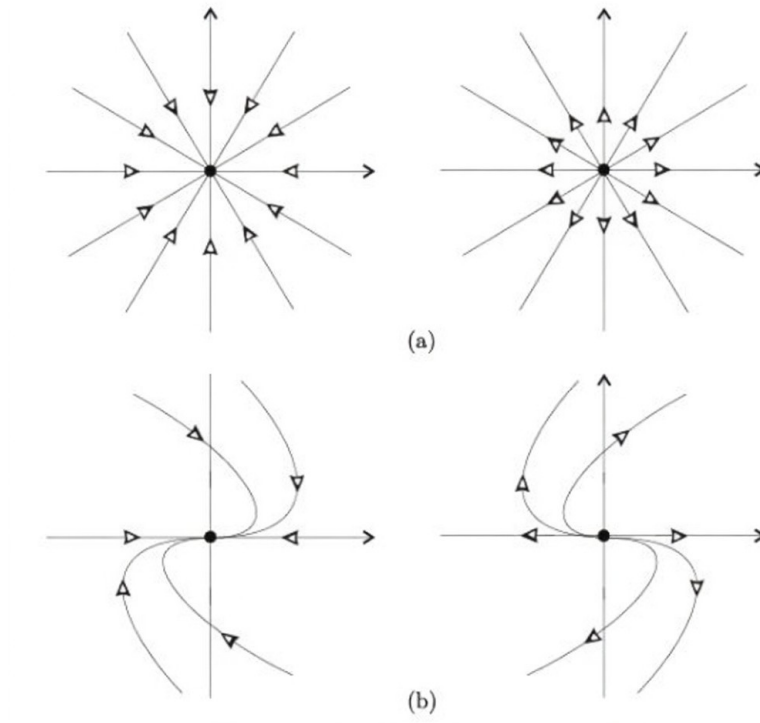


Figure 2.4: (a) Degenerated diagonal nodes (left $t < 0$; right $t > 0$); (b) Degenerated diagonal nodes (left $t < 0$; right $t > 0$) (Image adapted from Chapter 2, Section 2.5.3 in [10].)

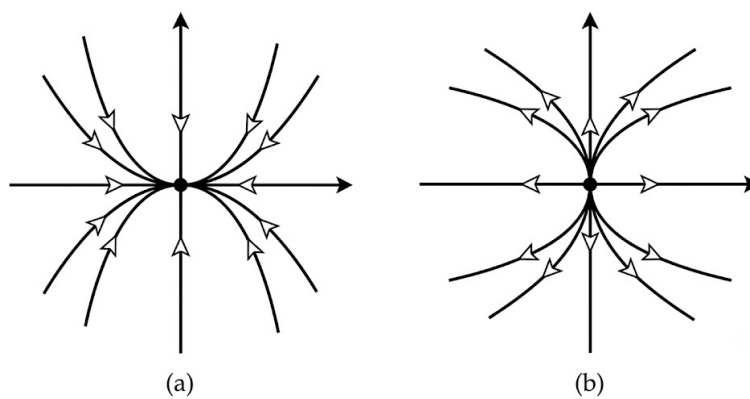


Figure 2.5: (a) Asymptotically stable node ($t < 0$); (b) Asymptotically unstable node ($t > 0$) (Image adapted from Chapter 2, Section 2.5.3 in [10])

Consider two different ODEs of the form (2.3), $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ and $\dot{\mathbf{y}} = \mathbf{g}(\mathbf{y})$ with locally Lipschitz functions on open subsets $U, V \subseteq \mathbb{R}^n$ and functions $\mathbf{f}: U \rightarrow \mathbb{R}^n$ and $\mathbf{g}: V \rightarrow \mathbb{R}^n$. Given their respective flows $\Phi(s, \mathbf{x}), \Phi^*(s, \mathbf{y})$, they are said to be **conjugate** if there exists a **conjugacy** bijection $\mathbf{h}: U \rightarrow V$ such that $\Phi^*(s, \mathbf{h}(\mathbf{x})) = \mathbf{h}(\Phi(s, \mathbf{x}))$, $\forall s \in \mathbb{R}, \forall \mathbf{x} \in U$. The conjugacy is said to be topological when $\mathbf{h}$ is a homeomorphism, differential when $\mathbf{h}$ is a diffeomorphism, and linear when it is a linear mapping.

Moreover, flows are called **equivalent** if there is an **equivalence** bijection $\mathbf{h}: U \rightarrow V$ such that γ is an orbit in one system if and only if $\mathbf{h}(\gamma)$ is also an orbit in the other system and with the same orientation. Notice how the concept of conjugacy is more restrictive than the equivalence because the first requires a coincidence in time while the latter admits time reparametrizations.

Definition 4. *The singular point $\mathbf{x}_0$ of a system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, with $\mathbf{f}$ a C^1 function in a neighbourhood of $\mathbf{x}_0$, is said to be **hyperbolic** if all the eigenvalues of $D\mathbf{f}(\mathbf{x}_0)$ have a non-zero real part.*

Theorem 6 (Hartman-Grobman). *Consider the autonomous ODE $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ with $\mathbf{f}: U \rightarrow \mathbb{R}^n$ a $C^1(U, \mathbb{R}^n)$ function, $\Phi(s, \mathbf{x})$ the flow of the ODE and $\mathbf{x}_0$ an hyperbolic singular point. Then, there exist two neighbourhoods W, V of the point $\mathbf{x}_0$ and the origin, respectively, an homeomorphism $\mathbf{h}: W \rightarrow V$ with $\mathbf{h}(\mathbf{x}_0) = \mathbf{0}$, and an interval $I \subseteq \mathbb{R}$ around the origin such that*

$$\mathbf{h} \circ \Phi(s, \mathbf{x}) = e^{sD\mathbf{f}(\mathbf{x}_0)} \mathbf{h}(\mathbf{x}), \quad \forall s \in I, \forall \mathbf{x} \in U.$$

A complete proof can be found in Theorem 9.9 in [18]. The Hartman-Grobman's Theorem asserts that under certain hypothesis, the system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ is topologically conjugated to the linearized system $\dot{\mathbf{x}} = D\mathbf{f}(\mathbf{x}_0)\mathbf{x}$ in a neighbourhood of an hyperbolic singular point $\mathbf{x}_0$ and a neighbourhood of the origin, respectively.

Let us evaluate in more detail the consequences of the Theorem 6 for autonomous systems of dimension one and two. In the case of unidimensional autonomous systems, the Jacobian matrix reduces to the derivative, so an hyperbolic critical point x_0 of $\dot{x} = f(x)$, with $f \in C^1(\mathbb{R})$, is asymptotically stable when $f'(x_0) < 0$ and unstable if $f'(x_0) > 0$. In the case of planar systems, the character of an hyperbolic equilibrium point $\mathbf{x}_0$ can be determined through the study of its Jacobian matrix following the development in Section 2.2.1. In particular, when both eigenvalues are real-valued and negative (positive) the equilibrium is asymptotically stable (unstable) and when they have different sign $\mathbf{x}_0$ is a saddle point. In the case of complex eigenvalues, $\mathbf{x}_0$ is a focus except for the case of a zero trace, when the Hartman-Grobman Theorem does not apply and thus does not distinguish between focus and center.

2.4 Piecewise Linear Systems

Piecewise dynamical systems are ODE systems for which the domain of definition is partitioned into sub-domains where a different analytical system is defined for each. In particular, the sub-systems that are considered in this work are linear systems. We refer to these systems as PWL systems. Even when being continuous, the differentiability at the borders of different sub-domains is not guaranteed. Consequently, many of the preceding results cannot be applied globally. Moreover, further concepts, like bifurcations

that will be treated in the following section, need to be redefined. Nonetheless, a rich variety of qualitative behaviour occurs anyway.

We present a general form for a three-zone planar PWL differential system with two parallel straight lines as switching curves. Observe that the representation is not restrictive since a change of variables allows to relocate any pair of vertical switching lines to $x = \pm 1$.

$$\dot{\mathbf{x}} = \begin{cases} A_L \mathbf{x} + \mathbf{b}_L, & \text{if } x < -1, \\ A_C \mathbf{x} + \mathbf{b}_C, & \text{if } |x| \leq 1, \\ A_R \mathbf{x} + \mathbf{b}_R, & \text{if } x > 1. \end{cases} \quad (2.6)$$

If we also consider the vector field in the previous system (2.6) to be continuous, the matrices must be of the following form

$$\begin{aligned} A_L &= \begin{pmatrix} a_{11}^L & a_{12} \\ a_{21}^L & a_{22} \end{pmatrix}, \quad A_C = \begin{pmatrix} a_{11}^C & a_{12} \\ a_{21}^C & a_{22} \end{pmatrix}, \quad A_R = \begin{pmatrix} a_{11}^R & a_{12} \\ a_{21}^R & a_{22} \end{pmatrix}, \\ \mathbf{b}_L &= \begin{pmatrix} b_1 + a_{11}^L - a_{11}^C \\ b_2 + a_{21}^L - a_{21}^C \end{pmatrix}, \quad \mathbf{b}_C = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad \mathbf{b}_R = \begin{pmatrix} b_1 - a_{11}^R + a_{11}^C \\ b_2 - a_{21}^R + a_{21}^C \end{pmatrix}. \end{aligned} \quad (2.7)$$

with $a_{i,j}^k, a_{i,j}, b_i \in \mathbb{R}$.

In the latter case, the PWL system is Lipschitz just as any linear system by considering the Lipschitz constant to be the maximum of the Lipschitz constants of all the subdomains of definition. Furthermore, Picard's Theorem 1 and Theorem 2 ensure the uniqueness of solutions defined on the whole of $\mathbb{R}$.

Taking an adequate change of variables (Proposition 2.2 in [15]) the preceding system can be reduced to the following normal form.

Proposition 3 (Liénard form). *The condition $a_{12} \neq 0$ is sufficient to write system (2.6) satisfying (2.7) in Liénard form:*

$$\begin{cases} \dot{x} = F(x) - y, \\ \dot{y} = g(x) - \lambda, \end{cases} \quad (2.8)$$

where

$$F(x) = \begin{cases} t_L(x+1) - t_C & \text{if } x < -1, \\ t_C x & \text{if } |x| \leq 1, \\ t_R(x-1) + t_C & \text{if } x > 1, \end{cases}$$

and

$$g(x) = \begin{cases} d_L(x+1) - d_C & \text{if } x < -1, \\ d_C x & \text{if } |x| \leq 1, \\ d_R(x-1) + d_C & \text{if } x > 1, \end{cases}$$

for $t_i, d_j, \lambda \in \mathbb{R}$.

Note that the Liénard form (2.8) can be a two-zone system if $t_C = t_L$ and $d_C = d_L$; and a three-zone odd symmetric function if $t_R = t_L$, $d_R = d_L$ and $\lambda = 0$.

2.5 Bifurcations

In this section we consider a parametric family of differential equations $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \lambda)$. When fixing an hyperbolic singular point, the Hartman-Grobman's Theorem (Theorem 6) allows us to study the dynamics of the system around the point through the linear-systems' theory. Moreover, since the local behaviour around the equilibrium is determined by the spectrum of a matrix, and it varies continuously with the parameter, small changes in λ will not produce significant qualitative changes in the flow. However, small changes in the parameter value around non-hyperbolic equilibria could cause qualitative changes in the local dynamics, thus producing a **bifurcation** (in a local way).

Even though some bifurcations may produce a change in the global dynamics, we will focus on a specific case of local bifurcation.

2.5.1 Hopf bifurcations

Following Section 2.10.2 in [10] we provide a topological description of Hopf bifurcation.

Definition 5. *The system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, $\mathbf{f}$ defined in $E \subseteq \mathbb{R}^n$ and Lipschitz, is said to present a **supercritical Hopf bifurcation** at the critical point $\mathbf{x}_0$ for the bifurcation value $\lambda = \lambda_0$ if:*

- i) *for $\lambda \leq \lambda_0$, there does not exist any more singular points in a given neighbourhood U of $\mathbf{x}$, which is stable (respectively, unstable);*
- ii) *for $\lambda = \lambda_0$, $\mathbf{x}_0$ is the unique singular point in U .*
- iii) *for $\lambda \geq \lambda_0$, there is only one singular point $\mathbf{x}_0$ and one limit cycle γ in U . The singular point has become unstable (respectively, stable) and the limit cycle, whose amplitude tends to 0 as λ tends to λ_0 is stable (respectively, unstable).*

*In the case the limit cycle appears for values $\lambda < \lambda_0$ and disappears for $\lambda > \lambda_0$ it is called a **subcritical Hopf bifurcation**.*

We can easily think of a general mechanism that leads to a Hopf bifurcation. Consider the matrix $D\mathbf{f}(\mathbf{x}_0, \lambda_0) \in L(\mathbb{R}^2)$ to have a pair of imaginary eigenvalues. When the eigenvalues cross the imaginary axis at λ_0 , changing their real parts from negative to positive, the equilibrium loses stability and gets surrounded by spiral orbits escaping from the origin. Under the assumption that the global dynamics do not change, an invariant limit cycle appears at the border of the local and global behaviour

The following version of the Hopf bifurcation's Theorem for planar systems taken from [19] makes the above precise.

Theorem 7 (Hopf). *Consider the planar system $\begin{cases} \dot{x} = f(x, y, \lambda) \\ \dot{y} = g(x, y, \lambda) \end{cases}$ where λ is a parameter, and $f, g \in C^3$. Suppose it has an equilibrium point $(x, y)^T = (x_0, y_0)^T$ and let $\mu(\lambda)$, $\bar{\mu}(\lambda) = \alpha(\lambda) \pm i\beta(\lambda)$ be the eigenvalues of the Jacobian matrix of the system. Suppose further that for $\lambda = \lambda_0$, the following conditions are satisfied:*

- i) $\alpha(\lambda_0) = 0$, $\beta(\lambda_0) = \omega \neq 0$ where $\text{sgn}(\omega) = \text{sgn}\left(\frac{\partial g_\lambda}{\partial x}\bigg|_{(x_0, y_0), \lambda=\lambda_0}\right)$

This is the non-hyperbolicity condition.

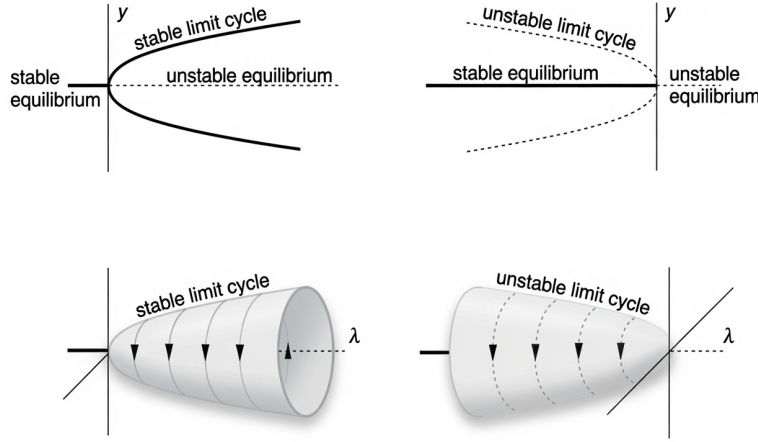


Figure 2.6: Supercritical (left) and subcritical (right) Hopf bifurcations (Image adapted from [7]).

$$ii) \left. \frac{d\alpha(\lambda)}{d\lambda} \right|_{\lambda=\lambda_0} := d \neq 0.$$

This is the transversality condition: it ensures that the eigenvalues cross the imaginary axis with non-zero speed.

iii) $a \neq 0$, where

$$a = \frac{1}{16} (f_{xxx} + f_{xyy} + g_{xxy} + g_{yyy}) + \frac{1}{16\omega} (f_{xy}(f_{xx} + f_{yy}) - g_{xy}(g_{xx} + g_{yy}) - f_{xx}g_{xx} + f_{yy}g_{yy})$$

This is the genericity condition.

Then a unique curve of periodic solutions bifurcates from the fixed point into the region $\lambda > \lambda_0$ if $ad < 0$, or $\lambda < \lambda_0$ if $ad > 0$.

The fixed point is stable for $\lambda > \lambda_0$ (resp. $\lambda < \lambda_0$) and unstable for $\lambda < \lambda_0$ (resp. $\lambda > \lambda_0$) if $d < 0$ (resp. $d > 0$), whilst the periodic solutions are stable (resp. unstable) if the fixed point is unstable (resp. stable) on the side of $\lambda = \lambda_0$ where the periodic solutions exist.

The amplitude of the periodic orbits grows like $\sqrt{|\lambda - \lambda_0|}$, whilst their periods tend to $\frac{2\pi}{|\omega|}$ as $\lambda \rightarrow \lambda_0$.

2.5.2 Hopf-like bifurcations

When we consider a PWL system with a parameter λ like in (2.8), the Jacobian is constant in each subdomain of definition of the system, meaning that changes in the parameter do not involve changes in the trace of the matrix. Therefore, no Hopf bifurcations are expected to happen within the subdomains of definition. Still, changes in the eigenvalues may produce changes of stability when transitioning through a switching line of the model. Even though the presented definition of Hopf bifurcation does

not require the system to be smooth, the concept of Hopf is usually related to that of Theorem 7. For the sake of convention, we will refer to **Hopf-like bifurcations** in the PWL case when the observed phenomena follows the topological definition of a Hopf.

Let us introduce the concept of a **Boundary Equilibrium Bifurcation (BEB)**. Consider a piecewise system defined

$$\dot{\mathbf{x}} = \begin{cases} \mathbf{f}_L(\mathbf{x}, \lambda), & \text{if } h(\mathbf{x}) \leq 0, \\ \mathbf{f}_R(\mathbf{x}, \lambda), & \text{if } h(\mathbf{x}) > 0, \end{cases}$$

where $\mathbf{f}_L, \mathbf{f}_R$ are functions in $\mathbb{R}^n$, function $\mathbf{h} : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is $\mathbf{h} \in C^k(D)$, $k \geq 1$ and $\lambda \in \mathbb{R}$. Notice how the definition involves a generic piecewise system rather than a PWL.

Definition 6. *Given an equilibria branch $\mathbf{x}(\lambda)$ such that $\mathbf{f}_L(\mathbf{x}(\lambda), \lambda) = 0$ for all λ in a neighbourhood of a critical value for the parameter, $\lambda = \lambda_0$, and $h(\mathbf{x}(\lambda_0)) = 0$, then the system exhibits a **BEB** at $\mathbf{x}(\lambda_0)$.*

BEB in systems with two zones

Let us consider a two-zone system in a form (2.8) with a parameter λ and centered at $x = 0$. That is:

$$\begin{cases} \dot{x} &= F(x) - y, \\ \dot{y} &= g(x) - \lambda, \end{cases} \quad (2.9)$$

where

$$F(x) = \begin{cases} t_L x & \text{if } x < 0, \\ t_R x & \text{if } x \geq 0, \end{cases}$$

and

$$g(x) = \begin{cases} d_L x & \text{if } x < 0, \\ d_R x & \text{if } x \geq 0, \end{cases}$$

for $t_i, d_j, \lambda \in \mathbb{R}$.

One clearly sees that the traces and determinants for the flow of the system match with the values $t = t_L, d = d_L$ for $x < 0$ and $t = t_R, d = d_R$ for $x \geq 0$. Thus, the behaviour of equilibria in each linear subsystem can be directly studied from the parameters of the system and the criteria in Section 2.2.1.

The following result (Theorem 8) illustrates how a limit cycle emerges from a BEB due to a change in stability. These phenomena are classified under the framework of **Hopf-like bifurcations** as they follow the topological definition provided in the previous section. The proof for this result can be found in Theorem 5.1 of Chapter 5 of the book [15].

Theorem 8. *Consider the system (2.9) with $d_L > 0, d_R > 0, t_L < 0$, and $t_R > 0$. Selecting λ as the bifurcation parameter, the following statements hold:*

i) (Supercritical node-focus transition with one limit cycle)

If $t_L^2 - 4d_L \geq 0$ and $t_R^2 - 4d_R < 0$, then for $\lambda < 0$ the equilibrium point is a globally attracting node while for $\lambda > 0$ the equilibrium becomes an unstable focus surrounded by one stable limit cycle. For $\lambda = 0$ the equilibrium point is a global attracting node-focus.

ii) **(Subcritical node-focus transition with one limit cycle)**

If $t_L^2 - 4d_L < 0$ and $t_R^2 - 4d_R \geq 0$, then for $\lambda < 0$ the equilibrium point is a stable focus surrounded by one unstable limit cycle, while for $\lambda > 0$ the equilibrium becomes an unstable node, which is then a global repeller. For $\lambda = 0$ the equilibrium point is a repelling node-focus.

iii) **(Supercritical focus-focus transition with one limit cycle)**

If $t_L^2 - 4d_L < 0$, $t_R^2 - 4d_R < 0$, and furthermore

$$\frac{t_L}{\sqrt{d_L}} + \frac{t_R}{\sqrt{d_R}} < 0, \quad (5.4)$$

then for $\lambda \leq 0$ the equilibrium point is a globally attracting focus while for $\lambda > 0$ the equilibrium becomes an unstable focus surrounded by one stable limit cycle.

iv) **(Subcritical focus-focus transition with one limit cycle)**

If $t_L^2 - 4d_L < 0$, $t_R^2 - 4d_R < 0$, and furthermore

$$\frac{t_L}{\sqrt{d_L}} + \frac{t_R}{\sqrt{d_R}} > 0, \quad (5.5)$$

then for $\lambda < 0$ the equilibrium point is an attracting focus surrounded by one unstable limit cycle while for $\lambda \geq 0$ the equilibrium becomes an unstable focus, which is then a global repeller.

v) **(Degenerate focus-focus transition or 'center' BEB)**

If $t_L^2 - 4d_L < 0$, $t_R^2 - 4d_R < 0$, and furthermore

$$\frac{t_L}{\sqrt{d_L}} + \frac{t_R}{\sqrt{d_R}} = 0, \quad (5.6)$$

then for $\lambda < 0$ the equilibrium point is a globally attracting focus, a global center for $\lambda = 0$, and an unstable focus for $\lambda > 0$; in this last case, the point is a global repeller.

In all the cases, when there exists a limit cycle, its size grows linearly with $|\lambda|$.

See how cases i), iii) are equivalent for a Hopf supercritical bifurcation and cases ii), iv) for a Hopf subcritical bifurcation in the smooth case. Also, it is important to notice that in the PWL context the growth of the limit cycle is linearly proportional to the size of the parameter, meaning the distance to the critical point of the parameter, while in the smooth case the growth is faster and proportional to the square root of the distance to the bifurcation value.

Hopf-like bifurcations in systems with three zones

While the BEB bifurcation replicates the topological description of a Hopf with a parameter λ determining different critical points, the Hopf-like bifurcation presented below reproduces the mechanism of change of stability involved in the classical Hopf phenomenon by maintaining the same equilibrium point.

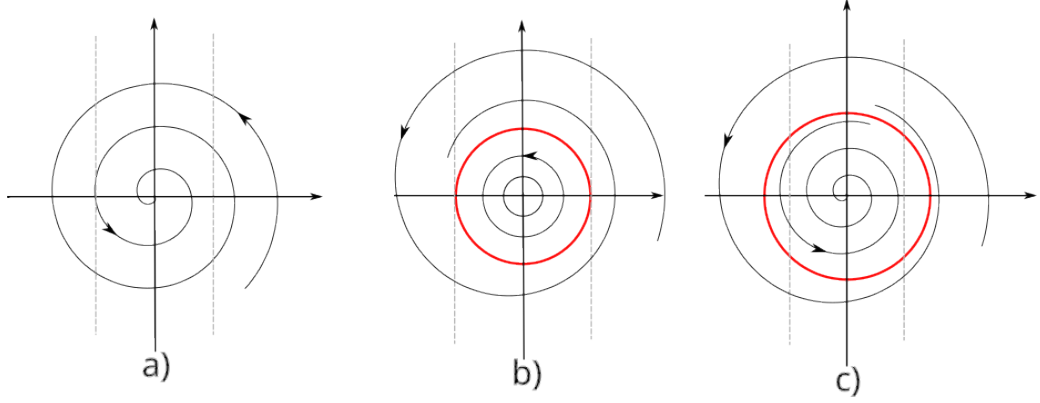


Figure 2.7: Cases described in Theorem 9 according to value of t_C . Specifically, $t_C < 0$ in a), $t_C = 0$ in b) and $t_C > 0$ in c).

Consider the symmetrical system of three zones (2.8) in Liénard form, meaning $t_L = t_R$ and $d_L = d_R$, where parameter λ is omitted. In this specific case, the bifurcation parameter is the trace t_C , thus it illustrates how changes in dynamics are produced exclusively by changes in the eigenvectors of the central subsystem containing an equilibrium.

Theorem 9. Assume for the above system that $t_L = t_R < 0$, $d_L = d_R > 0$ and $4d_C - t_C^2 > 0$. The following statements hold:

- i) The system has only one equilibrium point at the origin.
- ii) If $t_C < 0$ the origin is a globally stable equilibrium.
- iii) If $t_C = 0$ the system has an attractive center restricted to the zone $|x| < 1$, that gives place to a unique stable limit cycle for $t_C > 0$.

A proof for the result can be found in Theorem 1 in [6]. Figure 2.7 illustrates the birth of a limit cycle from the outer boundary of a period annulus through the mechanism described in Theorem 9.

THE CIRCUIT AND ITS MODELS

Following a comprehensive introduction to dynamical systems, we now introduce the specific models central to this project. Our objective is to evaluate two parallel mathematical approaches that explain the behaviour of the same system. In particular, we present a smooth and a PWL system of differential equations modelling a custom electronic circuit. Because this circuit implements the FitzHugh-Nagumo neuronal model, both theoretical models are framed within this same context.

This chapter first presents the circuit diagram and derives its state equations, subsequently defining the smooth and PWL models derived from them.

The starting point is a simple electronic circuit that resembles a slow-fast system (See Chapter 4) like:

$$\begin{cases} \dot{x} = f(x, y, \lambda, \varepsilon) - y, \\ \dot{y} = \varepsilon(x - a - P \sin(2\pi f t)), \end{cases} \quad (3.1)$$

where $0 < \varepsilon \ll 1$. The oscillatory term might be left out depending on the phenomena to be studied.

The profile of the function f is quite definitory, and will be determined by a non-linear resistor. When resembling a specific cubic polynomial, $f(x) = x - \frac{x^3}{3}$, the system (3.1) matches the schema of a **FitzHugh-Nagumo** type model of a neuron.

Just as a remark, the FitzHugh-Nagumo model usually comes with the forcing term written in the first equation, like

$$\begin{cases} \dot{x} = x - \frac{x^3}{3} - y + I(t), \\ \dot{y} = \varepsilon(x - a). \end{cases}$$

By considering $I(t) = -\frac{\varepsilon P}{2\pi f} \cos(2\pi f t)$, the change of variables $\hat{y} = y - I(t)$ transforms previous system into (3.1).

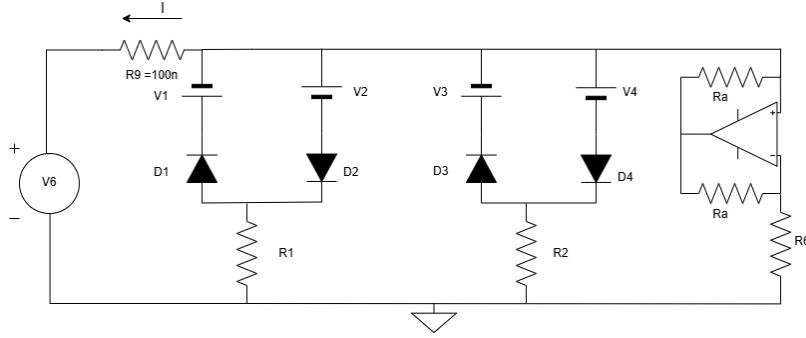


Figure 3.1: Schematic of the non-linear resistor used in the circuit. The diodes structure, along with the non-linear resistor makes the component work as a Chua's diode.

3.1 The non-linear resistor

We claim that the characteristic equation of the non-linear resistor shown in the Figure 3.1, where $V_1 \geq V_3$, $V_2 \geq V_4$, $V_1 + V_2 > -2V_\gamma$ and $V_3 + V_4 > -2V_\gamma$ is like follows:

$$I(V_6) := I(v) = \begin{cases} -v \left(-\frac{1}{R_6} + \frac{1}{R_1} + \frac{1}{R_2} \right) - \left(\frac{V_1 + V_\gamma}{R_1} + \frac{V_3 + V_\gamma}{R_2} \right) & \text{if } v < -V_\gamma - V_1, \\ \frac{v}{R_6} \left(1 - \frac{R_6}{R_2} \right) - \frac{V_3 + V_\gamma}{R_2} & \text{if } -V_\gamma - V_1 \leq v < -V_\gamma - V_3, \\ \frac{v}{R_6} & \text{if } -V_\gamma - V_3 \leq v < V_4 + V_\gamma, \\ \frac{v}{R_6} \left(1 - \frac{R_6}{R_2} \right) + \frac{V_4 + V_\gamma}{R_2} & \text{if } V_4 + V_\gamma \leq v < V_2 + V_\gamma, \\ -v \left(-\frac{1}{R_6} + \frac{1}{R_1} + \frac{1}{R_2} \right) + \frac{V_1 + V_\gamma}{R_1} + \frac{V_3 + V_\gamma}{R_2} & \text{if } V_2 + V_\gamma \leq v. \end{cases} \quad (3.2)$$

This characteristic equation makes the circuit resemble a **Chua's diode**.

A starting point for deriving the function $I(v)$ might be finding the Thevenin's equivalent circuit to the rightmost part of the circuit (see Figure 3.2) involving the operational amplifier, to check that it works as a negative resistor. Considering it to be an ideal operational amplifier, and following the notation in Figure 3.2, then $V_+ = V_- = V_{in}$. Moreover, since there is no current flowing through the terminals of the operational amplifier, then $I_2 = \frac{V_x - V_{in}}{R_a} = \frac{V_{in}}{R_6}$ so $V_x = V_{in} \left(\frac{R_a + R_6}{R_6} \right)$. Finally, $I_{in} = \frac{V_{in} - V_x}{R_a} = -\frac{V_{in}}{R_6}$, and by Ohm's Law, $R_{in} = \frac{V_{in}}{I_{in}} = -R_6$.

Following the conventional method to study a circuit involving diodes, we might consider different hypothesis on the states of the diodes and see under which conditions they are satisfied. It is important to remark that we will consider for the diode the ideal model in series with a voltage source of V_γ , so it behaves as an open circuit if the voltage drop from anode to cathode is less than V_γ and as a voltage source of V_γ , otherwise.

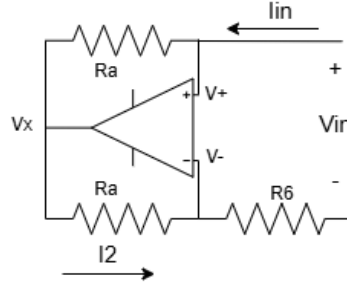


Figure 3.2: Circuit equivalent to a negative resistor.

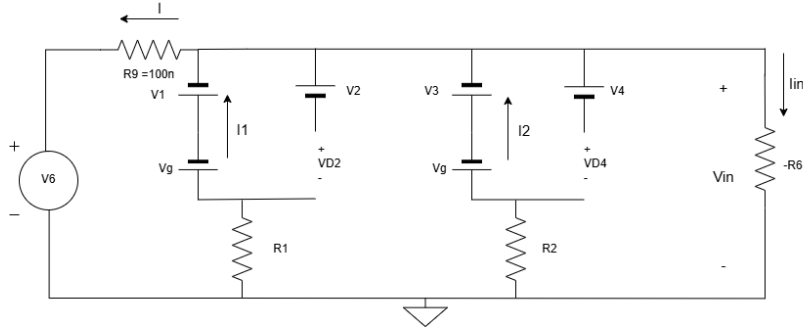


Figure 3.3: First case of study in the non-linear resistor characteristic equation.

Case 1: D_1, D_3 **ON**, D_2, D_4 **OFF** (see Figure 3.3): By Kirchoff's Current Law (KCL), $I + I_{in} = I_1 + I_2$, and by Ohm's Law,

$$\frac{V_{in} - V_6}{R_9} + \frac{V_{in}}{-R_6} = -\frac{V_{in} + V_1 + V_\gamma}{R_1} - \frac{V_{in} + V_3 + V_\gamma}{R_2},$$

and consequently,

$$V_{in} = \frac{(R_6 R_9) \left(\frac{V_1 + V_\gamma}{R_1} + \frac{V_3 + V_\gamma}{R_2} \right) - R_6 V_6}{R_9 - R_6 - \frac{R_6 R_9}{R_1} - \frac{R_6 R_9}{R_2}}. \quad (3.3)$$

Since $I = \frac{V_{in} - V_6}{R_9}$,

$$I = \frac{R_6 \left(\frac{V_1 + V_\gamma}{R_1} + \frac{V_3 + V_\gamma}{R_2} \right)}{R_9 - R_6 - \frac{R_6 R_9}{R_1} - \frac{R_6 R_9}{R_2}} - \frac{V_6}{R_9} \left(\frac{R_6}{R_9 - R_6 - \frac{R_6 R_9}{R_1} - \frac{R_6 R_9}{R_2}} + 1 \right).$$

Because R_9 is negligible, we might consider $\lim_{R_9 \rightarrow 0} I$. By using l'Hôpital-Bernoulli's rule, one finally gets

$$I = -\left(\frac{V_1 + V_\gamma}{R_1} + \frac{V_3 + V_\gamma}{R_2} \right) - V_6 \left(\frac{1}{R_1} + \frac{1}{R_2} - \frac{1}{R_6} \right).$$

3. THE CIRCUIT AND ITS MODELS

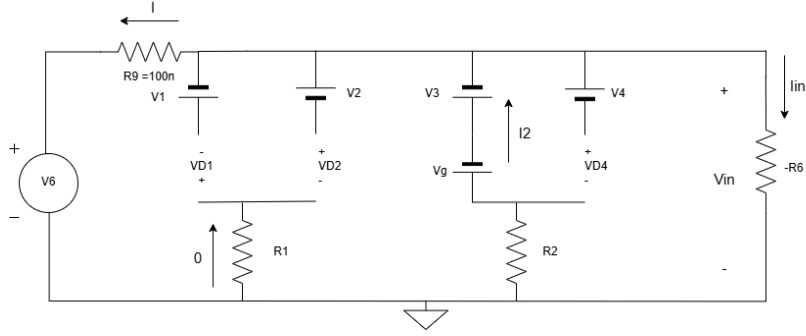


Figure 3.4: Second case of study in the non-linear resistor characteristic equation.

It is necessary to verify the conditions under which the proposed hypotheses hold true. On the one hand, for D_1, D_3 to be conducting it is necessary that $I_1 = -\frac{V_{in} + V_1 + V_\gamma}{R_1} > 0$, $I_2 = -\frac{V_{in} + V_3 + V_\gamma}{R_2} > 0$. Indeed, as $V_1 \geq V_3$, is it enough to check that $V_{in} + V_\gamma + V_1 < 0$. Considering R_9 to be negligible in (3.3), it must hold that $V_6 < -V_\gamma - V_1$. On the other hand, for D_2, D_4 to be in cut-off, $V_{D_2} < V_\gamma$, $V_{D_4} < V_\gamma$. However, $V_1 + V_\gamma = -V_{D_2} - V_2$ so $V_{D_2} = -(V_1 + V_2 + V_\gamma) < V_\gamma$, and equivalently for V_{D_4} .

It is easy to verify that this development is analogous for the **case** D_1, D_3 **OFF**, D_2, D_4 **ON** where the relation is given by

$$I = \frac{V_1 + V_\gamma}{R_1} + \frac{V_3 + V_\gamma}{R_2} - V_6 \left(\frac{1}{R_1} + \frac{1}{R_2} - \frac{1}{R_6} \right),$$

for $V_6 > V_2 + V_\gamma$.

Case 2: D_3 **ON**, D_1, D_2, D_4 **OFF** (see Figure 3.4): From $I + I_{in} = I_2$ and by Ohm's Law,

$$\frac{V_{in} - V_6}{R_9} + \frac{V_{in}}{-R_6} = -\frac{V_{in} + V_3 + V_\gamma}{R_2},$$

hence

$$V_{in} = \frac{(R_6 R_9) \left(\frac{V_3 + V_\gamma}{R_2} \right) - R_6 V_6}{R_9 - R_6 - \frac{R_6 R_9}{R_2}}. \quad (3.4)$$

Just like before, as $I = \frac{V_{in} - V_6}{R_9}$,

$$I = \frac{R_6 \left(\frac{V_3 + V_\gamma}{R_2} \right)}{R_9 - R_6 - \frac{R_6 R_9}{R_2}} - \frac{V_6}{R_9} \left(\frac{R_6}{R_9 - R_6 - \frac{R_6 R_9}{R_2}} + 1 \right),$$

and taking limit $R_9 \rightarrow 0$,

$$I = -\frac{V_3 + V_\gamma}{R_2} + \frac{V_6}{R_6} \left(1 - \frac{R_6}{R_2} \right).$$

Once again, we need to set the conditions for the hypothesis to hold true. On the one hand, $I_2 > 0$ if and only if $V_{in} + V_3 + V_\gamma < 0$ and considering $R_9 \approx 0$ in (3.4) $I_2 > 0$ if

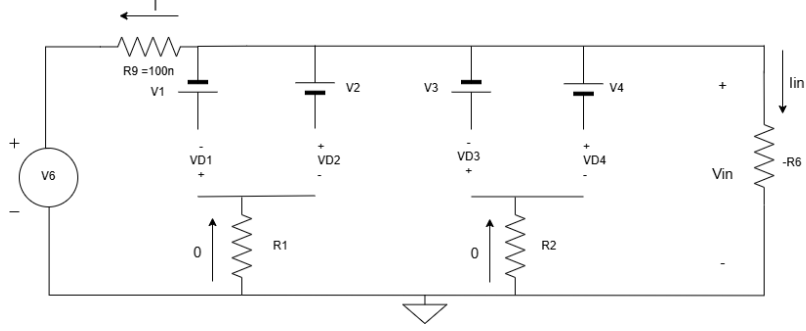


Figure 3.5: Third case of study in the non-linear resistor characteristic equation.

and only if $V_6 < -V_\gamma - V_3$. On the other hand, V_{D1}, V_{D2}, V_{D4} need to be lower than V_γ . While $V_{D4} = -(V_3 + V_4 + V_\gamma) < V_\gamma$, the value for $V_{D1} = -V_1 - V_{in} < V_\gamma$ only in the case $V_6 > -V_1 - V_\gamma$. Finally, $V_{D2} = V_{in} - V_2 < V_\gamma$ if and only if $V_6 < V_2 + V_\gamma$.

It is easy to verify that this development is analogous for the **case** D_4 **ON**, D_1, D_2, D_3 **OFF** where the function is given by:

$$I = \frac{V_4 + V_\gamma}{R_2} + \frac{V_6}{R_6} \left(1 - \frac{R_6}{R_2} \right),$$

for $V_4 + V_\gamma < V_6 < V_2 + V_\gamma$.

Case 3: D_1, D_2, D_3, D_4 **OFF** (see Figure 3.5): The scenario reduces to $I + I_{in} = 0$ and by Ohm's Law,

$$\frac{V_{in} - V_6}{R_9} = \frac{V_{in}}{R_6},$$

thus

$$V_{in} = \frac{-R_6}{R_9 - R_6} V_6.$$

Moreover, $I = \frac{V_{in} - V_6}{R_9} = -\frac{V_6}{R_9 - R_6}$, and because $R_9 \approx 0$, then $I = \frac{V_6}{R_6}$.

In this case all diodes need to be in cut-off. Since $V_1 + V_{D1} = V_3 + V_{D3} = -V_{in}$, and $V_1 \geq V_3$ then $-V_3 - V_\gamma < V_6$. Similarly, $V_2 + V_{D2} = V_4 + V_{D4} = V_{in}$ implying $V_6 < V_4 + V_\gamma$.

3.2 The circuit

The schematic in Figure 3.6 represents the complete electronic circuit that will be used in the project. Having found the characteristic equation of the non-linear resistor (3.2) and using basic knowledge in circuit analysis one can easily find the system of coupled differential equations that govern the behaviour of the circuit.

Applying KCL and Ohm's law it follows that:

$$i_C = i_n + i_s - i_L, \quad i_s = -\frac{v}{R_s}.$$

Moreover

$$v = L \frac{di_L}{dt} + E + P \sin(2\pi f t),$$

so

$$\frac{di_L}{dt} = \frac{1}{L} (v - E - P \sin(2\pi f t)),$$

and

$$i_C = C \frac{dv}{dt}$$

meaning

$$\frac{dv}{dt} = \frac{1}{C} i_C.$$

All together we get the system:

$$\begin{cases} \frac{dv}{dt} = \frac{1}{C} (i_n + i_s - i_L), \\ \frac{di_L}{dt} = \frac{1}{L} (v - E - P \sin(2\pi f t)). \end{cases}$$

In order to do a further development, let us express the current i_n by the response of the non-linear resistor, $i_n = f(v)$. What is more, in order for R_1, R_2, R_6 to have values of $\sim 100\Omega$, we end up having $i_n = \frac{f(v)}{100}$. So, we will also consider $R_s = 100\hat{R}_s$. In this way, we get:

$$\begin{cases} \frac{dv}{dt} = \frac{1}{C} \left(\frac{1}{100} \left[f(v) - \frac{v}{\hat{R}_s} \right] - i_L \right), \\ \frac{di_L}{dt} = \frac{1}{L} (v - E - P \sin(2\pi f t)). \end{cases}$$

Taking the change of variables $100i_L = i'_L$ (do not mistake the name of the new variable i'_L with the derivative $\frac{di_L}{dt}$) we obtain:

$$\begin{cases} \frac{dv}{dt} = \frac{1}{100C} \left(f(v) - \frac{v}{\hat{R}_s} - i'_L \right), \\ \frac{di'_L}{dt} = \frac{100}{L} (v - E - P \sin(2\pi f t)). \end{cases}$$

We can normalize the equations by changing time $\tau = \frac{t}{100C}$. So, we eventually obtain:

$$\begin{cases} \frac{dv}{d\tau} = f(v) - \frac{v}{\hat{R}_s} - i'_L, \\ \frac{di'_L}{d\tau} = 10^4 \frac{C}{L} (v - E - P \sin(2\pi(f100C)\tau)). \end{cases} \quad (3.5)$$

3.3 The models

Two different theoretical models will be considered in order to study the circuit, one of them smooth and the other one PWL. Both of them will be developed in the framework of FitzHugh-Nagumo models.

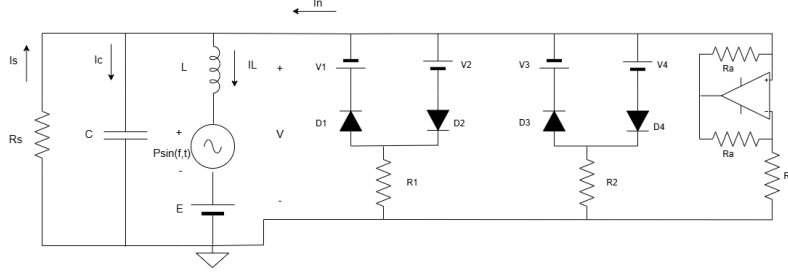


Figure 3.6: Schematic of the electronic circuit.

3.3.1 Smooth model

In order to resemble a classic FitzHugh-Nagumo formulation, the non-linear resistor will be considered to respond with a polynomial smooth function $f(v) = v - \frac{v^3}{3}$. Consequently, the model is given by

$$\begin{cases} \frac{dx}{dt} = x(1 - \varepsilon) - \frac{x^3}{3} - y, \\ \frac{dy}{dt} = c(x - a - P \sin(2\pi f t)). \end{cases} \quad (3.6)$$

Note that in our model, the singular perturbation parameter of the system is expressed by c . In contrast, the ε parameter, which is commonly used as the singular perturbation parameter in the literature, is reserved for the coefficient of x governing the negative feedback rate.

3.3.2 PWL model

In this case, the resistor's response is considered to be a PWL function. In order to be close to a FitzHugh-Nagumo, the non-linear resistor characteristic equation's parameters will be adjusted to resemble the smooth function $f(x) = x - \frac{x^3}{3}$. To this end, we propose using a PWL function g that presents the following features:

- For the pieces of g near the extremes of f to have a width proportional to square root of the slow-fast c parameter. Specifically, to be constrained to intervals $(x_- - \sqrt{c}, x_- + \sqrt{c})$ and $(x_+ - \sqrt{c}, x_+ + \sqrt{c})$, where x_- , x_+ represent the abscissa of the minimum and maximum of f . This restriction is backed by Remark 1 in Subsection 4.1.1.
- For g to have the same slope γ for the pieces near the extremes of f , only for simplification purposes.
- To maintain a relation $-1/2$ between the slopes of the outer lines and the one in the center, since $\frac{f'(0)}{f'(x_0)} = -\frac{1}{2}$ where x_0 is a root of f distinct from the origin. Having different slope relations would make g approximate other type of polynomials rather than a cubic.

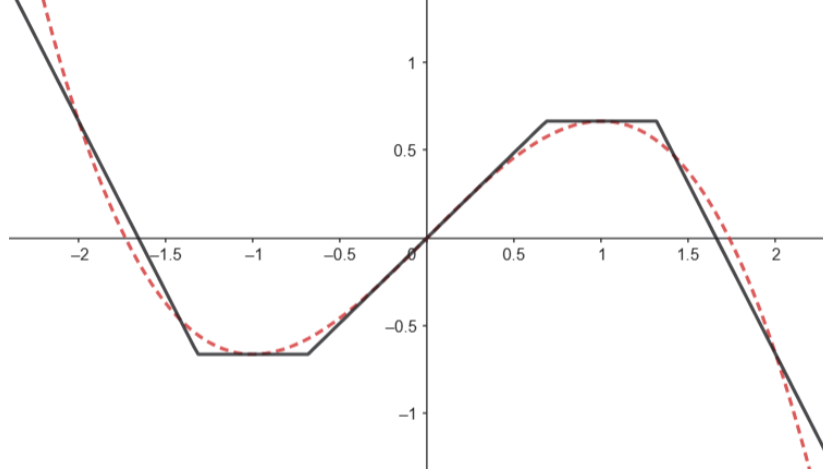


Figure 3.7: Comparison of the functions $f(x) = x - \frac{x^3}{3}$ (in dashed red line) and g (in solid line) defined in (3.8) for $\gamma = 0$ and $c = 0.1$.

With all this in mind, we present the model

$$\begin{cases} \frac{dx}{dt} = g(x) - \epsilon x - y, \\ \frac{dy}{dt} = c(x - a - P \sin(2\pi f t)), \end{cases} \quad (3.7)$$

where

$$g(x) = \begin{cases} -2m(x + 1 + \sqrt{c}) - \gamma\sqrt{c} - \frac{2}{3} & \text{if } x < -1 - \sqrt{c}, \\ \gamma(x + 1) - \frac{2}{3} & \text{if } -1 - \sqrt{c} \leq x < -1 + \sqrt{c}, \\ mx & \text{if } -1 + \sqrt{c} \leq x < 1 - \sqrt{c}, \\ \gamma(x - 1) + \frac{2}{3} & \text{if } 1 - \sqrt{c} \leq x < 1 + \sqrt{c}, \\ -2m(x - 1 - \sqrt{c}) + \gamma\sqrt{c} + \frac{2}{3} & \text{if } 1 + \sqrt{c} \leq x, \end{cases} \quad (3.8)$$

being $m = \frac{\frac{2}{3} - \gamma\sqrt{c}}{1 - \sqrt{c}}$. The similarity between the functions f and g is shown in Figure 3.7.

By comparing the expressions of each of the pieces in functions (3.8) and (3.2), and taking into account the formulation of the system (3.5) we get the following relation between each of their parameters:

$$\begin{aligned} R_1 &= \frac{100}{2m + \gamma}, \quad R_2 = \frac{100}{m - \gamma}, \quad R_6 = \frac{100}{m}, \quad R_s = \frac{100}{\epsilon}, \\ \hat{f} &= \frac{f}{100C}, \quad V_1 = V_2 = 1 + \sqrt{c} - V_\gamma, \quad V_3 = V_4 = 1 - \sqrt{c} - V_\gamma, \end{aligned} \quad (3.9)$$

where C denotes the capacitance, $\hat{f}$ the frequency of the source in the circuit and f the frequency term in the model (3.7).

SLOW-FAST SYSTEMS

This chapter presents the main theoretical results for the smooth (3.6) and PWL (3.7) models introduced previously. We first review essential tools for slow-fast systems, such as Fenichel theory, and then analyse the models' dynamics to reveal their underlying properties. Following an approach analogous to [9] and [2], we prove the occurrence of a canard explosion in both systems. Specifically, the formal analysis from [9] is adapted for the smooth system (3.6), whereas the corresponding results for the PWL system (3.7) are rigorously proven by extending the framework of [2]. Finally, the presence of canard phenomena paves the way to explore and report other complex behaviours at the end of the section, such as bursting due to a slow passage through a Hopf bifurcation and period-doubling bifurcations.

The main systems of study in this project belong to the class of the so-called **slow-fast** systems, where processes changing at different time scales interact together and are able to produce a great variety of complex behaviour.

Definition 7. A *slow-fast system* is an ODE autonomous system of the form

$$\begin{cases} \varepsilon \frac{dx}{dt} := \varepsilon x' = f(x, y, \lambda, \varepsilon), \\ \frac{dy}{dt} := y' = g(x, y, \lambda, \varepsilon), \end{cases} \quad (4.1)$$

where $0 < \varepsilon \ll 1$, $f, g \in C^k$, $k \geq 3$ and $(x, y)^T \in \mathbb{R}^2$. The variable x is called the **fast variable** and y is the **slow variable**.

The intuition from which the name of this type of system comes from is quite simple: far from the x -nullcline the norm of solutions' derivative is big so solutions evolve quickly, while near the x -nullcline the norm is small and solutions evolve very slow. Indeed, the norm of the velocity vector $x'^2 + y'^2 = \left(\frac{f(x, y)}{\varepsilon}\right)^2 + g(x, y)^2$ has order $O(1)$ when $f(x, y) \sim \varepsilon$ and $O(\frac{1}{\varepsilon}) \gg O(1)$ otherwise.

In (4.1) one can consider the change in time $t = \frac{\tau}{\varepsilon}$ (**fast time**) to obtain an equivalent system to (4.1)

$$\begin{cases} \dot{x} = f(x, y, \lambda, \varepsilon), \\ \dot{y} = \varepsilon g(x, y, \lambda, \varepsilon). \end{cases} \quad (4.2)$$

In fact, systems (4.2) and (4.1) share the same trajectories, but differ in their time parametrization. System (4.1) is known as a **slow system**, while (4.2) is a **fast system**.

As ε is considered to be such a small value, one expects to consider the pass to the limit $\varepsilon \rightarrow 0$.

Definition 8. *On the one hand, the singular limit $\varepsilon \rightarrow 0$ of the planar slow system (4.1) yields the semi-explicit Differential Algebraic Equation **reduced problem** or **slow-subsystem**:*

$$\begin{cases} 0 = f(x, y, \lambda, 0), \\ y' = g(x, y, \lambda, 0). \end{cases} \quad (4.3)$$

*On the other hand, the singular limit $\varepsilon \rightarrow 0$ of the planar fast system (4.2) yields the **layer problem** or **fast-subsystem**:*

$$\begin{cases} \dot{x} = f(x, y, \lambda, 0), \\ \dot{y} = 0. \end{cases} \quad (4.4)$$

Fenichel theory will allow to take conclusions on the system (4.1) by working with the limit case of the two presented systems.

Definition 9. *The **critical manifold**, S , of the slow-fast system is the set of equilibria of the layer problem (4.4), $S = \{(x, y)^T \in \mathbb{R}^2 : f(x, y, \lambda, 0) = 0\}$.*

One can view the layer problem (4.4) as a one-dimension problem over each straight line that is a solution of $\dot{y} = 0$. Thus, the stability of each equilibrium point of the unidimensional problem can be derived from the sign of $\frac{\partial f}{\partial x} := f_x$ by Hartman-Grobman's Theorem (Theorem 6). As a result, one defines the following sets.

Definition 10. *The **attracting branch**, S_a , of the slow-fast system is the set, $S^a = \{(x, y)^T \in S : f_x(x, y) < 0\}$, the **repelling branch** is $S^r = \{(x, y)^T \in S : f_x(x, y) > 0\}$ and the **fold points' set** is $L = \{(x, y)^T \in S : f_x(x, y) = 0\}$.*

By differentiating the first equation of the problem (4.3) one obtains the **reduced equation**

$$f_x x' + f_y y' = 0 \rightarrow x' = -\frac{f_y}{f_x} g. \quad (4.5)$$

Thus it becomes clear that fold points can lead to a singularity.

Figure 4.1 gives insight to how the analysis of the reduced (4.5) and layer (4.4) problems help understand the dynamics of a slow-fast system (4.2) with a cubic-shaped x -nullcline and a straight line as a y -nullcline. Notice that this is the case for the smooth model (3.6). One computes the attracting and repelling branches for the case $\varepsilon = 0$ in order to determine the dynamics far from the x -nullcline, while analyses the sign of g from the reduced problem to define the dynamics over the nullcline.

Fenichel theory asserts that compact subsets $S_0 \subseteq S$ made up of normally hyperbolic equilibria of the critical manifold persist in the original problem as locally invariant manifolds S_ε , known as **slow manifolds**, whose stability properties are equal to those

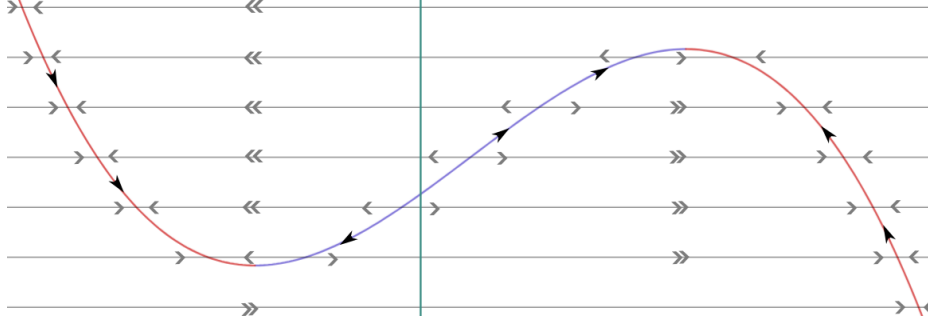


Figure 4.1: Diagram of the dynamics of a slow-fast system like (4.2) with a cubic-shaped x -nullcline and a straight line, in green, as a y -nullcline. Red sections indicate the attracting branch S^a and blue section the repelling branch S^r . Arrows indicate the direction of the flow on the x -nullcline and far from it for $\varepsilon = 0$, while dynamics for $\varepsilon > 0$ would be similar due to Fenichel Theorem 10.

of S_0 . Local invariance means that S_ε can have boundaries through which trajectories enter or leave. Furthermore, the flow over S_ε is a regular perturbation of the flow defined by the reduced problem (4.3). A specific version of this result, the one that is presented below, can be found in [4]. Demonstrations for the items in the Theorem 10 can be found as several results along articles [5] and [8].

Definition 11. An equilibrium point $(x, y)^T$ of the slow-fast system (4.1) is said to be **normally hyperbolic** if the Jacobian matrix has exactly one eigenvalue with non-zero real-value. A set is called **normally hyperbolic** if it is entirely constituted by normally hyperbolic equilibria.

When dealing with manifolds containing non-equilibrium points, the concept of normal hyperbolicity becomes more general and its definition is difficult to verify on slow-fast systems. It has to do with the power of attraction (repulsion) from the manifold to be stronger than the flow over the manifold. See discussion in page 215 in [4].

Theorem 10 (Fenichel). Suppose S_0 is a compact normally hyperbolic submanifold (possibly with boundary) of the critical manifold S of the slow-fast system (4.2) and that $f, g \in C^r, r < \infty$. Then for $\varepsilon > 0$ sufficiently small the following hold:

- i) There exists a locally invariant manifold S_ε diffeomorphic to S_0 .
- ii) S_ε has a Hausdorff distance of $O(\varepsilon)$ from S_0 .
- iii) The flow on S_ε converges to the slow flow as $\varepsilon \rightarrow 0$.
- iv) S_ε is C^r -smooth.
- v) S_ε is normally hyperbolic and has the same stability properties with respect to the fast variables as S_0 (attracting, repelling, or saddle type). This is, when $S_0 \subseteq S^a$, it is attracting and when $S_0 \subseteq S^r$, it is repelling.

- vi) S_ε is usually not unique. In regions that remain at a fixed distance from the boundary of S_ε , all manifolds satisfying i) – v) lie at a Hausdorff distance $O\left(e^{-\frac{K}{\varepsilon}}\right)$ from each other for some $K > 0$ with $K = O(1)$.

The normally hyperbolic manifold S_0 has associated local stable and unstable manifolds

$$W_{loc}^s(S_0) = \bigcup_{p \in S_0} W_{loc}^s(p) \quad \text{and} \quad W_{loc}^u(S_0) = \bigcup_{p \in S_0} W_{loc}^u(p),$$

where $W_{loc}^s(p)$ and $W_{loc}^u(p)$ are the local stable and unstable manifolds of p as a hyperbolic equilibrium of the layer equations (4.4), respectively. These manifolds also persist for $\varepsilon > 0$ sufficiently small: there exist local stable and unstable manifolds $W_{loc}^s(S_\varepsilon)$ and $W_{loc}^u(S_\varepsilon)$, respectively, for which conclusions i)–v) hold if we replace S_ε and S_0 by $W_{loc}^s(S_\varepsilon)$ and $W_{loc}^s(S_0)$ (or similarly by $W_{loc}^u(S_\varepsilon)$ and $W_{loc}^u(S_0)$).

Now it is clear that, thanks to Fenichel Theorem 10, the system's dynamics in Figure 4.1 for $\varepsilon > 0$ will be similar to those of the illustration.

Now it is clear that, thanks to Fenichel Theorem 10, the dynamics for $\varepsilon > 0$ of a system like (4.1) with a cubic x -nullcline will be similar to those of Figure 4.1, as the flow over the slow manifold will be analogous to that of the slow flow over the critical manifold in the illustration, and the fast flow will remain the same.

Slow-fast systems in the PWL framework are treated in an analogous way. One can equally consider the reduced and layer problem, and extract from them qualitative information of the overall system. When limiting the study to one of the subdomains of definition, the above theory applies as the system is C^∞ in the region. What is more, the slow manifold S_ε reduces to a unique affine variety that is directly computed from the eigenvectors of the system, usually from those associated to the real-values eigenvalues of order $O(\varepsilon)$ in system's matrix.

When treating the whole space, the critical manifold is defined as the union of the critical manifolds in each subdomain of definition of the layer problem. Similarly, the union of the slow manifolds of different regions of the PWL system results in the slow manifold S_ε . See Theorem 1 in [16] for a formal development of an analogous result to the Fenichel theorem for PWL systems of two zones.

4.1 Canard Explosion

Fenichel theory presented in the previous introduction to slow-fast systems describes how critical manifolds perturb to slow manifolds made up of normally hyperbolic equilibria. In contrast, this section deals with dynamics around non-normally hyperbolic points. Neighbourhoods of fold points in the smooth framework and in some border-points in the PWL case use to contain fragments from both attracting and repelling branches. The interaction of these latter objects in that neighbourhood drives the canard explosion phenomenon.

A fold point $\mathbf{p} = (p_x, p_y)^T$ of the critical manifold of (4.1) is called a **canard point** when satisfies $g(p_x, p_y, \lambda, 0) = 0$. In this situation, the reduced equation (4.5) becomes $\dot{x} = \frac{0}{0}$ and can be solved through a desingularisation process. As a result, the reduced problem (4.3) has a solution passing through $\mathbf{p}$ and the attracting branch S_ε^a of the slow manifold stays close to the repelling branch S_ε^r for a time $O(1)$, even having the

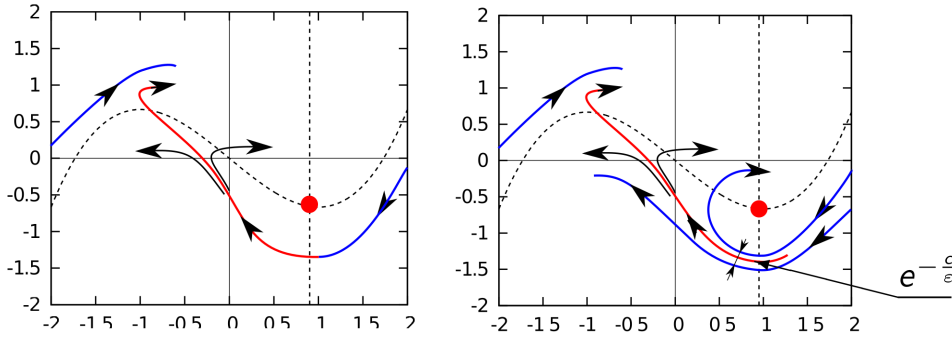


Figure 4.2: In red, a canard point. Slow manifold is represented in red as the repelling branch and in blue as the attracting branch, while arrows indicate the direction of the flow. Left image shows a connection between the branches scenario and right image shows scenarios where the attracting branch stays exponentially close to the repelling branch. Note that these scenarios are derived from exponentially small perturbations of the parameter value provoking the connection between the branches of the slow manifold.

possibility to connect for a specific value of the parameter λ . Border points in PWL systems having an attracting and repelling branches on each side act analogously.

Images in Figure 4.2 represent the previously described behaviour. Assuming the existence of a parameter value allowing the connection between slow manifolds around a canard point (situation represented by the left image), small value perturbations lead to scenarios where the attracting and repelling branches stay exponentially close in a neighbourhood of the point (situation represented by the right image). Thus, a flow coming sufficiently close to the attracting branch will continue close to the repelling branch leading to canard orbits, as can be seen in Figure 4.3.

What is more, when a Hopf or Hopf-like bifurcation (Definition 5) happens around a canard point, the Hopf cycles involved transition to canard cycles, see Figure 4.3, and finally to relaxation oscillations, see Figure 4.6. In this context, **relaxation oscillations**

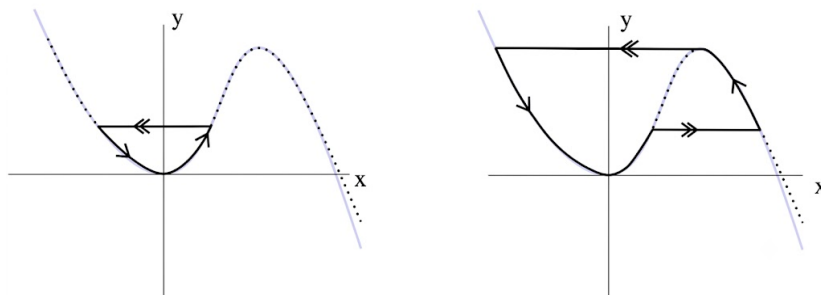


Figure 4.3: Canard Cycles. The central slope of the S-shaped critical manifold belongs to the repelling branch S^r . (Image adapted from [9]).

refer to orbits that present a long phase of slow evolution that leads to a stage of very quick evolution. This phenomenon is related to a **canard explosion**.

We will say that a canard explosion takes place when a value of the parameter λ makes a fold point in the smooth case and a border point in the PWL one become a canard point.

4.1.1 Smooth framework

Article [9] provides a full review and characterization of canard explosion in smooth slow-fast systems of type (4.1). In particular, system (3.6) with no oscillatory component can be studied within the framework that will be presented.

Throughout the whole section, results directly taken from [9] will be followed by an adaptation to the smooth model (3.6) we are considering. In order to be able to apply them, we convert the system (3.6) to a canonical form (4.6) presented in the article [9] by taking a rigid movement that translates the origin of the plane to the minimum of the x -nullcline.

$$\begin{cases} \dot{x} = -yh_1(x, y, \lambda, c) + x^2h_2(x, y, \lambda, c) + ch_3(x, y, \lambda, c), \\ \dot{y} = c(xh_4(x, y, \lambda, c) - \lambda h_5(x, y, \lambda, c) + yh_6(x, y, \lambda, c)), \end{cases} \quad (4.6)$$

$$h_3(x, y, \lambda, c) = O(x, y, \lambda, c); \quad h_j(x, y, \lambda, c) = 1 + O(x, y, \lambda, c), \quad j = 1, 2, 4, 5, 6.$$

Considering a translation and a reparametrization

$$\begin{aligned} (x, y) &= \left(x + \sqrt{1-\varepsilon}, y + \frac{2}{3}(1-\varepsilon)\sqrt{1-\varepsilon} \right), \\ \lambda &= a + \sqrt{1-\varepsilon}, \end{aligned} \quad (4.7)$$

one obtains the subsequent equivalent system:

$$\begin{cases} \dot{x} = x^2\sqrt{1-\varepsilon} - \frac{x^3}{3} - y, \\ \dot{y} = c(x - \lambda), \end{cases} \quad (4.8)$$

where $0 < c \ll 1$ is the singular perturbation parameter and λ the critical parameter which unfolds the canard explosion. System (4.8) presents an S-shaped critical manifold having both an attracting branch and a repelling branch given by

$$\begin{aligned} S^a &= \left\{ \left(x, x^2\sqrt{1-\varepsilon} - \frac{x^3}{3} \right)^T : x < 0 \text{ or } x > 2\sqrt{1-\varepsilon} \right\}, \\ S^r &= \left\{ \left(x, x^2\sqrt{1-\varepsilon} - \frac{x^3}{3} \right)^T : 0 < x < 2\sqrt{1-\varepsilon} \right\}. \end{aligned}$$

By considering the system (4.8) expressed in the canonical form (4.6), and following the notation introduced in Sections 3.2 and 3.3 in [9], we obtain:

$$h_i(x, y, \lambda, c) = 1, \quad i \in \{1, 4, 5, 6\}; \quad h_j(x, y, \lambda, c) = 0, \quad j \in \{3, 6\}; \quad h_2(x, y, \lambda, c) = \sqrt{1-\varepsilon} - \frac{x}{3}.$$

Thus, we obtain the values for the following constants

$$\begin{aligned} a_1 &:= \frac{\partial h_3}{\partial x}(0, 0, 0, 0), & a_2 &:= \frac{\partial h_1}{\partial x}(0, 0, 0, 0), & a_3 &:= \frac{\partial h_2}{\partial x}(0, 0, 0, 0), \\ a_4 &:= \frac{\partial h_4}{\partial x}(0, 0, 0, 0), & a_5 &:= h_6(0, 0, 0, 0), & A &= -a_2 + 3a_3 - 2a_4 - 2a_5, \end{aligned}$$

to be

$$a_k = 0, k \in \{1, 2, 4, 5\}; \quad a_3 = -\frac{1}{3}; \quad A = -1.$$

A first result (taken from Theorem 3.1 in [9]) is useful to locate the Hopf bifurcation in the space of parameters. The set V mentioned in this result, and consequently appearing in our following Theorem 11 and Theorem 12, refers to a neighbourhood of the origin where the non-trivial local canard dynamics take place. It is obtained by a pair of charts that involve specific *blow-up* transformations, see Section 3.2 in [9] for more details.

Remark 1. Among other properties, V has a size of $O(\sqrt{c})$ along the x axis.

Note that we require the region around the minimum and maximum in the PWL model, see (3.8), to have an analogous size to the neighbourhood V where the study in the smooth framework is carried out.

Theorem 11. Suppose we have a system with an S-shaped critical manifold, just like model (4.8). Then there exist $c_0 > 0$, $\lambda_0 > 0$ such that for each $0 < c < c_0$, $|\lambda| < \lambda_0$ system (4.1) has precisely one equilibrium $\mathbf{p}_e \in V$ which converges to the canard point as $(c, \lambda) \rightarrow 0$. Moreover, there exists a curve $\lambda_H(\sqrt{c})$ such that $\mathbf{p}_e$ is stable for $\lambda < \lambda_H(\sqrt{c})$ and loses stability through a Hopf bifurcation as λ passes through $\lambda_H(\sqrt{c})$. The curve $\lambda_H(\sqrt{c})$ has the expansion

$$\lambda_H(\sqrt{c}) = -\frac{a_1 + a_5}{2}c + O\left(c^{\frac{3}{2}}\right).$$

The Hopf bifurcation is non-degenerate if the constant A is nonzero. It is supercritical if $A < 0$ and subcritical if $A > 0$.

Consequently, the system (4.8) presents a supercritical Hopf bifurcation at $\lambda_H(\sqrt{c}) = 0 + O\left(c^{\frac{3}{2}}\right)$, which translates due to (4.7) to a supercritical Hopf bifurcation in the original system (3.6) at $a_H(\sqrt{c}) = -\sqrt{1-\varepsilon} + O\left(c^{\frac{3}{2}}\right)$.

Many authors relate the criticality of bifurcations with the stability of the limit cycles involved rather than their position with respect to the critical bifurcation parameter value. They call the bifurcation supercritical if there is involved a stable limit cycle around an unstable equilibria branch, and subcritical when there is an unstable limit cycle around a stable equilibrium branch. However, notice that as we are dealing with the Hopf bifurcation around the minimum of the S-shaped nullcline, this nomenclature matches the topological description that we provided in Definition 5.

The Figure 4.4 shows the canard explosion exhibited by the smooth model (4.8) with parameter values $\varepsilon = 0.25$, $c = 0.1$, and obtained through the software xppaut. These kind of diagrams represent the changing parameter of the system in the horizontal axis and the value of the variable y in the vertical one. In them we represent both

the parameter a versus the variable y of the invariant objects appearing in the system at such parameter value. These type of diagrams are called **bifurcation diagrams**. Moreover, they plot the equilibria of the system, in red colour when being stable and in dashed line when being unstable; as well as the maximum and minimum value of y for a limit cycle in green when being stable or dashed blue when being unstable. In our case, it shows that there is a supercritical Hopf bifurcation at $a = -0.866$ value along with the growth in the amplitude of the attracting limit cycle of the system. According to the previous formula, we expected the Hopf value to be around $a_H \approx -0.866$, as the software has numerically computed.

The next step is to find the parameter value that allows us to create a connection between the slow and the fast manifolds. See Theorem 3.2 in [9].

Theorem 12. *Suppose we have a system with an S-shaped critical manifold, just like model (4.8). Then there exists a smooth function $\lambda_c(\sqrt{c})$ such that a solution starting in S_a^c connects to S_r^c if and only if $\lambda = \lambda_S(\sqrt{c})$. The function λ_c has the expansion*

$$\lambda_S(\sqrt{c}) = -\left(\frac{a_1 + a_5}{2} + \frac{1}{8}A\right)c + O\left(c^{\frac{3}{2}}\right). \quad (4.9)$$

In our case of study, the connection is established at $\lambda_S(\sqrt{c}) = \frac{c}{8} + O\left(c^{\frac{3}{2}}\right)$, and equivalently for (3.6) at $a_S(\sqrt{c}) = -\sqrt{1-\varepsilon} + \frac{c}{8} + O\left(c^{\frac{3}{2}}\right)$.

As can be seen in Figure 4.5, when λ passes through $\lambda_c(\sqrt{c})$, the manifolds S_c^a and S_c^r change their relative position. Before the connection between branches is established, one stays above the other and "far away". This means that the flow approaching the left part of the attracting branch follows its manifold far above the repelling branch, leading to Hopf cycles. As the branches approach each other, a connection is made. Consequently, the flow coming from the left part of the attracting branch remains close to the repelling one for some time before escaping the manifold, thus yielding canard cycles. When the connection breaks again and the left part of the attracting branch falls far below the repelling one, the flow drops far below the manifold, resulting in a relaxation oscillation.

In brief, the curve $\lambda_S(\sqrt{c})$ plays a central role in the description of canard explosion, as given in Theorem 13, which can be found as Theorems 3.3 and 3.5 in [9].

Define $(x_M, y_M)^T$ as the coordinates for the maximum of the critical manifold. The notation for cycles $\Gamma(s)$ can be found in Chapter 3.1 in [9], but essentially:

- $\Gamma(s)$ for $s \in (0, y_M)$ refer to *headless canards* just as the left cycle in Figure 4.3, where $(0, s)^T$ designates its intersection with the y axis apart from the origin.
- $\Gamma(s)$ for $s \in (y_M, 2y_M)$ refer to *headed canards* just as the right cycle in Figure 4.3, where $(x_M, 2y_M - s)^T$ designates its intersection with the line $\{x = x_M\}$ apart from $(x_M, y_M)^T$.

Theorem 13. *Fix c_0 sufficiently small and $v \in (0, 1)$. Suppose we have a system with an S-shaped critical manifold, just like model (4.8), and $A < 0$ (respectively, $A > 0$). For $c \in (0, c_0)$ there exists a family of periodic orbits $s \rightarrow (\lambda(s, \sqrt{c}), \Gamma(s, \sqrt{c}))$, $s \in (0, 2y_M)$, which is C^k -smooth in $(s, \sqrt{c})$, and such that:*

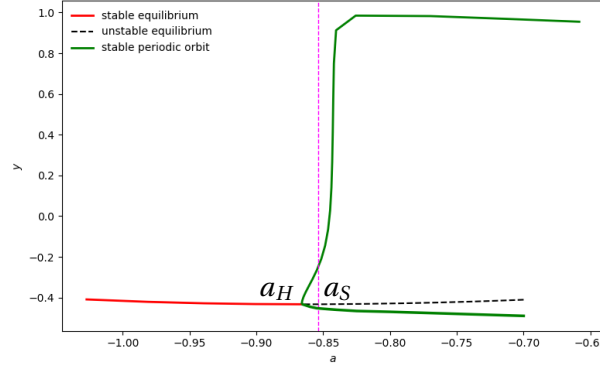


Figure 4.4: Hopf bifurcation value $a_H(\sqrt{c}) = -0.8660\dots$ and critical value $a_S(\sqrt{c}) = -0.8535\dots$ (both with an error term of $\pm 0.1^{\frac{3}{2}} \approx \pm 0.0316$) of the canard explosion for smooth system (3.6) with $\varepsilon = 0.25$, $c = 0.1$.

- i) for $s \in (0, c^v)$ the orbit $\Gamma(s, \sqrt{c})$ is attracting (respectively, repelling) and uniformly $O(c^v)$ close to the canard point and $\lambda(s, \sqrt{c})$ is strictly increasing (respectively, decreasing) in s ,
- ii) for $s \in (2y_M - c^v, 2y_M)$ the orbit $\Gamma(s, \sqrt{c})$ is a relaxation oscillation and $\lambda(s, \sqrt{c})$ is strictly increasing in s ,
- iii) for $s \in [c^v, 2y_M - c^v]$: $|\lambda(s, \sqrt{c}) - \lambda(\sqrt{c})| \leq e^{-1/c^{1-v}}$
- iv) as $c \rightarrow 0$ the family $\Gamma(s, \sqrt{c})$ converges uniformly in Hausdorff distance to $\Gamma(s)$
- v) any periodic orbit passing sufficiently close to the critical manifold S is a member of the family $\Gamma(s, \sqrt{c})$, or a relaxation oscillation.

The preceding result can be understood as the canard explosion, meaning the formation of canard cycles, happening at a distance $O\left(e^{-\frac{1}{c^{1-v}}}\right)$, which can be between $O\left(e^{-\frac{1}{c}}\right)$ and $O(e^{-1})$ because $v \in (0, 1)$, of the parameter value $\lambda_S(\sqrt{c})$ that allows a connection between the slow manifolds.

In brief, we have characterized the behaviour of the model (3.6) as it is summarized in the bifurcation diagram of Figure 4.4. For parameter values $a < a_H \approx -0.866$, the system presents a single stable equilibrium that is placed above the attracting branch of the slow manifold. Then, it transitions through a supercritical Hopf bifurcation to a situation where it appears an attracting relaxation oscillation that surrounds an unstable equilibrium, see Figure 4.6 for an example of such case. The relaxation oscillation flows near the attracting branch until the maximum and minimum of the nullcline while near the fold points it gets away from the repelling branch. In between those scenarios a brief canard explosion takes place, meaning the appearance of attracting limit cycles that go along both the attracting and repelling branches of the slow manifold.

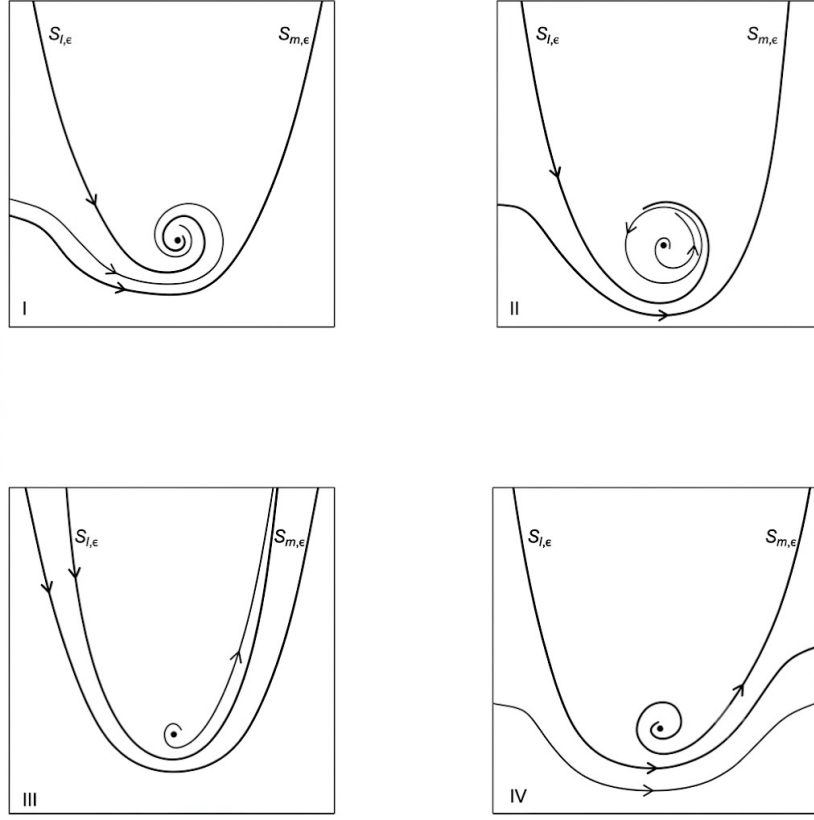


Figure 4.5: Image showing the relative position of the left part of the attracting (here named as $S_{l,\epsilon}$) and repelling (here named as $S_{m,\epsilon}$) branches of the slow manifold in the supercritical case $A < 0$ around the canard point according to the evolution of parameter λ . Region *I* refers to $\lambda \in (\lambda_0, \lambda_H)$, *II* to $\lambda \in (\lambda_H, \lambda_{sc})$, *III* to $\lambda \in (\lambda_{sc}, \lambda_S)$ and *IV* to $\lambda > \lambda_S$. λ_{sc} indicates the starting parameter value for which all orbits originating in V leave V . See Section 4.2 in [9] for an analogous review of the subcritical case. (Image adapted from [9]). These images give a more detailed representation of the scenario presented in Figure 4.2.

4.1.2 PWL framework

Following the same structure as in the smooth framework, we want to prove the presence of a canard explosion in the PWL model (3.7) when the forcing term is zero, that is $P = 0$. The outline consists on first proving the appearance of a Hopf-like bifurcation, which is necessary for the appearance of small limit cycles, and then finding a parameter value that allows a connection between the branches of the slow manifold. As it happens in the smooth case, the different relative positions of the slow manifolds (see Figure 4.5) explain the flow leading to different types of cycles, and specifically to canard cycles when it follows the repelling branch. This drives the explosive character of the transition to relaxation oscillation limit cycles.

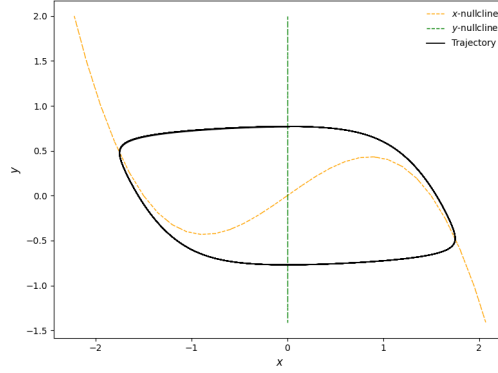


Figure 4.6: Example of a relaxation oscillation for the smooth system (3.6) when $c = 0.1, \varepsilon = 0.25, a = 0$. Dashed lines correspond to the nullclines and solid line to the actual orbit.

We start considering a change of variables and reparametrization

$$\begin{aligned} (x, y) &= (x - 1, y - \frac{2}{3} + \varepsilon), \\ a &= a + 1, \end{aligned} \quad (4.10)$$

in order to place the second piece of the PWL function of (3.7) in the region $\sigma_C = \{(x, y) : -\sqrt{c} \leq x < \sqrt{c}\}$ and around the origin (this means, in Figure 3.7, to place the middle point of the piece near the minimum, which represents the fold, of the solid black function in the origin). The change of variables produces a conjugated system according to the definition presented in Section 2.2.1, through the bijection $h : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $h(x, y) = (x - 1, y - \frac{2}{3} + \varepsilon)$, which is given by:

$$\begin{cases} \frac{dx}{dt} = g(x) - y, \\ \frac{dy}{dt} = c(x - a), \end{cases} \quad (4.11)$$

where

$$g(x) = \begin{cases} (-2m - \varepsilon)x - (2m + \gamma)\sqrt{c} & \text{if } x < -\sqrt{c}, \\ (\gamma - \varepsilon)x & \text{if } -\sqrt{c} \leq x < \sqrt{c}, \\ (m - \varepsilon)x - m + \frac{2}{3} & \text{if } \sqrt{c} \leq x < 2 - \sqrt{c}, \\ (\gamma - \varepsilon)x - 2\gamma + \frac{4}{3} & \text{if } 2 - \sqrt{c} \leq x < 2 + \sqrt{c}, \\ (-2m - \varepsilon)x + (2m + \gamma)\sqrt{c} + 4m + \frac{4}{3} & \text{if } 2 + \sqrt{c} \leq x, \end{cases}$$

$$\text{and } m = \frac{\frac{2}{3} - \gamma\sqrt{c}}{1 - \sqrt{c}}.$$

We will focus on the region of the system where $x \leq 2 - \sqrt{c}$ (this means, in Figure 3.7, to focus on the dynamics around the fold region in the minimum of the solid black function). After adding the definitions of the regions $\sigma_L = \{(x, y) : x < -\sqrt{c}\}$ and

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$\sigma_R = \{(x, y) : \sqrt{c} \leq x < 2 - \sqrt{c}\}$ to the previously defined σ_C , in matricidal form we get $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, where $\mathbf{F}(\mathbf{x}) = \mathbf{F}_i(\mathbf{x})$ if $\mathbf{x} \in \sigma_i$ for $i \in \{L, R, C\}$, and $\mathbf{F}_i(\mathbf{x}) = A_i \mathbf{x} + \mathbf{b}_i$ for:

$$A_L = \begin{pmatrix} -2m - \varepsilon & -1 \\ c & 0 \end{pmatrix}, \quad A_C = \begin{pmatrix} \gamma - \varepsilon & -1 \\ c & 0 \end{pmatrix}, \quad A_R = \begin{pmatrix} m - \varepsilon & -1 \\ c & 0 \end{pmatrix},$$

$$\mathbf{b}_L = \begin{pmatrix} -(2m + \gamma)\sqrt{c} \\ -ca \end{pmatrix}, \quad \mathbf{b}_C = \begin{pmatrix} 0 \\ -ca \end{pmatrix}, \quad \mathbf{b}_R = \begin{pmatrix} -m + \frac{2}{3} \\ -ca \end{pmatrix}.$$

The local behaviour of the dynamics of the system (4.11) in the indicated regions is determined by the trace t_i , the determinant $d = c$, the discriminant $\Delta_i = t_i^2 - 4c$, the slow λ_i^s and fast λ_i^q eigenvalues, the slow $\mathbf{v}_i^s = (\lambda_i^s, c)^T$ and fast $\mathbf{v}_i^q = (\lambda_i^q, c)^T$ eigenvectors and the equilibrium points $\mathbf{e}_i = -A_i^{-1} \mathbf{b}_i$. All the quantities are summarised in Tables 4.1 - 4.2.

Table 4.1: Significant quantities for the dynamics of system (4.11) in the regions σ_L, σ_R .

	L	R
t_i	$-2m - \varepsilon$	$m - \varepsilon$
Δ_i	$(-2m - \varepsilon)^2 - 4c$	$(m - \varepsilon)^2 - 4c$
λ_i^s	$\frac{-(2m + \varepsilon) + \sqrt{(2m + \varepsilon)^2 - 4c}}{2}$ $= -\frac{c}{\varepsilon + \frac{4}{3}} + O\left(c^{\frac{3}{2}}\right)$	$\frac{(m - \varepsilon) + \sqrt{(m - \varepsilon)^2 - 4c}}{2}$ $= \frac{c}{\frac{2}{3} - \varepsilon} + O\left(c^{\frac{3}{2}}\right)$
λ_i^q	$-2m - \varepsilon - \lambda_L^s = -\frac{4}{3} - \varepsilon + (2\gamma - \frac{4}{3})\sqrt{c} + O(c)$	$m - \varepsilon - \lambda_R^s = \frac{2}{3} - \varepsilon + (\gamma - \frac{2}{3})\sqrt{c} + O(c)$
$\mathbf{e}_i$	$\begin{pmatrix} a \\ -\gamma\sqrt{c} - 2m(a + \sqrt{c}) - a\varepsilon \end{pmatrix}$	$\begin{pmatrix} a \\ m(a - 1) - a\varepsilon + \frac{2}{3} \end{pmatrix}$

Table 4.2: Significant quantities for the dynamics of system (4.11) in the region σ_C .

	C
t_i	$\gamma - \varepsilon$
Δ_i	$(\gamma - \varepsilon)^2 - 4c$
λ_i^s, λ_i^q	$\frac{\gamma - \varepsilon}{2} \pm i \frac{\sqrt{4c - (\gamma - \varepsilon)^2}}{2}$
$\mathbf{e}_i$	$\begin{pmatrix} a \\ a(\gamma - \varepsilon) \end{pmatrix}$

In order to get analogous dynamics to the smooth system (3.6), we want the critical manifold to have an attracting branch in σ_L and a repelling branch in σ_R . In addition, we expect the system to exhibit a Hopf-like bifurcation (see Section 2.5.2), which can be either a subcritical BEB at $a = \sqrt{c}$ or a supercritical BEB at $a = -\sqrt{c}$.

Moreover, we aim to have imaginary eigenvalues only in the region σ_C , where the connection between the branches of the slow manifold has to take place. It is easy to check that for any of the following four conditions (4.12)-(4.15), the critical manifold's branch in σ_L is attracting (because $t_L < 0$) and the branch in σ_R is repelling (because $t_R > 0$) and $\Delta_L, \Delta_R > 0$ and $\Delta_C < 0$.

$$\gamma < \min \left\{ \frac{\frac{2}{3} - (1 - \sqrt{c})(\varepsilon + 2\sqrt{c})}{\sqrt{c}}, \frac{\frac{2}{3} + (1 - \sqrt{c})\frac{(\varepsilon - 2\sqrt{c})}{2}}{\sqrt{c}} \right\}, \quad (4.12)$$

$$\max \left\{ \frac{\frac{2}{3} - (1 - \sqrt{c})(\varepsilon - 2\sqrt{c})}{\sqrt{c}}, \frac{\frac{2}{3} + (1 - \sqrt{c})\frac{(\varepsilon - 2\sqrt{c})}{2}}{\sqrt{c}} \right\} < \gamma, \quad (4.13)$$

$$\frac{\frac{2}{3} - (1 - \sqrt{c})(\varepsilon - 2\sqrt{c})}{\sqrt{c}} < \gamma < \min \left\{ \frac{\frac{2}{3} + (1 - \sqrt{c})\frac{(\varepsilon - 2\sqrt{c})}{2}}{\sqrt{c}}, \frac{\frac{2}{3} - (1 - \sqrt{c})\varepsilon}{\sqrt{c}} \right\}, \quad (4.14)$$

$$\frac{\frac{2}{3} + (1 - \sqrt{c})\frac{(\varepsilon + 2\sqrt{c})}{2}}{\sqrt{c}} < \gamma < \frac{\frac{2}{3} - (1 - \sqrt{c})(\varepsilon + 2\sqrt{c})}{\sqrt{c}}. \quad (4.15)$$

From now on, we will assume for technical reasons that $0 < \varepsilon < \frac{2}{3}$, which ensures that condition (4.12) is satisfied for a sufficiently small c value. Indeed, the condition can be rewritten to

$$\gamma < \min \left\{ \frac{\frac{2}{3} - \varepsilon}{\sqrt{c}} + 2 - \varepsilon - 2\sqrt{c}, \frac{\frac{2}{3} + \frac{\varepsilon}{2}}{\sqrt{c}} - \left(\frac{\varepsilon}{2} + 1 \right) + \sqrt{c} \right\},$$

which translates to $\gamma < \infty$ as $c \rightarrow 0$ whenever ε stays in the mentioned range.

Proposition 4. *For $\gamma < \varepsilon$ (analogously, $\gamma > \varepsilon$), and considering that one of the conditions (4.12)-(4.15) holds, system (4.11) presents a subcritical (analogously, supercritical) Hopf-like bifurcation for the parameter value $a_H = \sqrt{c}$ (analogously, $a_H = -\sqrt{c}$).*

Proof. We will prove the case of the subcritical bifurcation. Let us obtain an equivalent system in Liénard form (2.8) for the system (4.11) in regions σ_C, σ_R . Consider the translation of variables $\hat{x} = x - \sqrt{c}$, $\hat{y} = y - (\gamma - \varepsilon)\sqrt{c}$ along with the reparametrization $\lambda = c(a - \sqrt{c})$. The outcome of the transformation is the following Liénard form centred at zero (see (2.9)):

$$\begin{cases} \dot{\hat{x}} = F(\hat{x}) - \hat{y}, \\ \dot{\hat{y}} = G(\hat{x}) - \lambda, \end{cases} \quad (4.16)$$

where

$$F(\hat{x}) = \begin{cases} (\gamma - \varepsilon)\hat{x} & \text{if } \hat{x} \leq 0, \\ (m - \varepsilon)\hat{x} & \text{if } \hat{x} > 0, \end{cases} \quad G(\hat{x}) = \begin{cases} c\hat{x} & \text{if } \hat{x} \leq 0, \\ c\hat{x} & \text{if } \hat{x} > 0. \end{cases}$$

Following the notation in Theorem 8 (take into account that now t_L, t_R, d_L, d_R refer to the system (4.16) and so to the same parameters as in the statement of the Theorem 8) the subsequent inequalities hold: $d_L = d_R = c > 0$, $t_R = m - \varepsilon > 0$ and $t_L = \gamma - \varepsilon < 0$ by

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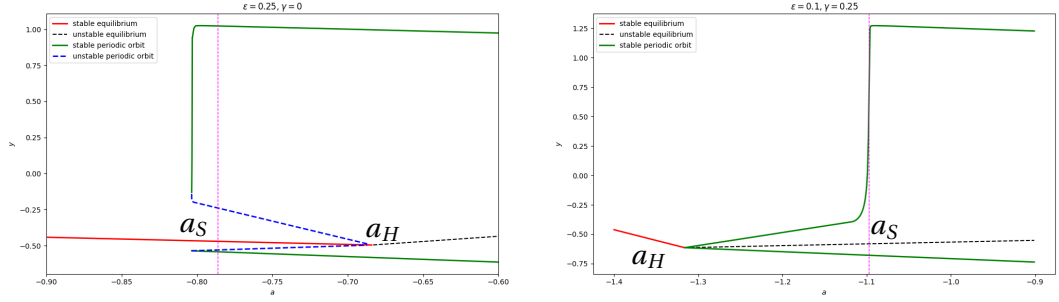


Figure 4.7: Bifurcation diagrams for the PWL model (3.7). Left image exposes a subcritical Hopf-like bifurcation when $\varepsilon = 0.25, \gamma = 0$, where $a_H = -0.6837\dots$ and $a_S = -0.78609\dots$ (with an error of ± 0.01). Right image shows a supercritical one case when $\varepsilon = 0.1, \gamma = 0.25$, where $a_H = -1.3162\dots$ and $a_S = -1.09766\dots$ (with an error of ± 0.01). Moreover, canard explosions can be seen. It is important to remark that two terms of the Taylor expansion of a_S have been used to compute the values on this figure, even though in Theorem 14 only the first coefficient is presented.

hypothesis. As, due to conditions (4.12)-(4.15), $t_L^2 - 4d_L = \Delta_C < 0$ and $t_R^2 - 4d_R = \Delta_R \geq 0$, by Theorem 8 *ii*), the system (4.16) presents a subcritical Hopf-like bifurcation for $\lambda = 0$. Because $\frac{d\lambda}{da} = c > 0$, the original system (4.11) exhibits the same type of bifurcation for $a = \sqrt{c}$.

The proof for the supercritical case follows in an analogous way by centring the system at $(x, y)^T = (-\sqrt{c}, g(-\sqrt{c}))^T$ and taking into account that when $\gamma > \varepsilon$ then $t_C > 0$. $\square$

By applying the previous proposition and undoing the change of variables (4.10) it easily follows that the original PWL system (3.7) presents a supercritical Hopf-like bifurcation at $a = -1 - \sqrt{c}$ when $\gamma > \varepsilon$ or a subcritical Hopf-like bifurcation at $a = -1 + \sqrt{c}$ when $\gamma < \varepsilon$. Figure 4.7 illustrates the specific cases of $\varepsilon = 0.25, \gamma = 0$ and $\varepsilon = 0.1, \gamma = 0.25$, both for $c = 0.1$.

In each of the regions σ_i for $i \in \{L, R\}$ the flow of the system is defined as $\varphi(s) = C_i^1 e^{s\lambda_i^s} \mathbf{v}_i^s + C_i^2 e^{s\lambda_i^q} \mathbf{v}_i^q$ for $C_i^1, C_i^2 \in \mathbb{R}$ (see Proposition 2 and consider the change of basis to the diagonal matrix given by the eigenvalues λ_i^s, λ_i^q and eigenvectors $\mathbf{v}_i^s, \mathbf{v}_i^q$), which suggests that the slow manifold can be computed directly from the slow eigenvalues. Lemma 7 in [17] directly proves the statement. In this way, the slow manifold S_c (note the change of notation to S_c for the slow manifold from the introduction of Chapter 4, where the slow manifold was designated as S_ε) of the system (4.11) in the designed regions is given by:

$$S_c = \begin{cases} \mu_L = \mathbf{e}_L + r \mathbf{v}_L^s & r \in \left[-\frac{\sqrt{c}+a}{\lambda_L^s}, \infty \right), \\ \mu_R = \mathbf{e}_R + r \mathbf{v}_R^s & r \in \left[\frac{a-2+\sqrt{c}}{\lambda_R^s}, \frac{a-\sqrt{c}}{\lambda_R^s} \right]. \end{cases}$$

And specifically, $S_c = S_c^a \cup S_c^r$, where $S_c^a = \mu_L$, $S_c^r = \mu_R$ are the attracting and repelling branches, respectively. The intersection of both branches with the switching lines

$\{x = -\sqrt{c}\}$ and $\{x = \sqrt{c}\}$ are given by:

$$\mathbf{q}_1^L = \begin{pmatrix} -\sqrt{c} \\ -\gamma\sqrt{c} - a\varepsilon + (\varepsilon + \lambda_L^s)(\sqrt{c} + a) \end{pmatrix}, \quad \mathbf{q}_0^R = \begin{pmatrix} \sqrt{c} \\ -m(1-a) - a\varepsilon + \frac{2}{3} - \lambda_R^q(a - \sqrt{c}) \end{pmatrix}, \quad (4.17)$$

respectively.

Through the following theorem (Theorem 14) we proof the existence of a parameter value a_S which allows a connection between S_c^a and S_c^r just as Theorem 12 does for the smooth framework. As region σ_L contains an attracting branch and σ_R a repelling branch, we expect a solution coming close to the slow manifold to reach the central region at $\mathbf{q}_1^L$. Thus, all we need to ensure is the existence of a solution in σ_C that connects points $\mathbf{q}_1^L$ and $\mathbf{q}_0^R$.

Theorem 14. Consider system (4.11) with $0 < \varepsilon < \frac{2}{3}$ and $\varepsilon - \gamma = k\sqrt{c}$ for $k \in (-2, 2) \setminus \{0, \pm \frac{\sqrt{2}}{2}\}$, and let $\mathbf{q}_1^L, \mathbf{q}_0^R$ be the points given in (4.17). There exists a value $0 < \mu \ll 1$ and two analytic functions A_S and η_S defined in $U = (0, 2) \setminus \{\frac{\sqrt{2}}{2}\} \times (0, \frac{2}{3}) \times (-\mu, \mu)$ in the subcritical case ($U = (-2, 0) \setminus \{-\frac{\sqrt{2}}{2}\} \times (0, \frac{2}{3}) \times (-\mu, \mu)$ in the supercritical case) such that, if $0 < c < \mu^2$ and $a = a_S(k, \varepsilon, c) := \sqrt{c}A_S(k, \varepsilon, \sqrt{c})$, then the orbit of system (4.11) starting at the point $\mathbf{q}_1^L$ reaches the switching line $\{x = \sqrt{c}\}$ at the point $\mathbf{q}_0^R$ with the time of flight $\tau_C^S(k, \varepsilon, c) := \frac{\eta_S(k, \varepsilon, \sqrt{c})}{\sqrt{c}} > 0$. In addition, the first terms of the expansion of a_S and τ_C^S in terms of $\sqrt{c}$ are given by

$$a_S(k, \varepsilon, c) = \frac{1 - e^{-\frac{\pi k}{\sqrt{4-k^2}}}}{1 + e^{-\frac{\pi k}{\sqrt{4-k^2}}}} \sqrt{c} + O(c^{\frac{3}{2}}), \quad (4.18)$$

and

$$\tau_C^S(k, \varepsilon, c) = \frac{2\pi}{\sqrt{4-k^2}} \frac{1}{\sqrt{c}} + \frac{18}{(3\varepsilon-2)(3\varepsilon+4)} + O(\sqrt{c}). \quad (4.19)$$

Proof. The conditions for a solution to go from $\mathbf{q}_1^L$ to $\mathbf{q}_0^R$ reduces to the existence of values $a_S \in \mathbb{R}$, $\tau_C^S > 0$ such that the following system of equations holds

$$\begin{cases} F_1(\tau_C^S, a_S, k, \varepsilon, c) = 0, \\ F_2(\tau_C^S, a_S, k, \varepsilon, c) = 0, \\ F_1(s, a_S, k, \varepsilon, c) \in (0, 2\sqrt{c}), \quad \forall s \in (0, \tau_C^S), \end{cases} \quad (4.20)$$

where functions F_1, F_2 represent the distance from each coordinate of the point $\mathbf{q}_0^R$ and the intersection point between $\{x = \sqrt{c}\}$ and the solution $(x^C(s; \mathbf{q}_1^L, a, k, \varepsilon), y^C(s; \mathbf{q}_1^L, a, k, \varepsilon))$ of system (4.11) starting at $\mathbf{q}_1^L$. The third expression ensures that the connection is established entirely in the region of definition σ_C over the time interval $(0, \tau_S)$. Therefore,

$$\begin{cases} F_1(s, a_S, k, \varepsilon, c) = x^C(s; \mathbf{q}_1^L, a, k, \varepsilon) - \sqrt{c}, \\ F_2(s, a_S, k, \varepsilon, c) = y^C(s; \mathbf{q}_1^L, a, k, \varepsilon) + \left(\left(\frac{2}{3(1-\sqrt{c})} + \frac{k\varepsilon - \sqrt{c}}{1-\sqrt{c}} \right) (1-a) + a\varepsilon - \frac{2}{3} + \lambda_R^q(a - \sqrt{c}) \right). \end{cases} \quad (4.21)$$

Having the particular solution $\mathbf{e}_C$, one can easily find the solution of system (4.11) in region σ_C by solving the homogeneous system $\dot{\mathbf{z}} = A_C \mathbf{z}$ through Proposition 2, undergoing the change $\mathbf{z} = \mathbf{x} - \mathbf{e}_C$ and adding $\mathbf{e}_C$. Thus,

$$\begin{cases} x^C(s; \mathbf{q}_1^L, a, k, \varepsilon) = & a + (a + \sqrt{c})e^{-s\frac{k\sqrt{c}}{2}} \left[-\cos\left(\frac{\sqrt{4-k^2}}{2}\sqrt{cs}\right) \right. \\ & \left. - \frac{1}{\sqrt{4-k^2}} \left(k + \frac{2}{\sqrt{c}}\lambda_L^s \right) \sin\left(\frac{\sqrt{4-k^2}}{2}\sqrt{cs}\right) \right], \\ y^C(s; \mathbf{q}_1^L, a, k, \varepsilon) = & -ak\sqrt{c} + (a + \sqrt{c})e^{-s\frac{k\sqrt{c}}{2}} \left[(k\sqrt{c} + \lambda_L^s)\cos\left(\frac{\sqrt{4-k^2}}{2}\sqrt{cs}\right) \right. \\ & \left. + \frac{1}{\sqrt{4-k^2}} (k^2\sqrt{c} - 2\sqrt{c} + k\lambda_L^s) \sin\left(\frac{\sqrt{4-k^2}}{2}\sqrt{cs}\right) \right]. \end{cases}$$

The change of variables given by the *blow-up* $(s, a, k, \varepsilon, c) = (\frac{\eta}{\delta}, A\delta, k, \varepsilon, \delta^2)$, valid for $c, \delta > 0$ enables to rewrite the equations $F_1 = 0, F_2 = 0$ into the equivalent form $\delta G_1(\eta, A, k, \varepsilon, \delta) = 0, \delta^2 G_2(\eta, A, k, \varepsilon, \delta) = 0$ where G_1, G_2 are given by:

$$\begin{cases} G_1(\eta, A, k, \varepsilon, \delta) = & A + (A + 1)e^{-\eta\frac{k}{2}} \left[-\cos\left(\frac{\sqrt{4-k^2}}{2}\eta\right) - \frac{1}{\sqrt{4-k^2}} \left(k + \frac{4\delta}{K_1} \right) \sin\left(\frac{\sqrt{4-k^2}}{2}\eta\right) \right] - 1, \\ G_2(\eta, A, k, \varepsilon, \delta) = & -Ak + (A + 1)e^{-\eta\frac{k}{2}} \left[\left(k + \frac{2\delta}{K_1} \right) \cos\left(\frac{\sqrt{4-k^2}}{2}\eta\right) \right. \\ & \left. - \frac{1}{\sqrt{4-k^2}} \left(k^2 - 2 + \frac{2k\delta}{K_1} \right) \sin\left(\frac{\sqrt{4-k^2}}{2}\eta\right) \right] + \frac{k + O(\delta)}{1 - \delta}, \end{cases}$$

where $K_1 := -\left(\frac{4}{3(1-\delta)} + \frac{\varepsilon(1-3\delta)+2k\delta^2}{1-\delta}\right) - \sqrt{\left(\frac{4}{3(1-\delta)} + \frac{\varepsilon(1-3\delta)+2k\delta^2}{1-\delta}\right)^2 - 4\delta^2}$. Notice how the value of λ_R^q has been reduced to its Taylor expansion given in Table 4.1.

System $(F_1, F_2) = (0, 0)$ is equivalent to system

$$E_q(\eta, A, k, \varepsilon, \delta) := (G_1(\eta, A, k, \varepsilon, \delta), G_2(\eta, A, k, \varepsilon, \delta)) = (0, 0), \quad (4.22)$$

for $\delta > 0$ Moreover, system (4.22) can be extended to $\delta = 0$. Indeed,

$$(\eta_0, A_0, k_0, \varepsilon_0, \delta_0) = \left(\frac{2\pi}{\sqrt{4-k^2}}, \frac{1 - e^{-\frac{k\pi}{\sqrt{4-k^2}}}}{1 + e^{-\frac{k\pi}{\sqrt{4-k^2}}}}, k, \varepsilon, 0 \right),$$

is already a solution. Willing to apply the Implicit Function Theorem, we compute

$$\begin{aligned} \frac{\partial G_1}{\partial \eta} \Big|_{(\eta_0, A_0, k_0, \varepsilon_0, \delta_0)} &= (A_0 + 1)k e^{-\frac{k}{2}\eta_0}, & \frac{\partial G_1}{\partial A} \Big|_{(\eta_0, A_0, k_0, \varepsilon_0, \delta_0)} &= 1 + e^{-\frac{k}{2}\eta_0}, \\ \frac{\partial G_2}{\partial \eta} \Big|_{(\eta_0, A_0, k_0, \varepsilon_0, \delta_0)} &= (A_0 + 1)(k^2 - 1)e^{-\frac{k}{2}\eta_0}, & \frac{\partial G_2}{\partial A} \Big|_{(\eta_0, A_0, k_0, \varepsilon_0, \delta_0)} &= -k \left(1 + e^{-\frac{k}{2}\eta_0} \right), \end{aligned}$$

so,

$$\det(D_{\eta, A} E_q(\eta_0, A_0, k_0, \varepsilon_0, \delta_0)) = -(A_0 + 1)e^{-\frac{k}{2}\eta_0} \left(1 + e^{-\frac{k}{2}\eta_0} \right) (2k^2 - 1) = 0 \Leftrightarrow k = \pm \frac{\sqrt{2}}{2}.$$

Consequently, by the Implicit Function Theorem, we conclude that there exist a value $\mu > 0$ and functions $A_S(k, \varepsilon, \delta)$ and $\eta_S(k, \varepsilon, \delta)$, analytic as a function of (k, ε, δ) defined in the open set $U = (0, 2) \setminus \left\{ \frac{\sqrt{2}}{2} \right\} \times (0, \frac{2}{3}) \times (-\mu, \mu)$ for the subcritical case and $U = (-2, 0) \setminus \left\{ -\frac{\sqrt{2}}{2} \right\} \times (0, \frac{2}{3}) \times (-\mu, \mu)$ for supercritical case and such that

$$E_q(\eta_S(k, \varepsilon, \delta), A_S(k, \varepsilon, \delta), k, \varepsilon, \delta) = (0, 0).$$

By undoing the change of variables one directly proves the existence of functions τ_C^S and a_S . Terms in expressions (4.18)-(4.19) can be obtained by the method of undetermined coefficients in system (4.21), having $\frac{\eta_0}{\sqrt{c}}$ as a starting point for τ_C^S and $A_0\sqrt{c}$ for a_S . A *Maxima* file verifying the preceding computations is available in this *GitHub* public repository.

We need to check that the third expression in (4.20) is fulfilled. Let us prove that $\mathbf{q}_0^R$ stays below the nullcline for a sufficient small c . Subtracting the y component of $\mathbf{q}_0^R$ to the y component of the intersection of the nullcline with $\{x = \sqrt{c}\}$ we get

$$(a_S - \sqrt{c})(\varepsilon + \lambda_R^q - m) = (a_S - \sqrt{c}) \left(\frac{2(\gamma - \frac{2}{3})\sqrt{c} + O(c)}{1 - \sqrt{c}} \right),$$

and since $(a_S - \sqrt{c}), (\gamma - \frac{2}{3}) < 0$ for a sufficiently small c value, the expression above is positive. Indeed, $\gamma = \varepsilon - k\sqrt{c} < \frac{2}{3} - k\sqrt{c}$ where k is bounded.

Notice how in system (4.11) the solution increases in the horizontal component x when lying below $\{g(x) = 0\}$. Therefore, as for $|a_S(k, \varepsilon, c)| < \sqrt{c}$ the point $\mathbf{q}_0^R$ stays below the nullcline, the connection is established from the left of the line $\{x = \sqrt{c}\}$. Moreover, $\sqrt{c(4 - k^2)}\tau_C^S(k, \varepsilon, c) < 2\pi$, so the angular displacement of the solution is below π , hence the path must be contained entirely inside the region σ_C . $\square$

We are already in the position to prove the existence of canard cycles and the canard explosion in the system (4.11), and consequently (3.7). However, let us first define a value $x_S(\sqrt{c})$ that sets a border between Hopf and canard cycles.

The solution of a linear system in the slow-fast scenario has been proven to depend on a term of order $e^{\lambda^s t}$ (here, t refers to fast time) in the direction of the slow manifold and on a term of order $e^{\lambda^q t}$ in the fast direction. Moreover, as $O(\lambda^q) = 1$, the flow evolves in the direction of $\mathbf{v}^q$ as e^{kt} , with $k \in \mathbb{R}$, which translates in slow time $\tau = tc$ to $e^{\frac{k}{c}\tau}$. Therefore, the flow moves away or approaches the slow manifold at a pace of $O\left(e^{\frac{1}{c}}\right)$. Hence, we expect the flow of a limit cycle to be at a distance similar to $O\left(e^{-\frac{1}{c}}\right)$ at the beginning of the repelling branch, $\mathbf{q}_0^R$, in order to remain over it. Analogously, the flow of the canard orbit needs to be exponentially close to $\mathbf{q}_1^L$ in order to eventually be close to $\mathbf{q}_0^R$ thanks to the theorem of flow continuity with respect to initial conditions on continuous and Lipschitz systems, which applies for the system in σ_C .

Proposition 5. *For $c_0 > 0$ sufficiently small, set $c < c_0$ and $-\sqrt{c} < a < \sqrt{c}$. If $x_0 \in (x_S, 1 - \sqrt{c})$, then an orbit Γ_{x_0} that goes through $(x_0, f(x_0))^T$ passes exponentially close to $\mathbf{q}_1^L$ and to $\mathbf{q}_0^R$. In the case of being a cycle, it is under canard regime. The endpoint of the canard regime x_S is defined as $x_S = v_S(\sqrt{c} - a_S) + a_S$, where v_S depends on whether the Hopf bifurcation is supercritical*

$$v_S = \exp \left[\frac{1}{\frac{2}{3} - \varepsilon} \left(\left(\frac{2}{3} - \varepsilon \right) \ln \left(1 - \frac{(\sqrt{c} + a_S)(\varepsilon + \frac{4}{3})}{(\sqrt{c} - a_S) \ln(c)(\frac{2}{3} - \varepsilon)} \right) - \frac{c}{\frac{2}{3} - \varepsilon} \ln \left(1 - \frac{(\sqrt{c} + a_S)(\varepsilon + \frac{4}{3})}{(\sqrt{c} - a_S)c \ln(c) \left(\frac{2}{3} - \varepsilon \right)} \right) \right) \right], \quad (4.23)$$

or subcritical

$$v_S = 1 - \frac{1}{\ln(c)}. \quad (4.24)$$

Proof. The proof is performed through the analysis of the divergence in a neighbourhood of both the repelling and attracting branches, and it is similar to that of Lemma 4.1 in [3].

Our goal is to find $x_S = \max(x_S^L, x_S^R)$, where x_S^L (equivalently x_S^R) is the ordinate of the intersection point of an orbit Γ with the critical manifold in the region σ_R such that its flow gets exponentially close to $\mathbf{q}_1^L$ (equivalently $\mathbf{q}_0^R$). Starting from the region σ_L , we consider reverse time $\tau = -t$ in order to follow the proof in the mentioned Lemma. Notice how this change does not affect the position of the orbits in the original system. We can rewrite the system as $\tilde{A}_L \mathbf{x} + \tilde{\mathbf{b}}_L$ where:

$$\tilde{A}_L = \begin{pmatrix} 2m + \varepsilon & 1 \\ -c & 0 \end{pmatrix}, \quad \tilde{\mathbf{b}}_L = \begin{pmatrix} (2m + \gamma)\sqrt{c} \\ ca \end{pmatrix},$$

so eigenvalues are $\tilde{\lambda}_L^s \approx \frac{c}{\varepsilon + \frac{4}{3}}$ and $\tilde{\lambda}_L^q \approx \varepsilon + \frac{4}{3}$.

As it is indicated in Lemma 4.1 in [3], $H_i(u_1, u_2) = |u_1 + \tilde{\lambda}_i^s u_2|^{\tilde{\lambda}_i^q} |u_1 + \tilde{\lambda}_i^q u_2|^{-\tilde{\lambda}_i^s}$ is a first integral of the system in region σ_i for $i \in \{L, R\}$, meaning that the function is constant over the flow.

We consider the Krylov basis $\{\mathbf{p}_L, \dot{\mathbf{p}}_L\}$, where the coordinates of vector $\mathbf{p}_L$ coincide with those of the intersection between the critical manifold and the switching line $\{x = -\sqrt{c}\}$. When written in the Krylov basis, $\mathbf{q}_1^L = \left(1, -\frac{1}{\tilde{\lambda}_L^q}\right)^T$, so we consider the point

$(1, -u_S^L(c))^T$ with $u_S^L(c) = \frac{1}{\tilde{\lambda}_L^q} - \frac{1}{\tilde{\lambda}_L^q} e^{-\frac{(\varepsilon + \frac{4}{3})^2}{c|\ln(c)|}}$ to be the border between canard and not canard phenomena. According to expression (41) in [3], $H_L(1, -u_S^L(c)) = \left(1 + \frac{1}{|\ln(c)|}\right)^{\varepsilon + \frac{4}{3}}$.

Since H_L is a first integral, we can find the *Poincaré Half Map*, meaning the second point of intersection of the flow with the switching line $\{x = -\sqrt{c}\}$, from $H_L(1, -u_S^L(c)) = H_L(1, v_S^L(c))$, which leads after taking approximations to $v_S^L(c) \approx \frac{1}{\tilde{\lambda}_L^s |\ln(c)|} \approx \frac{\varepsilon + \frac{4}{3}}{c|\ln(c)|}$. Indeed, by using the Taylor expansion of the eigenvalues for a small c value one gets

$$H(1, v_S^L(c)) = \frac{|1 + \tilde{\lambda}_L^s v_S^L(c)|^{\tilde{\lambda}_L^q}}{|1 + \tilde{\lambda}_L^q v_S^L(c)|^{\tilde{\lambda}_L^s}} \approx |1 + \tilde{\lambda}_L^s v_S^L(c)|^{\tilde{\lambda}_L^q}.$$

Hence, expression $H_L(1, -u_S^L(c)) = H_L(1, v_S^L(c))$ can be written as

$$\left(1 + \frac{1}{|\ln(c)|}\right)^{\varepsilon + \frac{4}{3}} = |1 + \tilde{\lambda}_L^s v_S^L(c)|^{\varepsilon + \frac{4}{3}}.$$

Then, solving for $v_S^L(c)$ we obtain

$$v_S^L(c) \approx \frac{\varepsilon + \frac{4}{3}}{c|\ln(c)|}.$$

Suppose that Γ crosses the whole region σ_C . Then, Γ intersects the switching lines at the points $(-\sqrt{c}, h)^T$ and $(\sqrt{c}, h + O(c^2))^T$. In fact, as the central band has a width of $O(\sqrt{c})$, then quick time takes $O(\sqrt{c})$ to go from one switching line to the other. This translates for the vertical displacement $\Delta y = \int_0^t c(x(t) - a) dt$ to have $t, a, x \sim O(\sqrt{c})$, so

$\Delta y = O(c^2)$. Consequently, we will consider both intersecting points of Γ with switching lines $\{x = -\sqrt{c}\}, \{x = \sqrt{c}\}$ to have the same ordinate.

The point $(1, v_S^L(c))^T$ in Krylov basis can be expressed as $\left(-\sqrt{c}, \varepsilon - \gamma - \frac{(\sqrt{c}+a)(\varepsilon+\frac{4}{3})}{\ln(c)}\right)^T$ in canonical basis. Denote by $\mathbf{p}_v^L$ the point on the switching line $\{x = \sqrt{c}\}$ with this same ordinate.

Moving to the region σ_R , we can carry out an analogous development to get the value $v_S^R(c) \approx \frac{\frac{2}{3}-\varepsilon}{c|\ln(c)|}$ as the second coordinate of the highest intersecting point with the switching line of an orbit going exponentially close to $\mathbf{q}_0^R$. We can express $\mathbf{p}_v^L$ in the Krylov basis of region σ_R , $\{\mathbf{p}_R, \dot{\mathbf{p}}_R\}$, as

$$\mathbf{p}_v^L = \begin{pmatrix} 1 \\ \frac{2(\varepsilon-\gamma)}{\sqrt{c}(\sqrt{c}-a)} - \frac{(\sqrt{c}+a)(\varepsilon+\frac{3}{4})}{(\sqrt{c}-a)c\ln(c)} \end{pmatrix}.$$

Suppose we are on the supercritical case ($\varepsilon - \gamma < 0$), then a should be near a_S which is much closer to $\sqrt{c}$ than to $-\sqrt{c}$. Thus, we neglect the term $\frac{2(\varepsilon-\gamma)}{\sqrt{c}(\sqrt{c}-a)}$ in the expression of $\mathbf{p}_v^L$,

$$\mathbf{p}_v^L \approx \begin{pmatrix} 1 \\ -\frac{(\sqrt{c}+a)(\varepsilon+\frac{3}{4})}{(\sqrt{c}-a)c\ln(c)} \end{pmatrix},$$

and easily see that its second coordinate is bigger than $v_S^R(c)$. Then, as its magnitude is also bigger than $O(c^2)$, we can ensure that the orbit going through $\mathbf{p}_v^L$ will cross from region σ_R to σ_L and will be exponentially close to points $\mathbf{q}_1^L$ and $\mathbf{q}_0^R$. Consequently, the intersection of the orbit Γ that goes through $\mathbf{p}_v^L$ in σ_R with the critical manifold will give the coordinate x_S . Since this latter point can be expressed in the Krylov basis as $(v_S, 0)^T$, we can obtain the value (4.23) for v_S from the first integral H_R .

Let us now consider the subcritical case ($\varepsilon - \gamma > 0$). The point $(1, v_S^R(c))^T$ can be expressed as $\left(-\sqrt{c}, \varepsilon - \gamma - \frac{(\sqrt{c}-a)(\frac{2}{3}-\varepsilon)}{\ln(c)}\right)^T$ in canonical basis, which turns out to have a greater second coordinate than $(1, v_S^L(c))^T$ when a_S is close to $-\sqrt{c}$. As a result, we can compute x_S for the subcritical case directly from the Krylov basis in the same region σ_R .

In order to Γ be exponentially close to $\mathbf{q}_0^R$ we consider $u_S^R(c) = \frac{1}{\lambda_R^q} - \frac{1}{\lambda_R^q} e^{-\frac{(\frac{2}{3}-\varepsilon)^2}{c|\ln(c)|}}$, and hence $H_R(1, -u_S^R(c)) = \left(1 + \frac{1}{|\ln(c)|}\right)^{\frac{2}{3}-\varepsilon}$. Moreover, $H_R(v_S, 0) = v_S^{\lambda_R^q - \lambda_R^s}$, which can be approximated in first order as $v_S^{\frac{2}{3}-\varepsilon}$, thus leading to (4.24).

Finally, the value for x_S comes from the first coordinate in $\mathbf{e}_R + v_S(\mathbf{p}_R - \mathbf{e}_R)$. □

Until now, the existence of parameter values for a leading to a connection of the slow manifolds has been proven in Theorem 14, while Proposition 5 characterises an interval around the repelling branch of the slow manifold where periodic orbits get exponentially close to the latter. It will be proven in the following result that the parameter a value that ensures the orbits going through that interval to be periodic is exponentially close to the connection value a_S , which eventually explains the explosive growth of the amplitude of the orbit. For the sake of simplicity we will state and proof

the result for headless canard cycles only. The case of headed canard cycles follows similarly but requires to introduce two extra subdomains of definition of the original PWL system that complicate the computations.

Theorem 15. *Consider $c_0 > 0$ to be sufficiently small, $0 < \varepsilon < \frac{2}{3}$ and $\varepsilon - \gamma = k\sqrt{c}$ for $k \in (-2, 2) \setminus \{0, \pm \frac{\sqrt{2}}{2}\}$. Given $x_0 \in (x_S, 1 - \sqrt{c})$ there exists a function $a_N(k, \varepsilon, c, x_0)$, C^∞ function of (k, c, x_0) defined in the open set $U = (0, 2) \setminus \{\frac{\sqrt{2}}{2}\} \times (0, \frac{2}{3}) \times (0, c_0) \times (x_S, 1 - \sqrt{c})$ in the subcritical case and $U = (-2, 0) \setminus \{-\frac{\sqrt{2}}{2}\} \times (0, \frac{2}{3}) \times (0, c_0) \times (x_S, 1 - \sqrt{c})$ in the supercritical case, such that for $(k, c, x_0) \in U$ and the parameter value $a = a_N(k, \varepsilon, c, x_0)$ the system (4.11) presents a canard limit cycle Γ_{x_0} going through $(x_0, f(x_0))^T$.*

Moreover,

$$|a_S(k\varepsilon, c) - a_N(k, \varepsilon, c, x_0)| = O\left(\frac{e^{-\frac{1}{c}}}{c}\right).$$

Proof. Given a point $(x_0, f(x_0))^T$, with $x_0 \in (x_S, 1 - \sqrt{c})$ we can consider the orbit Γ_{x_0} which will be proven to be periodic. Given the dynamics of the system, by integrating in forward and backward time we get to points $\mathbf{p}_1$ and $\mathbf{p}_0$, which stay exponentially close to $\mathbf{q}_1^L$ and $\mathbf{q}_0^R$, respectively, since $x > x_S$. Thus we can express them as:

$$\mathbf{p}_1 = \mathbf{q}_1^L + \begin{pmatrix} 0 \\ O(e^{-\frac{1}{c}}) \end{pmatrix}, \quad \mathbf{p}_0 = \mathbf{q}_0^R + \begin{pmatrix} 0 \\ O(e^{-\frac{1}{c}}) \end{pmatrix}.$$

As long as there exists a parameter value a such that the connection between $\mathbf{p}_1$ and $\mathbf{p}_0$ is established in region σ_C , the orbit Γ_{x_0} is a canard limit cycle. Since the whole system becomes linear in the region σ_C , according to the development in Section 2.2.1, the following equation must hold:

$$e^{sAc} (\mathbf{p}_0 - \mathbf{e}_C) + \mathbf{e}_C - \mathbf{p}_1 = \mathbf{0},$$

so

$$e^{sAc} (\mathbf{q}_0^R - \mathbf{e}_C) + \mathbf{e}_C - \mathbf{q}_1^L + e^{sAc} \begin{pmatrix} 0 \\ O(e^{-\frac{1}{c}}) \end{pmatrix} - \begin{pmatrix} 0 \\ O(e^{-\frac{1}{c}}) \end{pmatrix} = \mathbf{0}. \quad (4.25)$$

By performing the same change of variables as in the proof of Theorem 14, $(s, a, k, \varepsilon, c) = (\frac{\eta}{\delta}, A\delta, k, \varepsilon, \delta^2)$, and multiplying by the matrix M_δ defined as

$$M_\delta := \begin{pmatrix} \frac{1}{\delta} & 0 \\ 0 & \frac{1}{\delta^2} \end{pmatrix},$$

we obtain the following expression from equation (4.25):

$$E_p(\eta, A, k, \varepsilon, \delta) := E_q(\eta, A, k, \varepsilon, \delta) + \tilde{E}_q(\eta, A, k, \varepsilon, \delta) = \mathbf{0}, \quad (4.26)$$

where $E_q(\eta, A, k, \varepsilon, \delta)$ is the same as in (4.22). From this, we obtain

$$\tilde{E}_q(\eta, A, k, \varepsilon, \delta) := M_\delta \left(e^{\eta B} \begin{pmatrix} 0 \\ O(e^{-\frac{1}{\delta^2}}) \end{pmatrix} - \begin{pmatrix} 0 \\ O(e^{-\frac{1}{\delta^2}}) \end{pmatrix} \right),$$

being

$$B = \begin{pmatrix} k & -\frac{1}{\delta} \\ \delta & 0 \end{pmatrix}.$$

System $\tilde{E}_q(\eta, A, k, \varepsilon, \delta) = \mathbf{0}$ is well defined for $\delta > 0$. Note that it can be extended to $\delta = 0$ since the exponential terms $e^{-\frac{1}{\delta^2}}$ in the expression vanish when δ goes to 0. Thus, we can extend continuously the function $E_p(\eta, A, k, \varepsilon, \delta)$ to $E_q(\eta, A, k, \varepsilon, \delta)$ when $\delta = 0$. In the proof of Theorem 14 we have seen that there is a solution for $E_q(\eta, A, k, \varepsilon, \delta)$ when $\delta = 0$, where the Jacobian matrix of $E_q(\eta, A, k, \varepsilon, \delta)$ is not zero. Consequently, since

$$D_{\eta, A} E_p \Big|_{(\eta, A, k, \varepsilon, \delta)} = D_{\eta, A} E_q \Big|_{(\eta, A, k, \varepsilon, \delta)} + M_\delta D_{\eta, A} \left(e^{\eta B} \begin{pmatrix} 0 \\ O\left(e^{-\frac{1}{\delta^2}}\right) \end{pmatrix} - \begin{pmatrix} 0 \\ O\left(e^{-\frac{1}{\delta^2}}\right) \end{pmatrix} \right) \Big|_{(\eta, A, k, \varepsilon, \delta)},$$

and $D_{\eta, A} \begin{pmatrix} 0 \\ O\left(e^{-\frac{1}{\delta^2}}\right) \end{pmatrix}$ involves terms in $O\left(e^{-\frac{1}{\delta^2}}\right)$ which tend to zero as $\delta \rightarrow 0$, the second term in the right hand side of the previous expression is not meaningful for the sign of $D_{\eta, A} E_p \Big|_{(\eta, A, k, \varepsilon, \delta)}$. In fact, it equals to the sign of $D_{\eta, A} E_q \Big|_{(\eta, A, k, \varepsilon, \delta)}$ and we have already proven it not to be zero. Just as in the proof of Theorem 14, we can apply the Implicit Function Theorem to conclude that there exist functions $A_N(k, \varepsilon, \delta)$ and $\eta_N(k, \varepsilon, \delta)$, analytic as functions of (k, ε, δ) defined in the open set $U = (0, 2) \setminus \left\{ \frac{\sqrt{2}}{2} \right\} \times \left(0, \frac{2}{3}\right) \times (-\mu, \mu)$ for the subcritical case and $U = (-2, 0) \setminus \left\{ -\frac{\sqrt{2}}{2} \right\} \times \left(0, \frac{2}{3}\right) \times (-\mu, \mu)$ for supercritical case and such that $E_p(\eta_N(k, \varepsilon, \delta), A_N(k, \varepsilon, \delta), k, \varepsilon, \delta) = (0, 0)$.

Let us finally locate the value of function $a_N(k, \varepsilon, c, x_0)$ close to $a_S(k, \varepsilon, c)$. From the Mean Value Theorem in integral form for functions in several variables, one gets that

$$E_p(\eta_N(k, \varepsilon, \delta), A_N(k, \varepsilon, \delta), k, \varepsilon, \delta) - E_p(\eta_S(k, \varepsilon, \delta), A_S(k, \varepsilon, \delta), k, \varepsilon, \delta) = \bar{E}_p(\eta_N, A_N, \eta_S, A_S, k, \varepsilon, \delta) \begin{pmatrix} \eta_N(k, \varepsilon, \delta) - \eta_S(k, \varepsilon, \delta) \\ A_N(k, \varepsilon, \delta) - A_S(k, \varepsilon, \delta) \end{pmatrix},$$

where

$$\bar{E}_p(\eta_N, A_N, \eta_S, A_S, k, \varepsilon, \delta) = \int_0^1 D_{\eta, A} E_p(s\eta_N + (1-s)\eta_S, sA_N + (1-s)A_S, k, \varepsilon, \delta) ds.$$

By taking a sufficiently small value of δ in (4.26), it follows that $E_p(\eta_N, A_N, k, \varepsilon, \delta) \approx \mathbf{0}$, $E_q(\eta_S, A_S, k, \varepsilon, \delta) \approx \mathbf{0}$ and $\bar{E}_p(\eta_N, A_N, \eta_S, A_S, k, \varepsilon, \delta) \approx D_{\eta, A} E_q(\eta_S, A_S, k, \varepsilon, 0)$. Therefore,

$$\begin{aligned} & -M_\delta \left(e^{\eta_S B} \begin{pmatrix} 0 \\ O\left(e^{-\frac{1}{\delta^2}}\right) \end{pmatrix} - \begin{pmatrix} 0 \\ O\left(e^{-\frac{1}{\delta^2}}\right) \end{pmatrix} \right) \approx \\ & \begin{pmatrix} (A_S(k, \varepsilon, 0) + 1)k e^{-\frac{k}{2}\eta_S(k, \varepsilon, 0)} & 1 + e^{-\frac{k}{2}\eta_S(k, \varepsilon, 0)} \\ (A_S(k, \varepsilon, 0) + 1)(k^2 - 1)e^{-\frac{k}{2}\eta_S(k, \varepsilon, 0)} & -k(1 + e^{-\frac{k}{2}\eta_S(k, \varepsilon, 0)}) \end{pmatrix} \begin{pmatrix} \eta_N(k, \varepsilon, \delta) - \eta_S(k, \varepsilon, \delta) \\ A_N(k, \varepsilon, \delta) - A_S(k, \varepsilon, \delta) \end{pmatrix}. \end{aligned}$$

Hence,

$$\begin{pmatrix} \eta_N(k, \varepsilon, \delta) - \eta_S(k, \varepsilon, \delta) \\ A_N(k, \varepsilon, \delta) - A_S(k, \varepsilon, \delta) \end{pmatrix} \approx \frac{1}{D} \begin{pmatrix} \frac{k(1+E)}{\delta} & \frac{1+E}{\delta^2} \\ \frac{(A+1)(k^2-1)E}{\delta} & -\frac{(A+1)E}{\delta^2} \end{pmatrix} \left(e^{\eta_S B} \begin{pmatrix} 0 \\ O\left(e^{-\frac{1}{\delta^2}}\right) \end{pmatrix} - \begin{pmatrix} 0 \\ O\left(e^{-\frac{1}{\delta^2}}\right) \end{pmatrix} \right),$$

where

$$E := e^{-\frac{k}{2}\eta_S(k,\varepsilon,0)} = e^{-\frac{\pi k}{\sqrt{4-k^2}}}; \quad A := A_S(k,\varepsilon,0) = \frac{E-1}{E+1},$$

$$D := \det(D_{\eta,A}E_q(\eta(k,\varepsilon,0), A(k,\varepsilon,0), k,\varepsilon,0)) = -(A+1)E(1+E)(2k^2-1).$$

Isolating now the second coordinates in the equality above, we conclude that

$$A_N(k,\varepsilon,\delta) - A_S(k,\varepsilon,\delta) \approx$$

$$-\frac{(A+1)Ee^{\frac{k}{2}\eta_S}}{D\sqrt{4-k^2}\delta} O\left(e^{-\frac{1}{\delta^2}}\right) \left(-\left(2(k^2-1) + \frac{k}{\delta}\right) \sin\left(\frac{\sqrt{4-k^2}}{2}\eta_S\right)\right.$$

$$\left. + \frac{\sqrt{4-k^2}}{\delta} \left(\cos\left(\frac{\sqrt{4-k^2}}{2}\eta_S\right) - e^{-\frac{k}{2}\eta_S}\right)\right) \approx O\left(\frac{e^{-\frac{1}{\delta^2}}}{\delta^2}\right).$$

By undoing the change of variables we obtain

$$a_N(k,\varepsilon,c) - a_S(k,\varepsilon,c) = O\left(\frac{e^{-\frac{1}{c}}}{c}\right).$$

□

From Proposition 4, Theorem 14, Proposition 5 and Theorem 15, we conclude that the canard explosion phenomenon can be equally expected in the PWL framework. As soon as a connection between the different branches of the slow manifold is formed, limit cycles placed above the manifold and going exponentially close to the repelling branch appear in the system as the parameter a is varied. Moreover, the phenomenon has been proven to happen at a distance of order $O\left(\frac{e^{-\frac{1}{c}}}{c}\right)$ from where the connection is established.

In this subsection we have characterized the behaviour of the model (3.7). Different configurations of the parameters γ, ε have been proven to lead to either a supercritical or subcritical Hopf-like bifurcation. In the first case, the scenario is analogous to that of the smooth model, with some differences in the placement of the bifurcations on the parameter axis and their relative distances to the canard explosion. In the subcritical case, the canard explosion is expected to happen for parameter values below the bifurcation value, as the canard cycles transition from the Hopf cycles that appear to the left of the bifurcation. Figure 4.7 shows these scenarios for two specific combinations of parameters γ, ε .

4.2 Bursting and Slow Passage through a Hopf

In the previous section we have characterized the behaviour of models (3.6) and (3.7) when having depreciated the oscillatory component. Now, we take the sinusoid back into the equation to see how the systems behave. In some way, letting a model evolve with this new configuration is like combining in the same simulation all those scenarios from the previous section where the parameter value a is now considered as $k_F = a + P \sin(2\pi f t_i)$ for a specific time value $t = t_i$. We will denote as the **static system** k_F the system that has no oscillatory component and whose parameter value is $a = k_F$.

The frequency parameter value will be set to some order of magnitude below that of the slow-fast parameter c in order to let the solution of the ODE system respond to each static system k_F with $k_F \in [a - P, a + P]$. Conversely, being governed by a fast oscillatory movement would not allow the system to adjust to the geometry of each static system but would rather adopt a more unpredictable behaviour (see Section 4.3 and specially Figure 4.15 for configurations with f of order $O(c)$, where specific ranges of P values lead to chaotic behaviour).

We will set the oscillation to be centred around the minimum of the x -nullcline, this is near the value of the Hopf (Hopf-like) bifurcation in the static system, in order to obtain static systems with equilibria in both the repelling and attracting branches. In this way, the expected behaviour can be directly deduced from the situations exposed in the previous section. While the static system possesses a stable equilibrium, the solution should get attracted to it and thus be close to the left attracting branch presented in the section before. As the system evolves to situations with equilibria of the static system on the repelling branch, we expect the solution to follow the trace of relaxation oscillations. The latter case leads to **bursting** or several narrow spikes in the x and y component of the solution, before going back to a planer situation when static systems return to stable equilibria.

Simulations for the smooth and PWL case with parameters $c = 0.1$, $\varepsilon = 0.25$, $\gamma = 0$, $a = -0.9$, $P = 1$ and different frequency values f presented in Figure 4.8 show that the dynamics are consistent with the hypotheses of the preceding paragraph. Values of $f \in \{0.03, 0.04\}$ result in almost identical dynamics in both smooth and PWL models, which present the same number of spikes in each burst and with the same amplitudes. Note that this is not the case for $f = 0.02$, where there is a mismatch in the number of peaks. However, the most remarkable difference from a qualitative analysis perspective is the discrepancy in the onset of bursting. While the PWL model's first spike appears right after the Hopf-like bifurcation value is surpassed, the first spike of the smooth model is delayed when crossing the Hopf value. In fact, see how immediately thereafter crossing the Hopf, the smooth model's solution x and y components experience small variance compared to the spikes, thus suggesting that the flow follows the slow manifold's repelling branch. This effect is called **delay in the loss of stability**, and even though it can also be detected in the other frequency cases of the smooth model, it is not as noticeable as in the latter scenario.

4.2.1 Slow passage through a bifurcation

The delay effect can be related to a **slow passage through a Hopf**, where a slowly varying forcing replacing a static parameter induces the solutions of the resulting slow-fast system to feel the effect of the loss of stability at a Hopf bifurcation with a delay. In fact, it is not a new behaviour but rather a manifestation of the same processes involved in the canard explosion. Both systems (3.6) and (3.7) can be rewritten as

$$\begin{cases} \dot{x} = g(x) - \varepsilon x - y, \\ \dot{y} = c(x - a), \\ \dot{a} = 2P\pi f \cos(2\pi f t), \end{cases}$$

which can be approximated when analysing a single passage through the bifurcation like

4. SLOW-FAST SYSTEMS

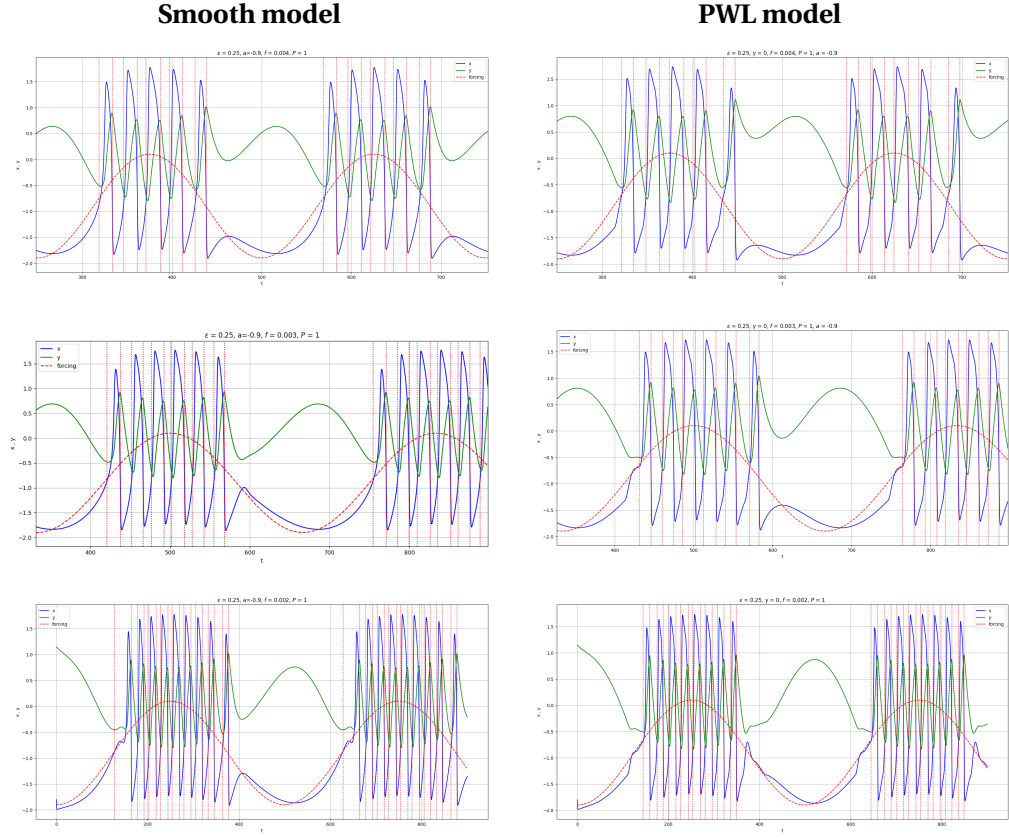


Figure 4.8: Traces of x (blue line) and y (green line) component of the solution of smooth model (3.6) (left column) and PWL model (3.7) (right column) with $c = 0.1$, $\varepsilon = 0.25$, $\gamma = 0$, $a = -0.9$, $P = 1$. Each row has a different frequency value, from top to bottom $f = 0.004$, $f = 0.003$, $f = 0.002$. Red dashed line plots the value of $k_F = a + P \sin(2\pi f t)$, vertical red lines mark the times when x equals the bifurcation value a_H according to Theorem 11 for the smooth case and to Proposition 4 for the PWL case.

$$\begin{cases} \dot{x} = g(x) - \varepsilon x - y, \\ \dot{y} = c(x - a), \\ \dot{a} = 2P\pi f, \end{cases} \quad (4.27)$$

with $0 < f \ll c \ll 1$ thus getting a multi-scale slow-fast system with two slow variables. Studying the PWL version of model (4.27), we will show that the slow manifold does not connect, so one does not expect the flow to follow the slow manifold once crossed the border between subdomains of definition. Said otherwise, one will expect the solution to follow the Hopf or relaxation cycles rightafter crossing the Hopf-like bifurcation.

Lemma 3 in [13] establishes the slope values, m , k for fixed border values ρ , μ in

order to ensure the connection of the slow manifold for a PWL system of the form

$$\begin{cases} \dot{x} = h(x) - y, \\ \dot{y} = x - a, \\ \dot{a} = \varepsilon, \end{cases} \quad (4.28)$$

where

$$h(x) = \begin{cases} -mx & \text{if } x < \rho, \\ lx + n & \text{if } \rho \leq x < \mu, \\ kx & \text{if } x \geq \mu, \end{cases}$$

with parameters l and n fixed as

$$l = \frac{m\rho + k\mu}{\mu - \rho}, \quad n = -\rho\mu \frac{k + m}{\mu - \rho}.$$

The resulting values for m and k are as follows

$$\begin{aligned} A &= \ln^2 \left| \frac{\rho}{\mu} \right| + \pi^2, \quad m = \frac{\mu - \rho}{\varepsilon} - k - \sqrt{A}, \\ k &= \frac{-1}{\mu - \rho} \left(\frac{2 \ln \left| \frac{\rho}{\mu} \right| (\rho - \mu)}{A} + \frac{\rho}{\varepsilon} (\mu - \rho - \varepsilon \sqrt{A}) \right). \end{aligned} \quad (4.29)$$

The proof of the Lemma 3 in [13] relies on solving a system of 6 equations with 6 unknowns that follows from making the solution of (4.28) connect the intersecting points of the slow manifold with the switching lines of h . Thus, as the system is determined, there is only one configuration that enables establishing a solution. In order to see that our PWL model does not adjust to the required values, we will rewrite it in the form of (4.28).

Consider the system (4.11) for regions σ_i , $i \in \{L, C, R\}$, that is equivalent to the original model (3.7), and rewrite it just like (4.27). By undergoing the change of variables $(x, y, a) = (x - D, y + T, a - D)$, where

$$D = \frac{(2c - 1)\gamma - \frac{2}{3}\sqrt{c}}{2 - 3\sqrt{c}\gamma}, \quad T = (m - \varepsilon)D - m + \frac{2}{3},$$

one gets

$$\begin{cases} \frac{dx}{d\tau} = g(x) - y, \\ \frac{dy}{d\tau} = c(x - a), \\ \frac{da}{d\tau} = 2P\pi f, \end{cases}$$

with

$$g(x) = \begin{cases} (-2m - \varepsilon)x & \text{if } x < -(D + \sqrt{c}), \\ (\gamma - \varepsilon)x & \text{if } -(D + \sqrt{c}) \leq x < D + \sqrt{c}, \\ (m - \varepsilon)x & \text{if } D + \sqrt{c} \leq x < 2 + D - \sqrt{c}. \end{cases}$$

Finally, through the change of variables $(x, y, a) = \left(x, \frac{y}{\sqrt{c}}, a\right)$ along with a time reparametrization $t = \frac{\tau}{\sqrt{c}}$ we get an equivalent system like (4.28):

$$\begin{cases} \dot{x} = \tilde{g}(x) - y, \\ \dot{y} = x - a, \\ \dot{a} = \frac{2P\pi f}{\sqrt{c}}, \end{cases} \quad (4.30)$$

where

$$\tilde{g}(x) = \begin{cases} \frac{(-2m-\varepsilon)}{\sqrt{c}} x & \text{if } x < -(D + \sqrt{c}), \\ \frac{(\gamma-\varepsilon)}{\sqrt{c}} x & \text{if } -(D + \sqrt{c}) \leq x < D + \sqrt{c}, \\ \frac{(m-\varepsilon)}{\sqrt{c}} x & \text{if } D + \sqrt{c} \leq x < 2 + D - \sqrt{c}. \end{cases} \quad (4.31)$$

It is now straightforward to prove that the slow manifold in the PWL model with the parameters considered in Figure 4.8 does not connect. Indeed, $k = m = \frac{\mu}{\varepsilon} - \frac{\pi}{2}$ when $\rho = -\mu$ in (4.29), but outer segments' slopes in $\tilde{g}$ are not equal.

One might expect to take advantage of the reported result in order to slightly modify the profile of the x -nullcline, meaning the g function, with the goal of observing the delay effect on the PWL model (3.7).

A first and low-disruptive approach consists on computing new border values $\tilde{\rho}$, $\tilde{\mu}$ and middle section segment $\tilde{l}x + \tilde{n}$ for function (4.31) while maintaining the remaining slopes. In this way, the profile of the nullcline will get stretched in the segment around the minimum but still conserves the relation $-1/2$ between the central and outer slopes. As already mentioned, Lemma 4.1 in [13] guarantees the connection of the slow manifold by finding border values ρ , μ from slopes m , k and not the other way around. However, as long as matrix $(m(\rho, \mu), k(\rho, \mu))$ has a non-zero Jacobian it is possible to find correspondent values $\rho(m, k)$, $\mu(m, k)$ through the Inverse Function Theorem.

Since we desire to compare the results with the situation presented in Figure 4.8, we will numerically compute values $\tilde{\rho}$, $\tilde{\mu}$ from parameter values $c = 0.1$, $\varepsilon = 0.25$, $\gamma = 0$, $f = 0.002$ and $P = 1$ in (4.30). Applying the *Newton-Rhapson* method to functions

$$\begin{aligned} r_1(\mu, \rho; \varepsilon, k, m) &= \frac{\mu - \rho}{\varepsilon} - k - \sqrt{A} - m, \\ r_2(\mu, \rho; \varepsilon, k, m) &= \frac{-1}{\mu - \rho} \left(\frac{2 \ln \left| \frac{\rho}{\mu} \right| (\rho - \mu)}{A} + \frac{\rho}{\varepsilon} (\mu - \rho - \varepsilon \sqrt{A}) \right) - k, \end{aligned}$$

where A is the same as in (4.29) and m, k are the outer slopes in (4.31), with initial guess $\rho = -0.5$, $\mu = 0.5$ we get values $\tilde{\rho} = -0.13355866032500455$, $\tilde{\mu} = 0.36508264072882396$. In order to ensure that at least twelve digits are correct, we may use the Poincaré-Miranda Theorem in two variables. We get that $r_1(\tilde{\mu} + 10^{-12}, \tilde{\rho} + 10^{-12}, \varepsilon, k, m) < 0$, $r_2(\tilde{\mu} + 10^{-12}, \tilde{\rho} + 10^{-12}, \varepsilon, k, m) < 0$ and $r_1(\tilde{\mu} - 10^{-12}, \tilde{\rho} - 10^{-12}, \varepsilon, k, m) > 0$, $r_2(\tilde{\mu} - 10^{-12}, \tilde{\rho} - 10^{-12}, \varepsilon, k, m) > 0$, meaning that at least the first twelve digits are shared with the actual root. A *Maxima* file verifying the preceding computations is available in this *GitHub* public repository.

Undoing the various changes of variables we get a modified version of the system

4.2. Bursting and Slow Passage through a Hopf

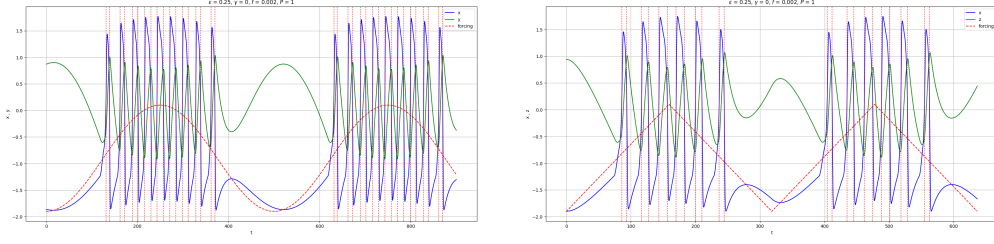


Figure 4.9: Traces of x (blue line) and y (green line) component of the solution of PWL (3.7) with $g_1 - \varepsilon x$ as (4.32) and parameters $\varepsilon = 0.25$, $c = 0.1$, $a = -0.9$, $f = 0.002$, $P = 1$ and $\gamma = 0$. Red dashed line plots the value of the forcing, which is $k_F = a + P \sin(2\pi f t)$ for the left image and $k_F = a + \text{triangular}(2\pi f t)$ for the right image. Vertical red lines mark the times when x equals the bifurcation value $-1 + D + \tilde{\mu}$.

(3.7) with g redefined as:

$$g_1(x) = \begin{cases} \begin{cases} -2m(x + 1 + \sqrt{c}) - \gamma\sqrt{c} - \frac{2}{3} \\ (-0.18483619206562535\sqrt{c} + \varepsilon)x \\ + 0.9044686632458744\sqrt{c} \\ + (1 - D)(-0.18483619206562535)\sqrt{c} \\ + T - 2/3 + \varepsilon \end{cases} & \text{if } x < -1 + D + \tilde{\rho}, \\ mx & \text{if } -1 + D + \tilde{\rho} \leq x < -1 + D + \tilde{\mu}, \\ \gamma(x - 1) + \frac{2}{3} & \text{if } -1 + D + \tilde{\mu} \leq x < 1 - \sqrt{c}, \\ -2m(x - 1 - \sqrt{c}) + \gamma\sqrt{c} + \frac{2}{3} & \text{if } 1 - \sqrt{c} \leq x < 1 + \sqrt{c}, \\ & \text{if } 1 + \sqrt{c} \leq x, \end{cases} \quad (4.32)$$

where parameters m , D and T are defined as usual.

Unfortunately, the results are not as expected and no delay can be seen in the outcome. Even by changing the forcing signal from a sinusoid to a triangular with slope $2P\pi f$ in order to minimize the modelling errors, the manifolds do not look like connecting, as there is not a preliminary and slow transition before the bursting starts in Figure 4.9.

Authors in article [13] found a similar behaviour and related the lack of delay with a numerical issue. What is more, in Figure 5 they associate a higher delay to higher numeric precision. Even by increasing the digits in computation we did not achieve to observe any delay, so we explore other options.

Both smooth and PWL ODE systems we are treating can be rewritten as the equation of an oscillator like $\ddot{x} = -cx + \dot{g} + c(a + P \sin(2\pi f t))$, where g is the equation of the x -nullcline. In Figure 4.10 we plot the second term $w := \dot{g} + c(a + P \sin(2\pi f t))$ for both the smooth model (3.6) and the latest presented PWL variant around the bifurcation parameter, along with the x component of the solution. It can be seen that the term w and the x component of the solution from both models (solid lines are for the smooth model and dashed lines for the PWL) greatly differ in the region of interest.

The oscillator equation gives insight into the importance of g 's derivative for the dynamics of the system, so we might consider a new model where the derivative of

the g function for both the PWL and smooth models are more closely aligned. We propose a new PWL model where we replace the segment around the minimum of the x -nullcline with three small-slope segments that satisfy the relation defined in (4.29) with the borders of the domains of definition. The idea is to define a region around the bifurcation value, say $\sigma_{\hat{C}} = \{(x, y)^T \mid \hat{K}_1 \leq x < \hat{K}_2\}$, involving segments with small slopes to ensure that the connection of the slow manifolds is successfully established.

Two more considerations need to be taken into account. In order to keep the global dynamics of the system as unaltered as possible, we maintain the segments outside $\sigma_{\hat{C}}$ with the same slopes as in the previous models. Moreover, this new region needs to have a width $\hat{K}_2 - \hat{K}_1$ sufficiently large to allow the solution of the system coming from the left of $\sigma_{\hat{C}}$ to get close to the slow manifold defined in $\sigma_{\hat{C}}$ in order for the delay to appear. Since the slow manifold in the PWL case coincides with the union of the slow manifolds in each subdomain of definition, it is usually not differentiable. Thus, as the solution of an ODE system is at least C^1 , a solution coming close to the slow manifold in the region before $\sigma_{\hat{C}}$ will need some time to approach the slow manifold in region $\sigma_{\hat{C}}$. Making the region $\sigma_{\hat{C}}$ large enough comes at the price of losing the feature of x -nullcline's profile to look like $x(1 - \varepsilon) - \frac{x^3}{3}$.

Function g for new model is redefined as follows, see blue curve in Figure 4.11 for a comparison with the x -nullcline of previously considered PWL systems.

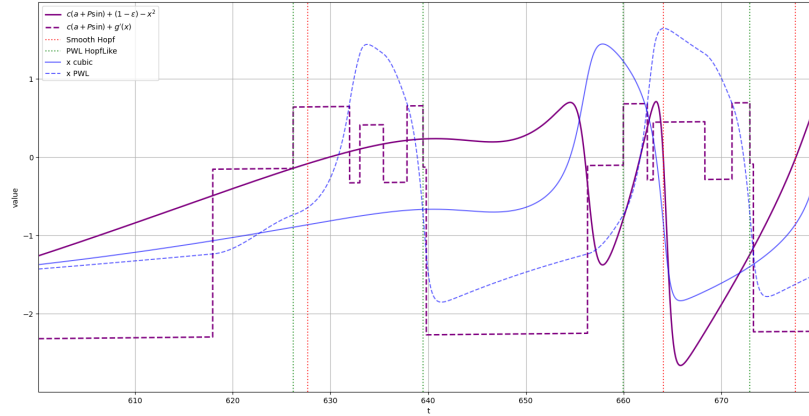


Figure 4.10: In blue, traces of x component of the solution and in purple traces of $w = c(a + P \sin(2\pi f t) + g')$, solid line for smooth model (3.6) and dashed for PWL model with g defined as (4.32). Vertical lines mark the times when x equals bifurcation values.

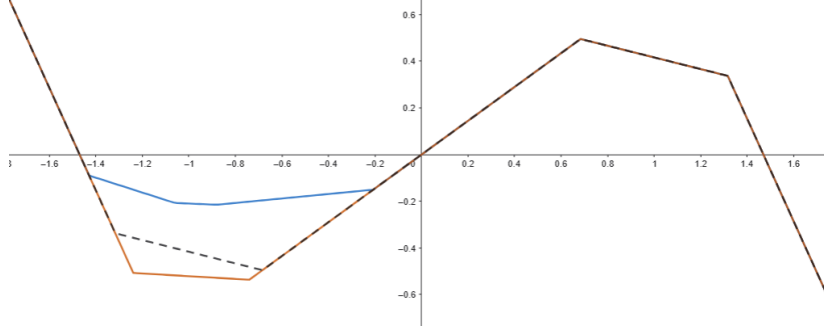


Figure 4.11: Comparison of the functions $g_1 - \varepsilon x$ (in solid orange line) and $g_2 - \varepsilon x$ (in solid blue line) with $g - \varepsilon x$ as originally defined in (3.7) (in dashed line) for $\varepsilon = 0.25$, $\gamma = 0$ and $c = 0.1$.

$$g_2(x) = \begin{cases} -2m(x+1+\sqrt{c}) - \gamma\sqrt{c} - \frac{2}{3} & \text{if } x < -1 + \hat{K}_1, \\ 1.0089264596122\sqrt{c}(x+1) + \varepsilon(1+x) - \frac{2}{3} + d & \text{if } -1 + \hat{K}_1 < x < -1 + \hat{\rho}, \\ -0.1339407411508\sqrt{c}(x+1) + 0.0524991431077\sqrt{c} & \\ \quad + \varepsilon(1+x) - \frac{2}{3} + d & \text{if } -1 + \hat{\rho} < x < -1 + \hat{\mu}, \\ 0.3035521180798\sqrt{c}(x+1) + \varepsilon(1+x) - \frac{2}{3} + d & \text{if } -1 + \hat{\mu} < x < -1 + \hat{K}_2, \\ mx & \text{if } -1 + \hat{K}_2 \leq x < 1 - \sqrt{c}, \\ \gamma(x-1) + \frac{2}{3} & \text{if } 1 - \sqrt{c} \leq x < 1 + \sqrt{c}, \\ -2m(x-1-\sqrt{c}) + \gamma\sqrt{c} + \frac{2}{3} & \text{if } 1 + \sqrt{c} \leq x, \end{cases} \quad (4.33)$$

where parameter m is considered as usual, values $\hat{\rho} = -0.06$, $\hat{\mu} = 0.12$, $d = 0.19$ have been set freely and

$$\hat{K}_1 = \frac{-d - (2m + \gamma)\sqrt{c}}{2m + \varepsilon - 1.0089264596122\sqrt{c}}, \quad \hat{K}_2 = \frac{m + d - \frac{2}{3}}{m - \varepsilon - 0.3035521180798\sqrt{c}}.$$

It is clear in Figure 4.12 that delay manifests in the x and y component of the solution around the bifurcation value $-1 + \hat{\mu}$ of the Hopf-like bifurcation, as solution experiences a long small preliminary variation before the bursting starts. The delay effect even lasts longer than in the smooth case. In addition, it is clear in Figure 4.13 that the component $w = \dot{g} + c(a + P \sin(2\pi f t))$ of the latest model (in dashed line) in the oscillator equation is now closer to that of the smooth model (in solid line).

Overall, the smooth (3.6) and PWL (3.7) models evolve very similarly for relatively small frequency values. However, a previously unnoticed phenomenon is observed in the smooth model when considering very small frequency values, the delay in the loss of stability, which does not manifest in the PWL model (3.7). Just as in the canard explosion, we relate this delay to the connection of the slow manifolds of the system. To verify it, we started proving that the slow manifold of the considered PWL model (3.7) does not connect. Then, we got a first unsuccessful try forcing the connection

4. SLOW-FAST SYSTEMS

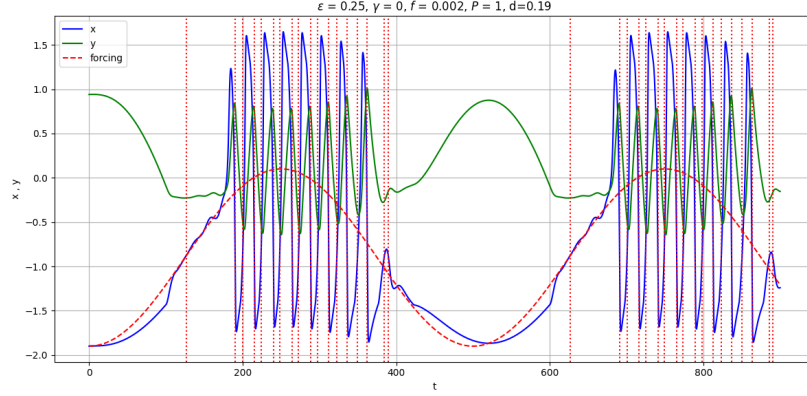


Figure 4.12: Traces of x (blue line) and y (green line) component of the solution of PWL (3.7) with $g - \epsilon x$ as (4.33) and parameters $\epsilon = 0.25$, $c = 0.1$, $a = -0.9$, $f = 0.002$, $P = 1$ and $\gamma = 0$. Red dashed line plots the value of the forcing $k_F = a + P \sin(2\pi f t)$. Vertical red lines mark the times when x equals the bifurcation value $-1 + \hat{\mu}$.

while also maintaining the profile of the x -nullcline. However, a last model where the nullcline function's profile gets clearly modified achieves to establish the connection, and finally the delay phenomena manifests. Thus, we can securely justify the presence of delay with the connection of the slow manifolds of the system. Consequently, a remarkable difference between the studied smooth (3.6) and PWL (3.7) models is the presence and absence of delay in the loss of stability, respectively, when considering a slow passage through a Hopf.

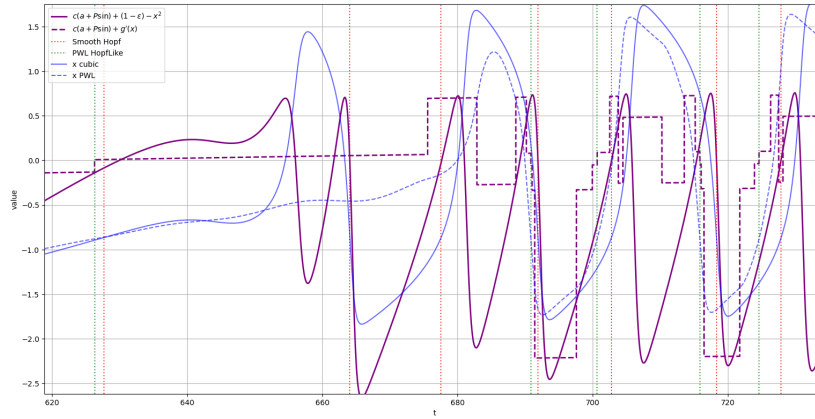


Figure 4.13: In blue, traces of x component of the solution and in purple traces of $w = c(a + P \sin(2\pi f t) + g')$, solid line for smooth model (3.6) and dashed for PWL model with g defined as (4.33). Vertical lines mark the times when x equals bifurcation values.

4.3 Period Doubling and Antimonotonicity

When in Section 4.2 we have reintroduced the sinusoidal component $P \sin(2\pi f t)$ back to models (3.6)-(3.7), there appear two different paths to stimulate the system: through changes in frequency f and through changes in amplitude P . While in Section 4.2 we discussed several cases involving different frequency values, this section deals with variations in amplitude P .

The approach of this section differs from the previous ones in the sense that the presented phenomenon will not be theoretically justified but rather reported and related to already assimilated concepts.

It is said that the creation and annihilation of periodic orbits is one of the most fundamental bifurcation processes in one-parameter families of maps. The value of the parameter at which an orbit is created as the parameter is increased is called an **orbit-creation value** and conversely an **orbit-annihilation value** refers to the value when an orbit does extinct as the parameter increases. The orbit-creation relates to **period-doubling bifurcations**, where a new periodic orbit appears from an existing one and having double the period of the original one. The paradigmatic example is the logistic family $f_\lambda(x) = \lambda - x^2$ which has been proven (see Theorem 13.1 in [11]) to only present orbit-creation values in a period-doubling cascade, and so it is said that orbits are created monotonically with the movement of the parameter. However, other systems are able to exhibit an antimonotonic behaviour (**antimonotonicity**) when presenting both orbit-creation and orbit-annihilation values. In the latter case, if we graph a bifurcation diagram with the parameter in the horizontal axis and with the maximum (or minimum) value of the periodic solution in the vertical axis, there appears a bubble shape.

Figure 4.14 illustrates the presence of period-doubling bubbles, that is antimonotonicity, for parameter values $c = 0.1, a = -1.3, f = 0.02, \gamma = 0$ and $\varepsilon \in \{0.3, 0.4, 0.5\}$. In Figure 4.15 more complex bifurcation diagrams, even presenting chaotic regions, with antimonotonic behaviour can be seen, with slightly different parameter configurations $f = 0.1, \varepsilon \in \{0, 0.15, 0.25\}$.

Even though the precise mathematical tools required to characterize the phenomena lie beyond the scope of this work, it is interesting to relate the concepts of canard explosion and slow manifolds to a specific case of the PWL model (3.7). In order to remark that the phenomenon can be exclusively related to changes in amplitude P we will substitute the sinusoidal function by a triangular signal of slope $2\pi f$. In this way, there are no changes in the absolute value of the velocity of the forcing signal.

First image in Figure 4.16 shows the bifurcation diagram for PWL model (3.7) with triangular forcing and parameters $c = 0.1, a = -1.3, f = 0.02, \gamma = 0, \varepsilon = 0.3$, while the following images show the x component of the solution along with the trace of the triangular forcing. Vertical lines mark times when forcing $k_F = a + P \cdot \text{triangular}(2\pi f)$ equals the value a_S for which the connection between slow manifolds of the static systems is established according to Theorem 14. The horizontal line marks the value a_S as well. From Theorem 15, the canard explosion in the PWL case for the static system k_F occurs for parameter values exponentially close to a_S . We will see that it is crucial for the solution to check if k_F lies above or below a_S when $x = a_S$. In reality, the subsequent statements are true whether comparing k_F with a_N , but our assumption follows from $a_S \approx a_N$. When $k_F < a_S$, we can consider the static system k_F to only have

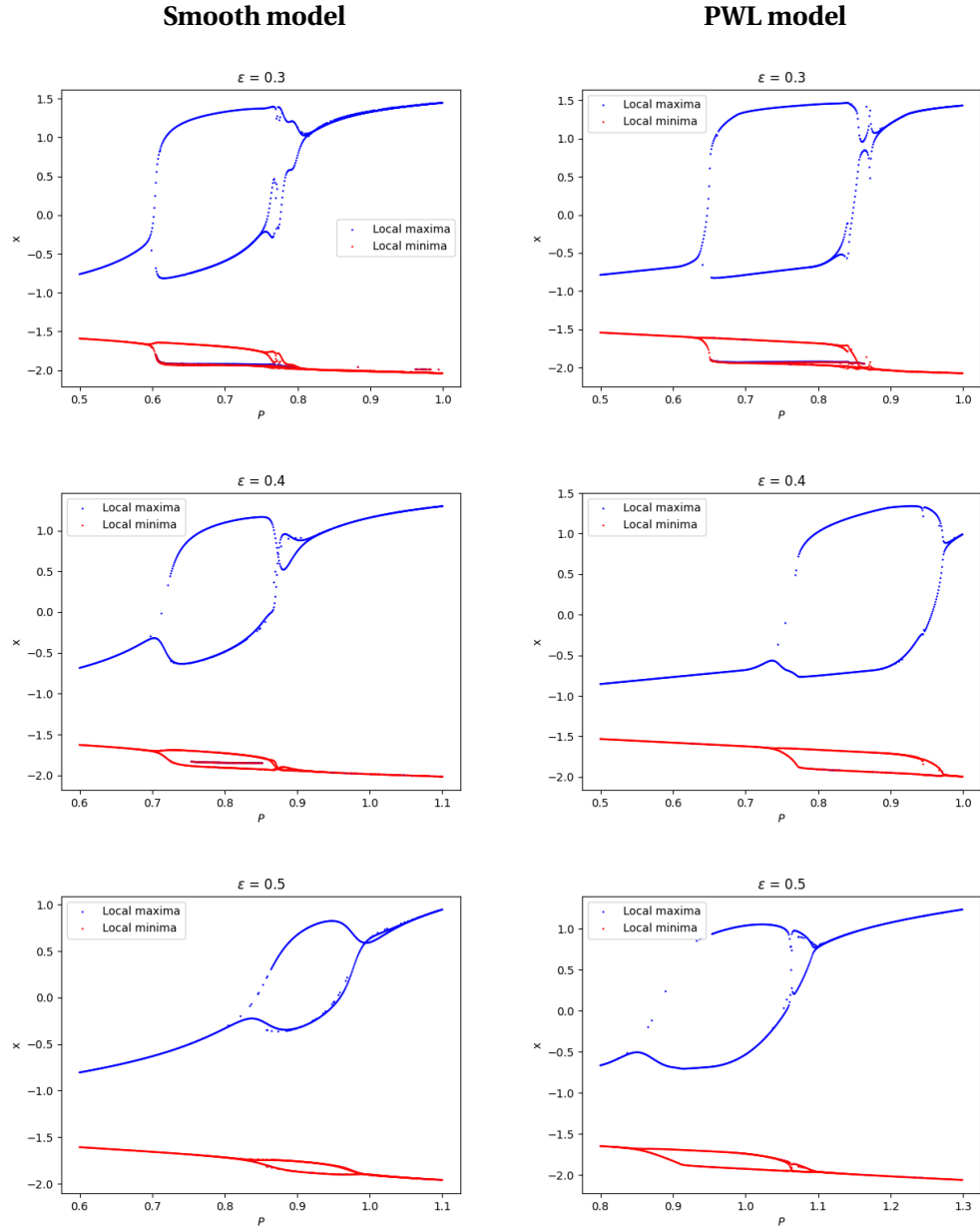


Figure 4.14: Bifurcation diagrams having amplitude parameter P in the horizontal axis and values for local minima (red line) and local maxima (blue line) of periodic solutions of smooth system (3.6) (left column) and PWL system (3.7) with parameters $c = 0.1, a = -1.3, f = 0.02, \gamma = 0$ and $\varepsilon \in \{0.3, 0.4, 0.5\}$. In all images we can observe regions with chaotic behaviour.

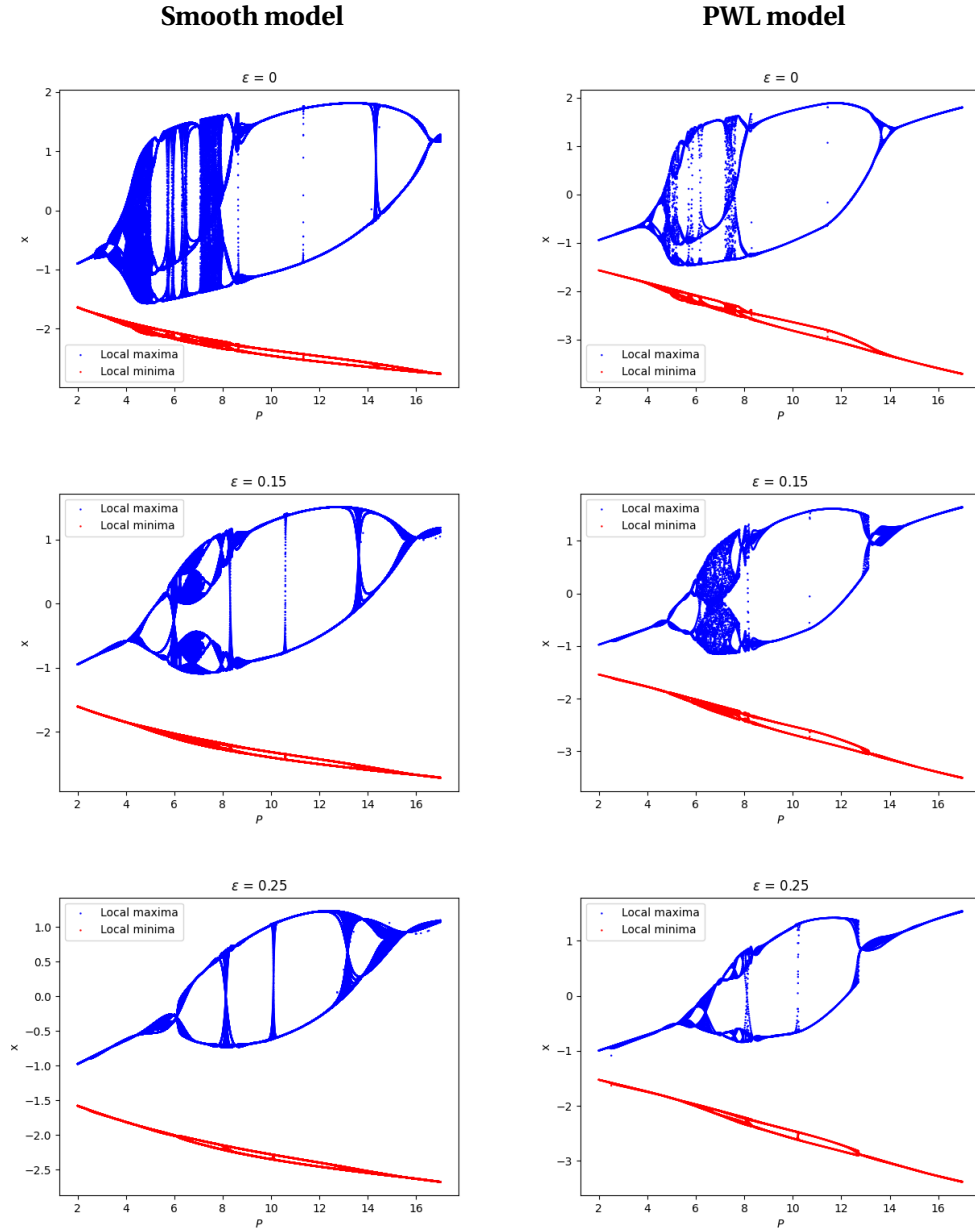


Figure 4.15: Bifurcation diagrams having amplitude parameter P in the horizontal axis and values for local minima (red line) and local maxima (blue line) of periodic solutions of smooth system (3.6) (left column) and PWL system (3.7) with parameters $c = 0.1, a = -1.3, f = 0.1, \gamma = 0$ and $\varepsilon \in \{0, 0.15, 0.25\}$.

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a stable equilibrium on the attracting branch, so the solution of the system will shift back towards the attracting branch without producing a burst. Conversely, if $k_F > a_S$ and because we are treating the subcritical case since $\varepsilon - \gamma > 0$, the static subsystem presents a stable relaxation orbit, possibly an unstable canard orbit surrounding the Hopf-like bifurcation, and an equilibrium that is stable in the presence of the unstable orbit and unstable otherwise. As a result, the solution crossing $x = a_S$ will take a longer path before going back to the attracting branch, or even produce a burst if it follows the relaxation orbits.

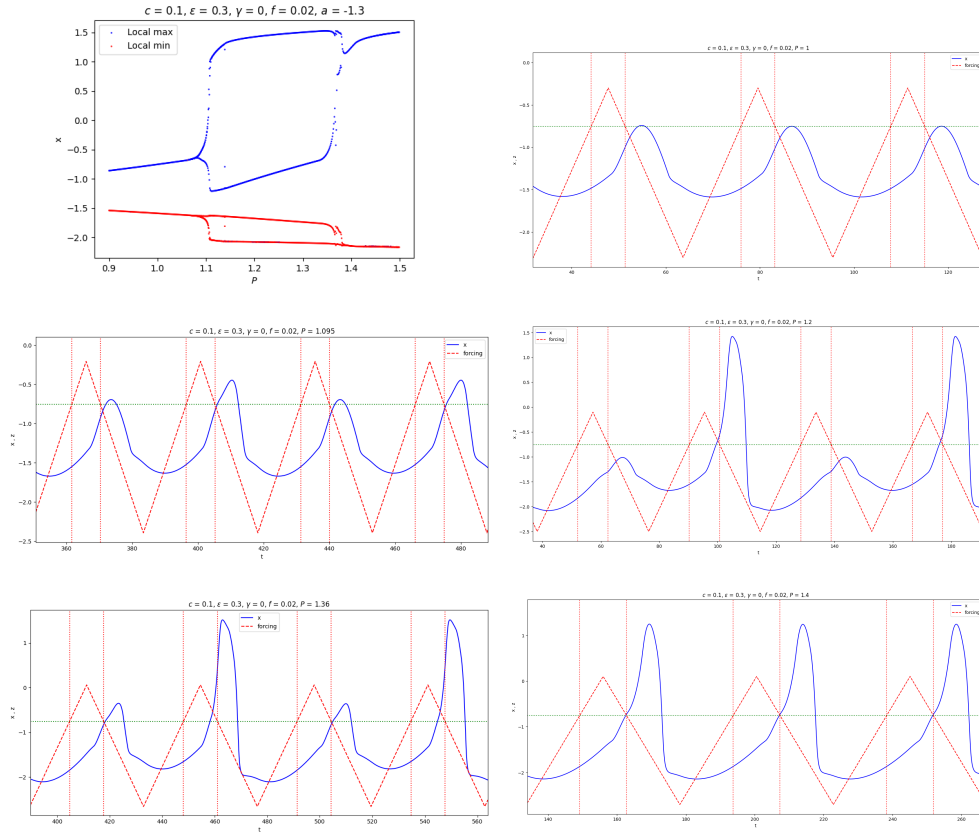


Figure 4.16: First image, bifurcation diagram for PWL model (3.7) with triangular forcing of slope $2\pi f$ and parameters $c = 0.1$, $a = -1.3$, $f = 0.02$, $\gamma = 0$, $\varepsilon = 0.3$. Subsequent images show x component of the solution for specific cases of $P = 1, 1.095, 1.2, 1.36, 1.4$ (from left to right and top to bottom). Red dashed line represents the value of the forcing $k_F = a + P \cdot \text{triangular}(2\pi f)$, vertical dashed lines mark times when $k_F = a_S$ and horizontal lines equal values a_S , where a_S is the value for the connection between slow manifolds in the static system to establish, according to Theorem 14.

This precise behaviour can be seen in the graphics of Figure 4.16 by checking if the blue line (x component of the solution) crosses the green horizontal line (meaning $x = a_S$) before (this means $k_F > a_S$) or after (this means $k_F < a_S$) the closest red vertical line (meaning $k_F = a_S$). In more detail, one sees that for $P = 1$ the crossing of $x = a_S$ always happens with $k_F < a_S$, so only small spikes are expected to happen. For $P = 1.095$

some of the crossings occur with $k_F \approx a_S$ provoking some bigger spikes related to canard orbits in the static subsystems and thus leading to period-doubling. For $P = 1.2$ it is clear that the solution follows relaxation orbits of the static systems when $k_F > a_S$ coming after small spikes that do not even cross $x = a_S$. As P gets near the orbit-annihilation value, like in $P = 1.36$, all the crossings $x = a_S$ tend to coincide with $k_F < a_S$, which finally occurs when $P = 1.4$. In the latter case, all spikes are alike and related to relaxation orbits.

CIRCUIT RESULTS

The aim of this chapter is to check the validity of findings in the previous chapter on models (3.6)-(3.7) of the circuit with schematic represented in Figure 3.6. Previous to the real analogue implementation of the circuit we will simulate the circuit on a SPICE software, where the noise and uncontrolled interference factors gets minimized. First measurements will consist on obtaining the nullcline of the system and the bifurcation diagram, which will give us the necessary information to modify the parameters on the forcing of the system in order to observe the bursting and antimonotonicity phenomena.

5.1 Simulation Results

A first step to know about the circuit behaviour will be to perform a simulation with a SPICE software (see [1] for a guide on SPICE simulations of chaotic circuits), and specifically *LTSpice*. This kind of software implements the various components' current and voltage equations considering the specifications from manufacturer's spreadsheets. After describing the circuit elements and their connections in a text netlist, it translates the document into equations to be solved using various numerical methods, see [20] for a detailed explanation. Even though the behaviour of the real electronic circuit may differ in some aspects, the description presented by SPICE uses to be accurate.

In order for the models we have been working with to resemble the equations of the circuit, we will directly deduce the circuit components' parameters from the PWL model (3.7) parameters c, γ, ε following the expressions in (3.9).

It is interesting to obtain the x -nullcline of the circuit, the one related to the variable v in system (3.5), because it has been a crucial object in Section 4.1 for undertaking the mathematical development. Considering the contents of that section one can estimate the position of the attracting and repelling branches and the Hopf bifurcation directly from the graph of the nullcline.

So as to obtain the x -nullcline in the SPICE software one may delete the capacitor

from the circuit description in order for $i_C = C \frac{dv}{dt}$ to be zero, and replace the circuit branch containing the inductor and the voltage source by a single voltage source. By doing a DC sweep to the new voltage source, which means simulating voltage values in a specific range, and monitoring the current value through that same source one gets the equivalent value for i'_L in the original system (3.5) for the v values considered in the sweep. Said otherwise, one gets the i'_L values for specific v values when $\frac{dv}{dt} = 0$, which is the same as the x -nullcline. Figure 5.1 shows in blue the circuit x -nullclines obtained using this method compared to the x -nullclines of the theoretical PWL model (3.7), in dashed black line. While the left image shows the theoretically subcritical case $\varepsilon = 0.25, \gamma = 0$, the right image corresponds to the theoretically supercritical case $\varepsilon = 0.1, \gamma = 0.25$, both for $c = 0.1$.

Remark 2. *Rather than the circuit current through the inductor, i_L , the variable of the studied system (3.5) is $i'_L = 100i_L$. All the plots will represent the variable i'_L . Do not mistake i'_L with $\frac{di_L}{d\tau}$.*

It can be seen that the blow-up regions around the minimum and maximum, where the interaction between the manifolds is expected to take place, differ between the circuit simulation and the PWL model (3.7). Moreover, the circuit's nullcline is not a PWL function, as it presents smooth transitions between its theoretical boundaries. Consequently, it is interesting to extract the position of the Hopf bifurcation in the circuit case. The simplest approach is to apply a numeric differentiation method to the data from the SPICE simulation of the nullcline in order to identify the fold points. Since the data points are equally-spaced, the five-points differentiation method given by

$$f'(x_0) \approx \frac{1}{12h} (f(x_0 - 2h) - 8f(x_0 - h) + 8f(x_0 + h) - f(x_0 + 2h)),$$

where $f(v)$ indicates a value of the nullcline and $h = 0.01$, is a good approach. In this way, the minimum is located around $x = -0.7159$ in the subcritical scenario, that is when $c = 0.1, \varepsilon = 0.25, \gamma = 0$ and around $x = -1.103$ in the supercritical scenario, when $c = 0.1, \varepsilon = 0.1, \gamma = 0.25$.

In order to acquire a general understanding of the global behaviour of the system for a range of the input voltage E parameter, we could plot a bifurcation diagram for the circuit. Moreover, we expect to do a comparison with Figures 4.4 - 4.7. Stable equilibria and limit cycles can be easily identified because the analysed voltage v and current i_L evolve towards them. By contrast, unstable objects cannot be tackled since back-in-time integration is not feasible beyond theory. Graphics in Figures 5.2 and 5.3 result from taking the local maxima and minima of the circuit variables v and i'_L as voltage E is varied and P is set to 0, meaning a short circuit.

Figure 5.2 deals with the theoretically supercritical case of parameters $c = 0.1, \varepsilon = 0.1, \gamma = 0.25$. The canard explosion phenomena can be clearly identified after the appearance of the first Hopf cycles. In comparison to the bifurcation diagrams of the smooth system (3.6) in Figure 4.4 and to the supercritical example of the PWL system (3.7) in Figure 4.7, the Hopf regime on Figure 5.2 lasts longer than in the smooth model and shorter than in the PWL model. Moreover, the growth of the cycles is closer to being linear like in the PWL case rather than the square rooted growth of the smooth case. In contrast to the theoretical models, where the oscillation appears right after the fold

5.1. Simulation Results

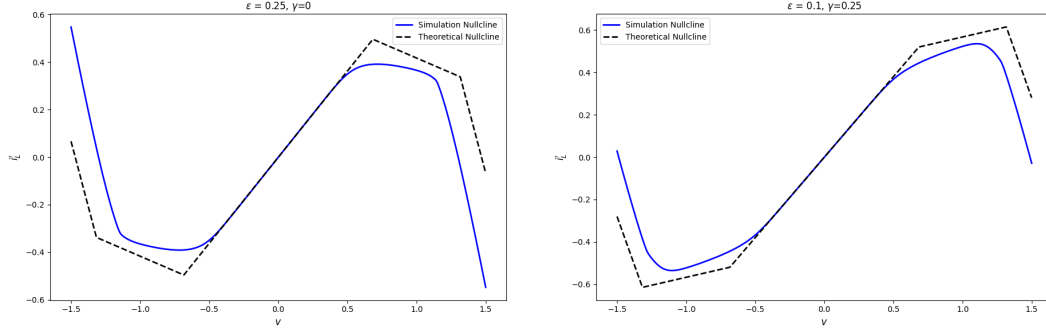


Figure 5.1: Blue lines correspond to the x -nullcline (corresponding to variable v) of the circuit related to system (3.5). Black dashed lines correspond to the x -nullclines of the PWL model (3.7). Parameters $c = 0.1, \varepsilon = 0.25, \gamma = 0$ have been considered for the left image and $c = 0.1, \varepsilon = 0.1, \gamma = 0.25$ for the right one.

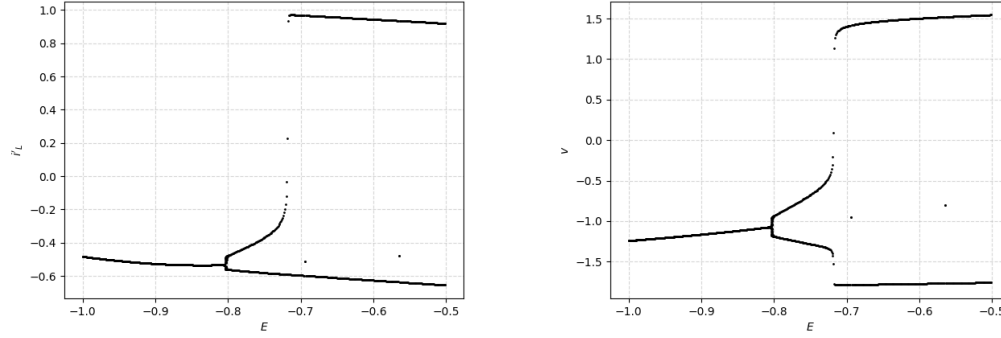


Figure 5.2: Bifurcation diagrams of the circuit for parameter values $c = 0.1, \varepsilon = 0.1, \gamma = 0.25$. Canard explosion, where the local maxima and minima of the solution is represented in the vertical axis. Since back-in-time simulation is not feasible, only stable objects are captured. Thus, the unstable equilibria branch after Hopf bifurcation does not appear. Left image deals with current variable i_L' and right image with voltage v .

point for the smooth case and the border point for the PWL case, the circuit oscillatory phase does not start right after the computed fold value $x = E = -1.103$. The underlying cause might have to do with the small slope values of the nullcline shown in Figure 5.1 around the fold point. A small slope value translates to an equilibrium point where the trace of the Jacobian matrix of the system (3.5) is negligible, meaning few repelling power and small orbits. The numerical treatment of the solutions of the SPICE software might be ignoring these oscillations. Indeed, increasing the numeric precision of the simulation has already brought forward the starting point of oscillations in comparison to early simulations that have been omitted in this document.

Figure 5.3 deals with the theoretically subcritical case of parameters $c = 0.1, \varepsilon =$

5. CIRCUIT RESULTS

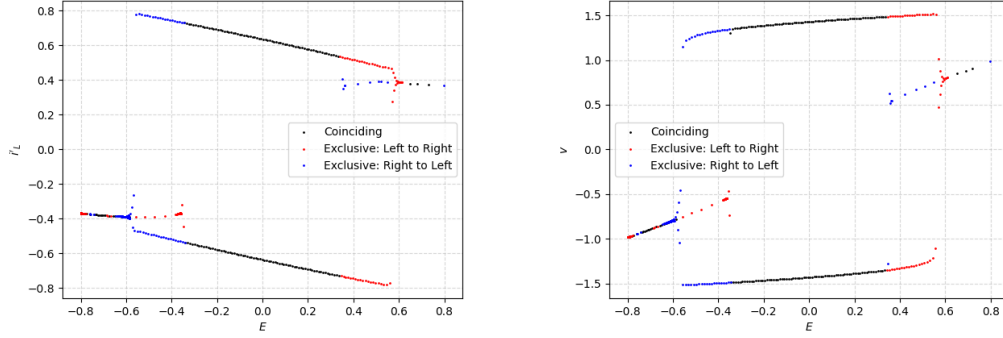


Figure 5.3: Bifurcation diagrams of the circuit for parameter values $c = 0.1$, $\varepsilon = 0.25$, $\gamma = 0$. Canard explosion, where the local maxima and minima of the solution is represented in the vertical axis. Since back-in-time simulation is not feasible, only stable objects are captured. Left image shows deals with current variable i_L' , right image with voltage v .

0.25, $\gamma = 0$. Colour legend is related to the direction in which the sweep of parameter E has been carried out: red colour dots correspond to extrema points that have only been encountered when sweeping from left to right (this is, increasing E value), blue dots relate to data values that only appeared when sweeping from right to left (meaning when decreasing the E value), while black dots correspond to maxima and minima values that appeared in both cases. As can be seen in Figure 4.4 and in the right image of Figure 4.7, in the supercritical bifurcation diagram there never coexist two different stable objects for the same parameter value. By contrast, there is a specific range for the bifurcation parameter in the subcritical bifurcation diagram that presents two simultaneous stable objects, what is known as **bistability**. Specifically, the repelling branch is surrounded by a stable relaxation orbit when presenting an unstable Hopf cycle in between, as can be seen in the left image of Figure 4.7. As a result, when increasing the bifurcation parameter (what we say going from left to right) a solution coming close to the attracting branch around the minimum will remain around it until the beginning of the repelling branch; while a solution coming close to the relaxation orbit will remain near it (instead of being near the attracting branch) until it vanishes, when considering a decrease in the bifurcation parameter (what we say going from the right to the left). The odd symmetry in the nullcline makes the situation around the local maxima of the nullcline be the inverse. This phenomenon, known as **hysteresis** clearly manifests itself in the bifurcation diagrams of Figure 5.3, and it is represented by the presence of structures of blue and red dots continuing the relaxation orbits or the attracting branch. Just as in the previous supercritical scenario, the diagram on Figure 5.3 differs from the bifurcation diagram of the PWL system (3.7) in Figure 4.7 in the sense that the bifurcation value is not placed around the fold point in $E = -0.7159$ but somewhere afterwards around $E = -0.36$.

5.1.1 Bursting

After having determined the general dynamics of the circuit simulation through the bifurcation diagrams in Figures 5.2-5.3, we expect to observe the bursting phenomenon described in Section 4.2. To do so, we will consider the circuit related to the same c, ε, γ parameters as in that section: $c = 0.1, \varepsilon = 0.25, \gamma = 0$. While the frequency value of the circuit will be matching the one of the theoretical experiment through the expression in (3.9), different values for voltage E (the equivalent parameter to the parameter a in the theoretical models) will be considered. Specifically, we will simulate the circuit for the exact same value as considered in Section 4.2, $E = a = -0.9$, but also $E = -0.6$. Circuit bifurcation diagrams showed that the oscillatory phase in the static system E (meaning the same concept as in Section 4.2) was somehow delayed from the fold point. Thus, as $a = -0.9$ was fixed to be certainly close to the value $a_S \approx -0.8$ of the PWL model, we select a new value $E = -0.6$ to be closer to the value where the circuit oscillations start to manifest.

Graphics in Figure 5.4 expose the solutions i'_L in green, v in blue and forcing term $P \sin(2\pi f t)$ in dashed red for the circuit simulation with values $c = 0.1, \varepsilon = 0.25, \gamma = 0, P = 1$. Left column corresponds to simulations with $E = -0.9$, just as in Figure 4.8 and right column to $E = -0.6$. Moreover, each of the rows relate to one of the frequency values $f \in \{0.002, 0.003, 0.004\}$. Even though configurations in both columns succeed in manifesting the bursting phenomenon, the theoretical behaviour exposed in Figure 4.8 perfectly matches the configuration in the right column. Indeed, the number of spikes with the theoretical PWL model coincides with that of the circuit simulation. Moreover, the behaviour of the signals before the first spike seems to be closer to the PWL model (3.7) than to that of the smooth model (3.6). Nonetheless, it is hard to determine if the delay phenomenon manifests in the circuit dynamics.

Figure 5.5 adds vertical dashed lines to the case $f = 0.002$ marking the moment when $v = -0.7159$ (left image) and $v = -0.36$ (right image), corresponding to the fold value -where the theoretical Hopf bifurcation should take place- and to the value where the actual oscillation begins before the repelling branch -see Figure 5.3- respectively. As already explained in Section 4.2.1, the delay phenomenon should be related to a later appearance of the bifurcation effect of the static system when considering the forced system. However, it makes sense to relate the delay to the later appearance of the oscillation phase of the static system on the forced one when one restricts to the analysis of the topological behaviour of the system. With this latter consideration in mind, the image on the right ratifies the lack of delay in the circuit behaviour, thus resulting in the PWL model (3.7) prediction of the circuit behaviour to be the most accurate.

5. CIRCUIT RESULTS

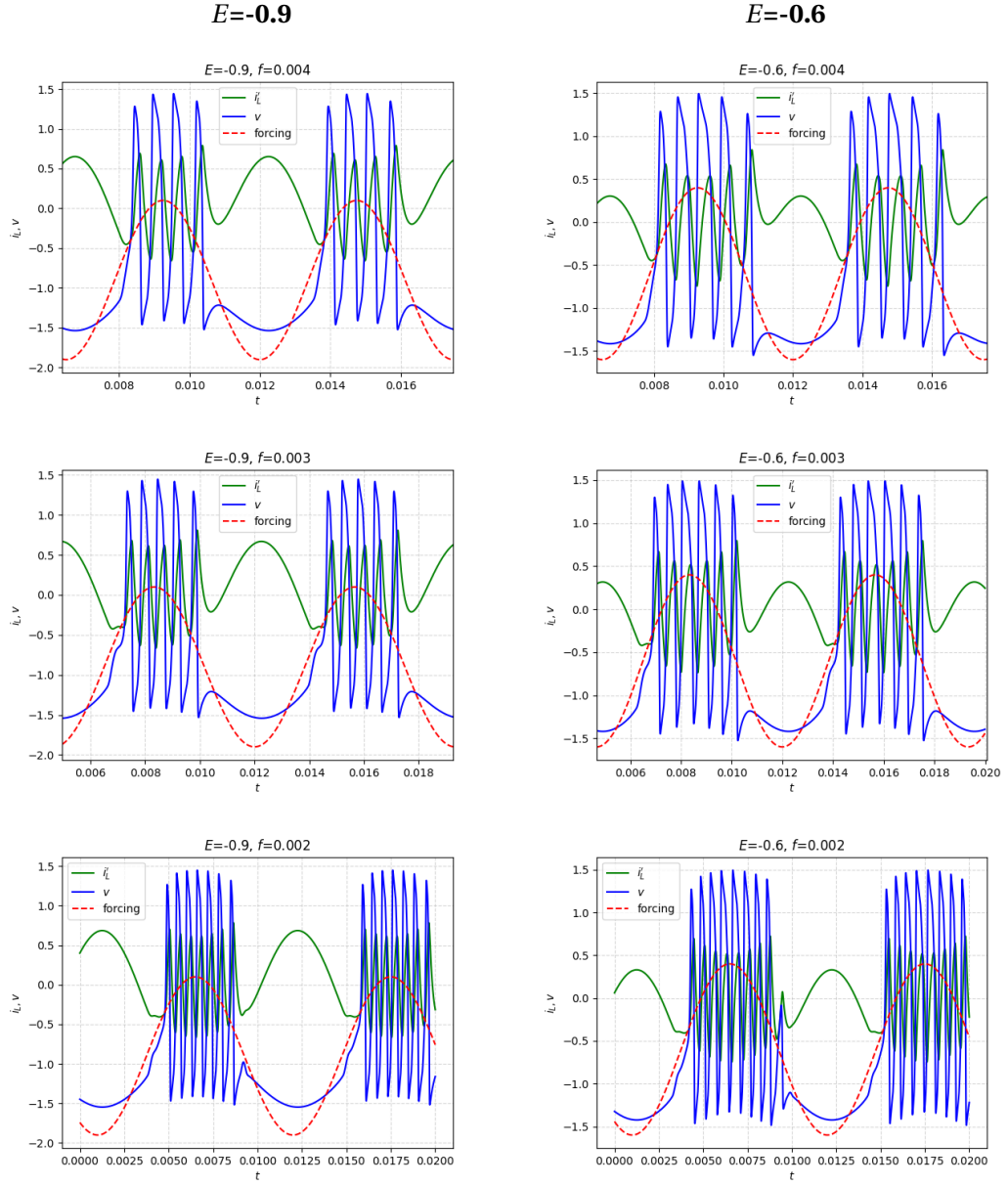


Figure 5.4: Traces of v (blue line) and i'_L (green line) component of the solution of circuit system (3.5), with $\varepsilon = 0.25$, $c = 0.1$, $a = -0.9$, $P = 1$ and $\gamma = 0$. Each line has a different frequency value, from top to bottom $f = 0.004$, $f = 0.003$, $f = 0.002$. Red dashed line plots the value of $k_F = E + P \sin(2\pi f t)$. Images in left column share the value $E = -0.9$ and right columns images have $E = -0.6$.

5.1.2 Antimonotonicity

The last phenomenon that has been observed in the theoretical smooth (3.6) and PWL (3.7) models is period-doubling and antimonotonicity, see Section 4.3. Specifically,

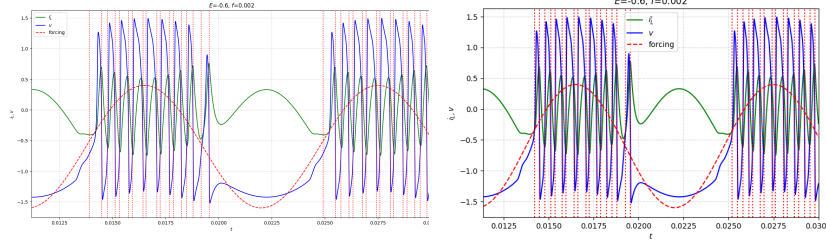


Figure 5.5: Same images of latest line in Figure 5.4 where added vertical dashed lines marking the moment when $v = -0.7159$ (left image) and $v = -0.36$ (right image).

these phenomena have been theoretically reported for combinations of parameters leading to subcritical Hopf bifurcations. These same combinations will be considered here.

Since the theoretical underlying causes of the mechanism have not been characterised, it is difficult to determine the expected output of the circuit simulation knowing the x -nullcline profile. However, a clear connection between period-doubling, the interaction of the slow manifolds of the system and the timing when the variable x crosses a specific value a_S has been reported (see Figure 4.16). As the circuit has shown to present a flatter region around the extrema in the x -nullcline for the subcritical case (see left image in Figure 5.1), timings for the v variable might differ from the x timings in the PWL model (3.6). Moreover, the bifurcation diagram for the circuit (Figure 5.3) has shown discrepancy with the theoretical result, hence the theoretical and circuitual a_S value is not expected to coincide.

Figure 5.6 shows bifurcation diagrams in the sense of Figure 4.14. Parameters $c = 0.1, \gamma = 0, f = 0.02$ and $\varepsilon \in \{0.3, 0.4, 0.5\}$ have been considered. Images in the left column share an $E = -1.3$ value, just as in Figure 4.14, while images on the right column share an $E = -1$ value. The aim of the second column is to consider an E value closer to the value where oscillations actually start in the circuit, just as it has been done in Figure 5.4. As it can be seen, changing the E value mainly affects the position of the bubble in the P parameter axis. All the considered parameter configurations result in bubble shapes with gradual transitions from period-doubling and to orbit-annihilation values. However, bubble-widths are narrower than in the outcomes of theoretical models (3.6) and (3.7) shown in Figure 4.14, which might have to do with the differences between circuit and theoretical model features mentioned in the preceding paragraph.

Overall, simulations showed that smooth (3.6) and (3.7) models fairly predicted the behaviour of the studied system, more in a qualitative than in a quantitative manner. Configuring the circuit with parameters matching theoretical models through (3.9) allowed to give a supercritical or subcritical character to bifurcation diagrams, though precise bifurcation values did not coincide. By adjusting the offset value of the forcing signal accordingly to the previously obtained diagrams, we managed to observe an analogous bursting and antimonotonicity phenomena to that of theoretical models, and specially to the PWL model as no delay was observed prior to the bursting.

5. CIRCUIT RESULTS

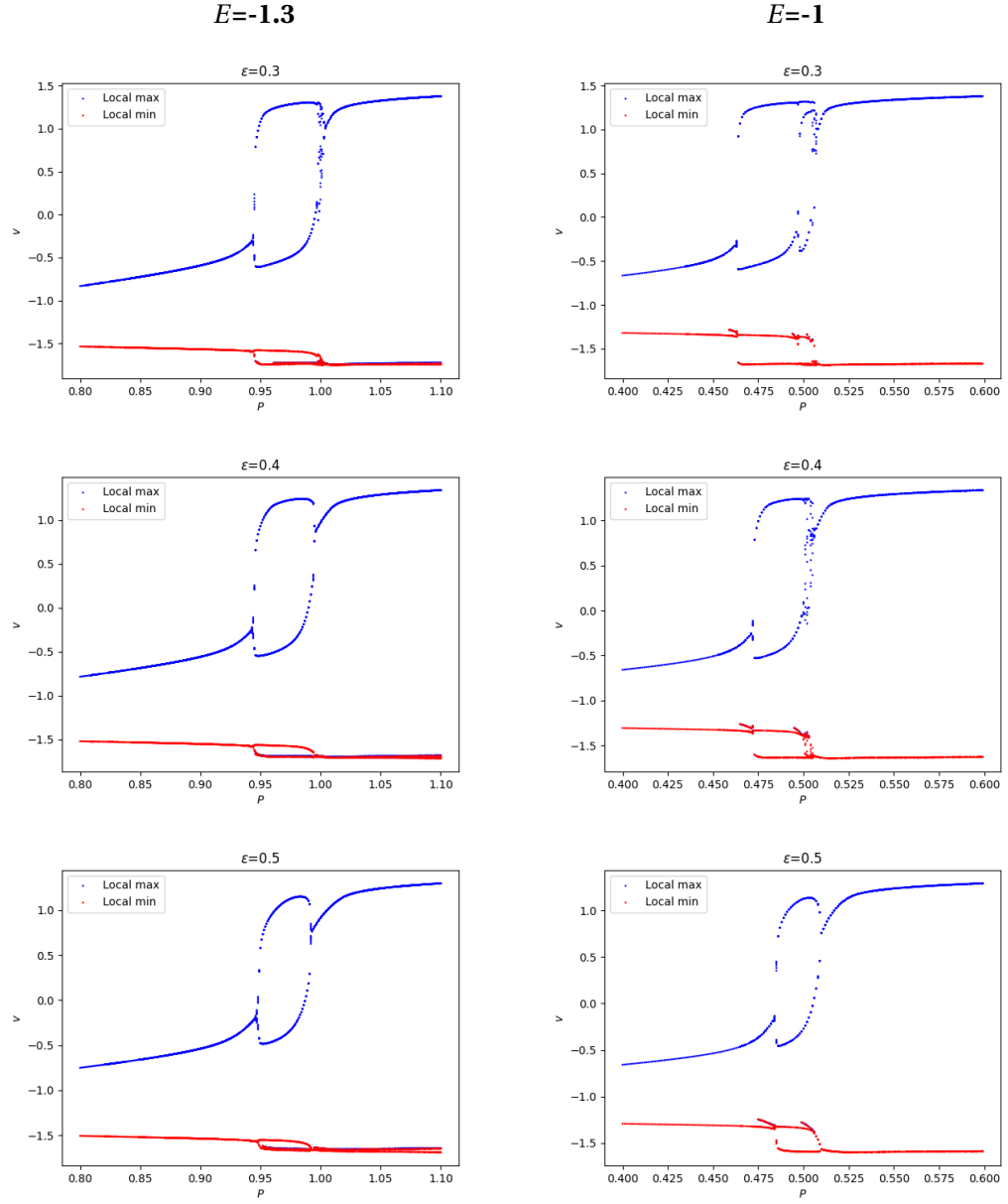


Figure 5.6: Bifurcation diagrams having amplitude parameter P in the horizontal axis versus values for local minima (red line) and local maxima (blue line) of v variable in the circuit's system (3.5). Parameters $c = 0.1, \gamma = 0, f = 0.02$ and $\epsilon \in \{0.3, 0.4, 0.5\}$ have been considered. Moreover images on the left column share an $E = -1.3$ value, and images on the right column share an $E = -1$ value.

5.2 Analogue measurements

In this section we provide a collection of the outputs obtained from experiments carried out on a real analogue implementation of the circuit design sketched in Figure 3.6, see Figure 5.7. Various potentiometers have been used in order to implement the variable values R_1, R_2, R_6 . Moreover, values for V_1, V_3 come from voltage dividers of $\pm 12V$ entries, while $V_2 = V_1$ and $V_4 = V_3$ are obtained through an inverting amplifier topology with an operational amplifier. An extra voltage inverter operational amplifier has been connected to the output of the forcing signal source, as placing this component solved the problem of the signal generator being unable to supply enough current for the circuit to properly work. Specifically, a 4-operational amplifier chip of model type *TL084* is involved in the obtaining of values $V_1, \dots, V_4$ and two operational amplifiers of model type *TL072* are used in the negative-resistor schematic of Figure 3.2 and in the inverter connected to the forcing source, respectively. Instrumentation that has been involved in the experimental phase includes the *E3631A* power supply, the *AFG320* function generator and the *TDS1002* oscilloscope.

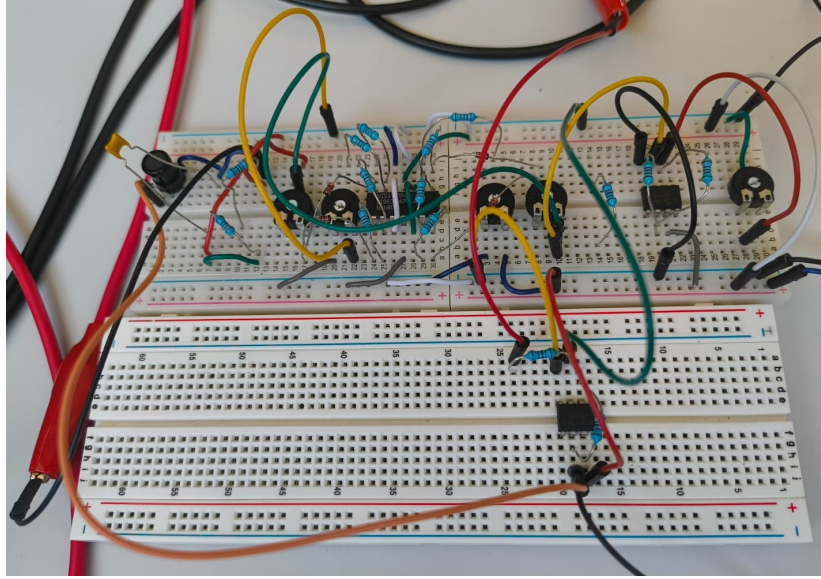


Figure 5.7: Implementation of the analogue circuit on a protoboard.

The first circuit configuration to be presented is the one derived from values $c = 0.1, \varepsilon = 0.25, \gamma = 0, V_\gamma = 0.7$ in expressions (3.9). We have used an inductor with spreadsheet value $L = 22$ mH and a capacitor with $C = 220$ nF, but real measurements showed that $L = 21.6$ mH and $C = 205$ nF thus leading to a $c = 0.093$ value in the circuit system (3.5). Moreover, theoretical values of $R_1 = 51\ \Omega, R_2 = 102\ \Omega, R_3 = 102\ \Omega$ and $V_1 = 0.616V, V_3 = -0.016V$ were measured in reality as $R_1 = 51\ \Omega, R_2 = 104\ \Omega, R_3 = 103\ \Omega$ and $V_1 = 0.65V, V_3 = -0.08V$. From now on, this setup will be called the **subcritical setup**. These differences in parameter values should not be the cause for any meaningful qualitative changes beyond minor quantitative modifications to the nullcline of the system. However, we can already advance that we will encounter a big difference

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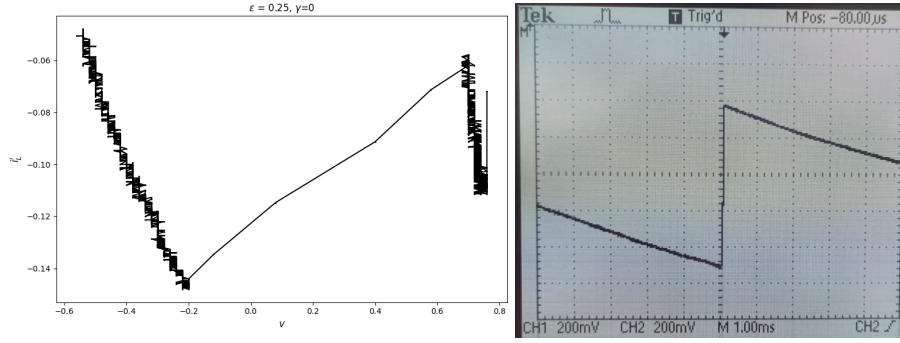


Figure 5.8: Nullcline for the analogue circuit with schema in Figure 3.6 and parameter configuration following (3.9) and $c = 0.1$, $\varepsilon = 0.25$, $\gamma = 0$. Image on the left corresponds to the x -nullcline of the system and image on the right was directly taken from the oscilloscope screen. Notice that the horizontal axis on the right image represents time values from which v values linearly depend, so the actual slopes slightly differ from those in the image, and coincide with the ones on the left plot.

between the actual nullcline of the circuit and that of the theoretical models, which will not be related to small changes in the parameter values.

Since theoretical background indicates that the x -nullcline (the v -nullcline for the circuit system (3.5) notation) of the system is determinant for the dynamics, we will first obtain this object representation from the analogue circuit. An analogous technique from the one presented in the previous SPICE analysis can be used. One substitutes the inductor and capacitor branches with a branch containing the source signal in series with a $1\text{ k}\Omega$ resistor. Then, by measuring the voltage drop on the resistor one may directly compute the current value from Ohm's Law. Figure 5.8 shows the output for the previous process. It is important to remark that $i'_l = 100i_l$ as indicated in system (3.5). However, the output presents one order of magnitude below that of the theoretical smooth (3.6) and PWL (3.7) models and SPICE simulations. As the plot for the x -nullcline on the left image is defined in a narrow range (the best that we could capture), image on the right shows a photography directly taken from the oscilloscope screen, where a broader range can be seen. Notice that the horizontal axis on the right image represents time values from which v values linearly depend, so the actual slopes slightly differ from those in the image, and coincide with the ones on the left plot. Also, the placement on the vertical axis is not representative.

Overall, Figure 5.8 shows that there is a great discrepancy between the expected nullcline according to SPICE simulations and theoretical smooth and PWL models (3.6)-(3.7), and the real subcritical setup's outcome. Among the most notable differences, the nullcline does present a considerable offset (as it does not cross the $(0,0)$ point) and contains three distinguished regions instead of five. This experiment already anticipates the great difference that we will encounter between models (3.6)-(3.7) and the correspondent analogue circuit following parameter values in (3.9). The origin of this discrepancy is not attributable to modelling flaws (as the SPICE simulation exhibits behaviour significantly closer to theoretical models); rather, it must be understood in terms of issues within component connections and interactions, as well as potential

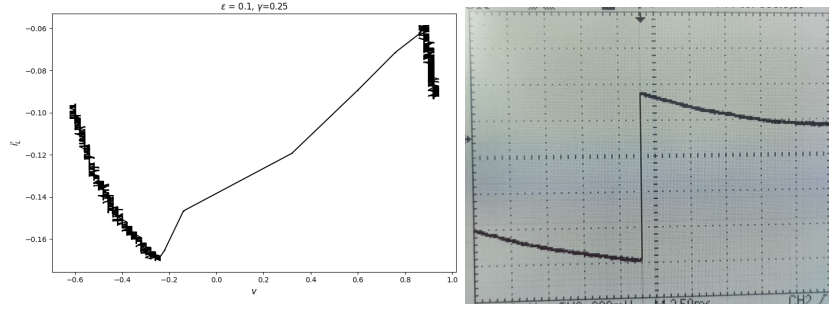


Figure 5.9: Nullcline for the analogue circuit with schema in Figure 3.6 and parameter configuration following (3.9) and $c = 0.1$, $\varepsilon = 0.1$, $\gamma = 0.25$. Image on the left corresponds to the x -nullcline of the system and image on the right was directly taken from the oscilloscope screen.

noise and unaccounted-for parasitic currents.

In Figure 5.9 we measure the x -nullcline for the theoretically supercritical system configuration with parameters $c = 0.1$, $\varepsilon = 0.1$, $\gamma = 0.25$. As the diagram is analogous to that of Figure 5.8, no supercritical bifurcation is expected to be found in reality.

Given that the real analogue circuit behaves in a more unpredictable manner than expected, and changes in $R_1, R_2, R_6, R_s, V_1, \dots, V_4$ do not have the effect that theoretical models and SPICE simulation predict, we propose using a modified circuit setup from which the previously mentioned parameter values are obtained by hands-on experience on the laboratory. This variation of the circuit, which will be called the **alternative setup** has been designed for it to present a x -nullcline with a shape that better resembles the ones in Figure 5.1. Later measurements showed that values $R_1 = 40\Omega$, $R_2 = 225\Omega$, $R_6 = 315\Omega$, $R_s = 400\Omega$, $V_1 = 1.69\text{ V}$, $V_2 = -0.05\text{ mV}$ were being used. Figure 5.10 show the x -nullcline of the circuit, which clearly presents five different regions of definition (as can be best seen on the right image) and a wider repelling branch.

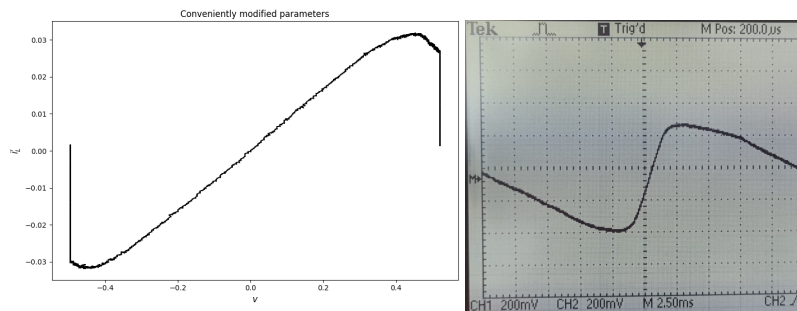


Figure 5.10: Nullcline for the analogue circuit with schema in Figure 3.6 and parameter configuration $R_1 = 40\Omega$, $R_2 = 225\Omega$, $R_6 = 315\Omega$, $V_1 = 1.69\text{ V}$, $V_2 = -0.05\text{ mV}$. Image on the left corresponds to the x -nullcline of the system and image on the right was directly taken from the oscilloscope screen.

5.2.1 Bifurcation diagrams

Let us now present the bifurcation diagrams for the circuit with the subcritical and alternative setups. The setup corresponding to the theoretically supercritical case has been left apart because, as the nullcline already advanced, it is very similar to that of the subcritical setup.

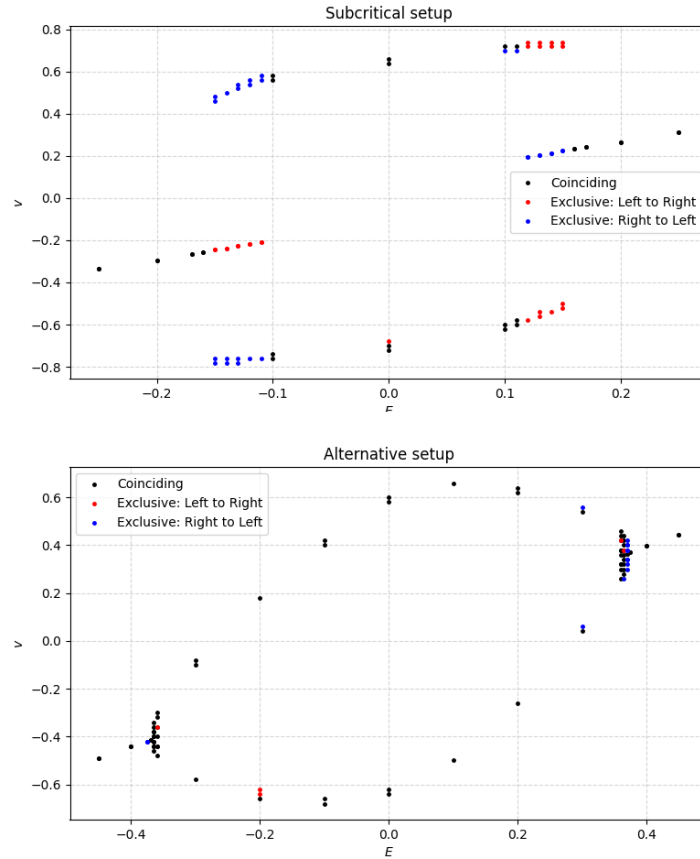


Figure 5.11: Bifurcation diagrams for the subcritical setup (top image) and alternative setup (bottom image). The bifurcation parameter E is represented in the horizontal axis versus the local maxima and minima in v in the vertical axis. Red dots exclusively appear when increasing the parameter value (going left to right), blue dots when decreasing the parameter value (going right to left), and black dots manifest in both cases.

Top image on Figure 5.11 corresponds to the bifurcation diagram of the subcritical setup and bottom image to the bifurcation diagram of the alternative setup. As usual, we represent parameter E [V] value in the horizontal axis versus local maxima and minima of the v solution [V] in the vertical one. Colour legend is related to the direction in which the sweep of parameter E has been carried out: red colour dots correspond to extrema that have only been encountered when sweeping from left to right (this is, increasing E value), blue dots relate to maxima and minima that only manifested when sweeping from right to left (meaning when decreasing the E value), while black

dots correspond to dots that appeared in both cases. Apart from the difference in shape, being one of them more oval-shaped than the other, top and bottom diagrams contain an essential difference that can be detected thanks to the colouring of dots, the hysteresis. Following the explanation done in the comment of Figure 5.3, the presence of structures of blue and red dots continuing the relaxation orbits or the attracting branch ensures the manifestation of an hysteresis in the subcritical setup of the circuit. Conversely, the presence of isolated red or blue dots in the alternative setup might be due to noise or due to giving insufficient time to transition. This fact, along with the gradual increase in the width of the orbits in the bottom image in contrast to the sharp transition to big orbits in the top case, ensures that the subcritical setup presents a subcritical bifurcation diagram and the alternative setup a supercritical one.

It is important to remark that, as already mentioned, there is a great discrepancy in the bifurcation value between the circuit subcritical setup and the theoretical bifurcation value of PWL model (3.7). In the first case, top image in Figure 5.11 suggests a bifurcation value around -0.1 V, while in Section 4.1 we determined a Hopf value of $-0.6837\dots$ V for the PWL model (3.7). Moreover, direct comparison of top image in Figure 5.11 and left image in Figure 4.7 suggests that the distance from bifurcation value to slow-manifolds connection value (what we denoted a_S in the theoretical case) in the PWL model (3.7) is twice the distance between the bifurcation and connection value for the circuit subcritical setup. In the alternative setup we achieved to obtain a wider oscillatory phase, as bifurcation parameter is nearly four times bigger (in absolute value) than in the subcritical setup. However, the growth of orbits seems to resemble a Hopf regime rather than a canard explosion.

5.2.2 Bursting

According to the SPICE simulation on the theoretical events of Figure 4.8, we would expect the circuit with the subcritical setup to manifest the bursting phenomenon when considering a sinusoidal with amplitude 1 and sufficiently small frequency forcing. Having obtained the bifurcation diagram in Figure 5.11 one can already anticipate a mismatch in the values of the offset E [V] and amplitude P [V] that provoke the bursting to happen. A narrower bifurcation value suggests using smaller amplitude and offset (in absolute value) parameters, even though the actual values have been obtained from experience on the laboratory. Indeed, Figure 5.12 shows several frequency configurations that allowed to observe bursting with different number of spikes when considering $E = -0.2$ V, $P = 0.4$ V. Frequency values $\hat{f} \in \{97, 146, 197\}$ for the circuit system (3.5) correspond to the theoretical values of $\hat{f} = \frac{f}{100C}$ with $f \in \{0.002, 0.003, 0.004\}$ from smooth and PWL models (3.6)-(3.7), which succeed in manifesting the bursting phenomena, with less spikes than expected, in Figure 4.8. However, last image shows that higher number of spikes can be obtained when considering even smaller frequency values, like $\hat{f} = 57$ Hz. Vertical dashed lines indicate the times when $v = -0.105$ V, which corresponds to the bifurcation value of the subcritical setup according to the bifurcation diagram in Figure 5.11. Thus, we are inclined to affirm that no delay manifests in the subcritical setup when considering a passage through the bifurcation with frequency values coinciding with the theoretical cases.

Experiments carried out with the circuit's alternative setup showed a better coincidence in the number of spikes per theoretical frequency value, as it can be seen in

5. CIRCUIT RESULTS

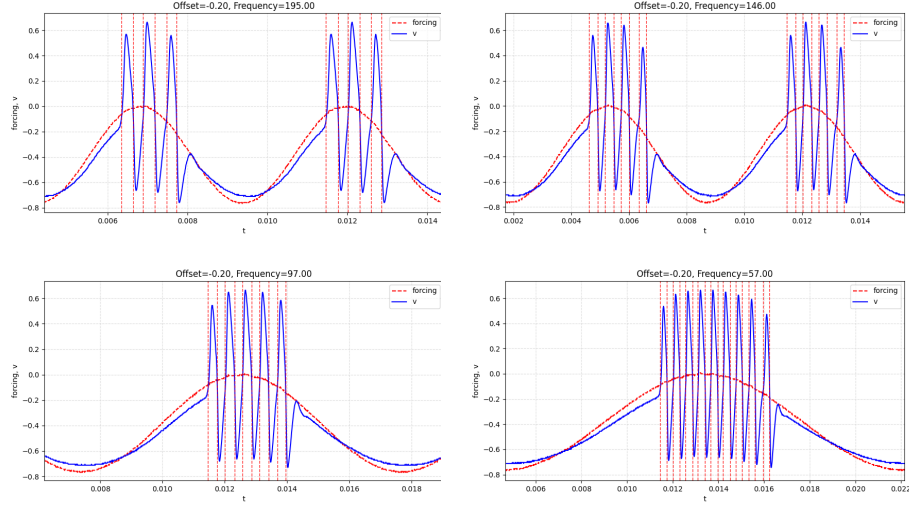


Figure 5.12: Traces of the v (blue line) component of the solution of circuit in the subcritical setup with $E = -0.2$ V, $P = 0.4$ V, along with the sinusoidal forcing in dashed red line. Each picture represents a scenario with a different frequency value for the forcing, as indicated in the title. Vertical dashed lines indicate when $v = -0.105$ V.

Figure 5.13. It is difficult to explain why spikes in this case occupy a smaller range on the vertical axis compared to other experiments and simulations. One may suggest relating it to the gradual growth of stable orbits in the supercritical bifurcation diagram, but other circuit experiments presenting narrower supercritical bifurcation diagrams have been able to manifest bursting with larger vertical spikes, as well as theoretical supercritical configurations of the PWL model (3.7). The graphics for these latter cases have been left apart since they not are meaningful beyond this observation.

Another possible explanation relates to the delay. In the images, vertical lines indicate the time when $v = -0.367$ V which is approximately the bifurcation value according to the bifurcation diagram in Figure 5.11. Notice how oscillations start to appear when approaching the first vertical line, and their amplitude gradually grows as time advances. This might suggest the manifestation of a strong delay effect, where the solution remains close to the repelling branch and, even when oscillating, fails to approach the relaxation orbit.

5.2.3 Antimonotonicity

Following the same line as in the previously presented outcomes, the antimonotonicity phenomenon can be reported when considering different forcing configuration than in the theoretical cases. Bubbles' shapes in Figure 5.15 indicate that the orbit-creation and annihilation process does not behave analogously to that of Figures 4.14-5.6. In Figure 5.15 we present bifurcation diagrams for the subcritical (image on the left) and alternative (image on the right) setups, with forcings containing an offset of -0.42 V or -0.9 V and a frequency of 675 Hz or 730 Hz, respectively. The circuital frequency value related to the frequency value of $f = 0.02$ Hz in the smooth and PWL models (3.6)-(3.7)

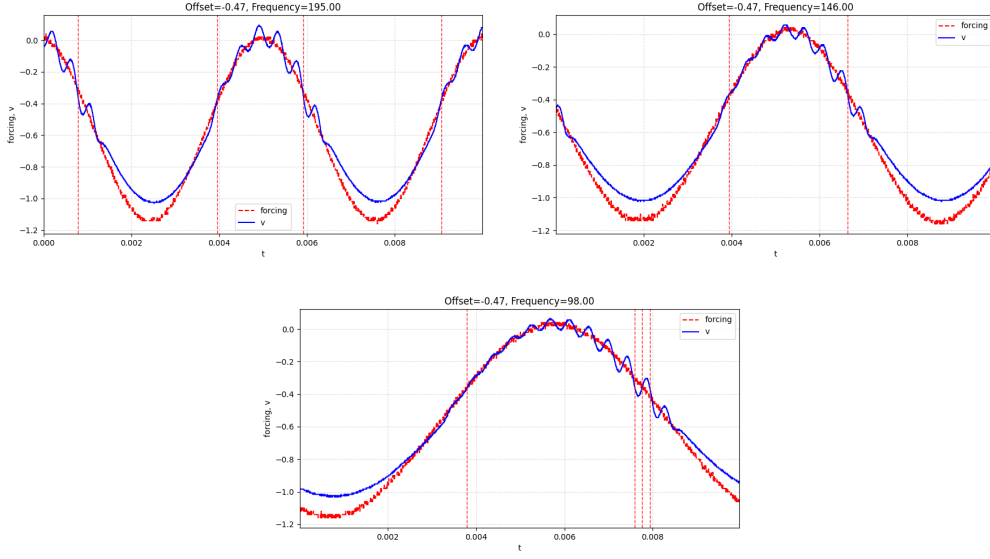


Figure 5.13: Traces of the ν (blue line) component of the solution of circuit in the alternative setup with $E = -0.47$ V, $P = 0.7$ V, along with the sinusoidal forcing in dashed red line. Each picture represents a scenario with a different frequency value for the forcing, as indicated in the title. Vertical dashed lines indicate when $\nu = -0.367$ V.

is $\hat{f} = \frac{0.02}{100C} \approx 975$ Hz, but we were not able to detect antimonotonicity regions with such value in any of the setups. Moreover, ε value has been maintained to 0.25 as no noticeable differences were detected when changing to $\varepsilon = 0.4$. The amplitude P [V] value is represented in the horizontal axis versus the value of the local maxima and minima in V of the oscillations in ν . Outcome bubbles in Figure 5.15 differ from those in 4.14-5.6 not only by width but also by the overall shape.

It is important to remark that the antimonotonicity phenomenon that is observed in the circuit behaviour does not relate to period-doubling in a strict sense. It is clear in Figure 4.16 that the signal coming after the orbit-creation value presents double the period of the preceding signal. However, raw signals from subcritical setup (left image) and alternative setup (right image) in Figure 5.14 indicate that the newly appeared mode of oscillation is combined within the same peaks of the preceding oscillation rather than forming alternated even-spaced peaks. What is more, while a new peak is born in the subcritical setup, the orbit-creation value in the alternate setup gives birth to a plateau.

In theoretical and simulation cases, orbit-creation processes produce a new solution presenting an oscillatory component with two peaks having local maxima above and below the preceding solution, and local minima of the same order and below the previous case. Analogously, the solution that arises after the orbit-annihilation value presents a local maximum in between those of the preceding solution, and a local minimum matching the value of the lowest local minimum. Images in Figure 5.15 do not match this behaviour. Even though the appearance of a new orbit in the subcritical setup matches the schema of new maxima values explained above, the local

5. CIRCUIT RESULTS

minima does not stay at the same level or below the past solution, but rather at the same level and above. By contrast, the birth of a new orbit in the alternative setup relates to the appearance of a plateau in the solution rather than an extra oscillation, as already seen. Notice how the newly appeared orbit presents the same value for local maxima and minima, until the plateau is again assimilated to the biggest orbit during the orbit-annihilation process. The case for the subcritical setup is not exactly the same, as the graphic suggests the appearance of an extra oscillation in the solution, because local maxima and minima does not match at first, that transitions to a plateau that is eventually assimilated to the principal oscillation, just as in the alternative setup.

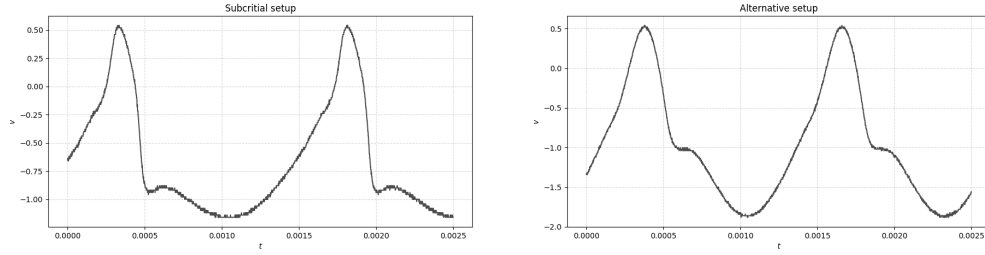


Figure 5.14: Raw signal from circuit's subcritical setup with forcing parameters $P = 0.75$ V, $\hat{f} = 675$ Hz, $E = -0.42$ V (left image) and alternative setup with forcing parameters $P = 1.6$ V, $\hat{f} = 730$ Hz, $E = -0.9$ V (right image).

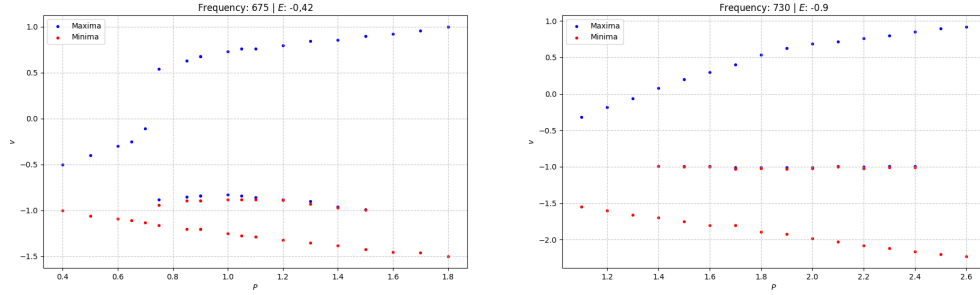


Figure 5.15: Bifurcation diagrams having amplitude parameter P [V] in the horizontal axis and values for local minima (red dots) and local maxima (blue dots) of v variable in the circuit's subcritical setup (left image) and alternative setup (right image). Subcritical setup forcing parameters are $\hat{f} = 675$ Hz, $E = -0.42$ V, and $\hat{f} = 730$ Hz, $E = -0.9$ V for the alternative setup.

Overall, experimentation performed on the real circuit scenario has not been as favourable for the theoretical models (3.6) and (3.7) as the SPICE simulations. We must emphasize that further optimization of the physical layout, specifically regarding

component interconnects to minimize parasitic phenomena such as noise and electromagnetic interference, might lead to a closer alignment with the idealized SPICE simulations. However, attending to the presented results, we must conclude that parameter values configurations following (3.9) do not translate equally to the theoretical models and to the circuit. In this way, when being restricted to theoretical parameter configurations the circuit did not always present the expected character in the bifurcation diagram and critical values, such as the bifurcation value or frequency values leading to bursting, greatly differed from the theoretical predictions. Nonetheless, the bursting and antimonotonicity behaviour manifested anyway, the latter presenting the appearance of a new oscillation but without period-doubling.

When we appropriately modified the circuit parameters, we managed to obtain a nullcline closer to that of the model's predictions, and wider bifurcation diagrams were produced. However, it was not possible to obtain a wide subcritical bifurcation diagram. Moreover, frequency values closer to the theoretical counterpart manifested the bursting and antimonotonicity effects, again with some differences. Bursting in this case appeared to exhibit a strong delay effect, and antimonotonicity was associated with the appearance and disappearance of a plateau component in the solution rather than a second oscillation.

A resume for significant values and qualitative characteristics of theoretical models and practical results can be encountered in Tables 5.1-5.2 at the end of the section.

Table 5.1: Quantitative and qualitative resume for smooth (3.6) and PWL (3.7) models.

	Smooth model (3.6)	PWL model (3.7)
Bifurcation diagram	Supercritical character. $c = 0.1, \varepsilon = 0.25 : a_H \approx -0.866, a_S \approx -0.853.$	$c = 0.1, \varepsilon = 0.25, \gamma = 0$: Subcritical character, $a_H \approx -0.683, a_S \approx -0.786.$ $c = 0.1, \varepsilon = 0.1, \gamma = 0.25$: Supercritical character, $a_H \approx -1.316, a_S \approx -1.098.$
Bursting	$c = 0.1, \varepsilon = 0.25, a = -0.9, P = 1.$ $f = 0.004 : 5 \text{ spikes (sp)}, f = 0.003 : 6 \text{ sp}, f = 0.002 : \text{delay} + 9 \text{ sp}.$	$c = 0.1, \varepsilon = 0.25, \gamma = 0, a = -0.9, P = 1.$ $f = 0.004 : 5 \text{ sp}, f = 0.003 : 6 \text{ sp}, f = 0.002 : 8 \text{ sp}.$
Antimonotonicity	$c = 0.1, \varepsilon \in \{0.03, 0.4, 0.5\}, a = -1.3, f = 0.02.$ Period-doubling.	$c = 0.1, \varepsilon \in \{0.03, 0.4, 0.5\}, \gamma = 0, a = -1.3, f = 0.02.$ Period-doubling.

Table 5.2: Quantitative and qualitative resume for SPICE simulations and analogue implementation results.

	SPICE simulation	Analogue circuit
Bifurcation diagram	$c = 0.1, \varepsilon = 0.25, \gamma = 0$: Subcritical character, $a_H \approx -0.36, a_S \approx -0.59$. $c = 0.1, \varepsilon = 0.1, \gamma = 0.25$: Supercritical character, $a_H \approx -0.8, a_S \approx -0.72$.	$c = 0.1, \varepsilon = 0.25, \gamma = 0$: Subcritical character, $a_H \approx -0.105, a_S \approx -0.15$. $c = 0.1, \varepsilon = 0.1, \gamma = 0.25$: Subcritical character. $c = 0.1, \varepsilon = 0.25, R_1 = 40\Omega, R_2 = 225\Omega, R_6 = 315\Omega, V_1 = 1.69V, V_2 = -0.05mV$: Supercritical character, $a_H \approx -0.35$.
Bursting	$c = 0.1, \varepsilon = 0.25, \gamma = 0, a = -0.9, P = 1$. $f = 0.004$: 4 spikes (sp), $f = 0.003$: 5 sp, $f = 0.002$: 7 sp. $c = 0.1, \varepsilon = 0.25, \gamma = 0, a = -0.6, P = 1$. $f = 0.004$: 5 sp, $f = 0.003$: 6 sp, $f = 0.002$: 8 sp.	$c = 0.1, \varepsilon = 0.25, \gamma = 0, a = -0.2, P = 0.4$. $f = 0.004$: 3 sp, $f = 0.003$: 4 sp, $f = 0.002$: 5 sp, $f = 0.0011$: 9 sp. $c = 0.1, \varepsilon = 0.25, R_1 = 40\Omega, R_2 = 225\Omega, R_6 = 315\Omega, V_1 = 1.69V, V_2 = -0.05mV, a = -0.47, P = 0.7$. $f = 0.004$: delay + 5 sp, $f = 0.003$: delay + 6 sp, $f = 0.002$: delay + 8 sp.
Antimonotonicity	$c = 0.1, \varepsilon \in \{0.03, 0.4, 0.5\}, \gamma = 0, a = -1.3, f = 0.02$. Period-doubling. $c = 0.1, \varepsilon \in \{0.03, 0.4, 0.5\}, \gamma = 0, a = -1, f = 0.02$. Period-doubling.	$c = 0.1, \varepsilon = 0.25, \gamma = 0, a = -0.42, f = 0.013$. Appearance of a new peak oscillation $c = 0.1, \varepsilon = 0.25, R_1 = 40\Omega, R_2 = 225\Omega, R_6 = 315\Omega, V_1 = 1.69V, V_2 = -0.05mV, a = -0.9, f = 0.015$. Appearance of a plateau

CONCLUSIONS

This project lies at the intersection of mathematics and circuit theory, focusing on the study and validation of theoretical models derived from an electronic circuit.

We first presented the schematic of an electronic circuit built upon a non-linear resistor, from which we derived the state equations and obtained a slow-fast system of ODEs. From them, we formulated two theoretical mathematical models, one of them smooth and the other PWL, within the framework of FitzHugh-Nagumo models.

We carried out an exhaustive theoretical analysis of both models, proving that the simpler PWL model closely resembles the dynamics of its smooth counterpart. Specifically, we adapted the results from [9] to the smooth model and obtained the parameter values for the Hopf bifurcation and the slow manifolds connection leading to the canard explosion. To prove that the canard explosion also occurs in the PWL case, we stated and proved several results from scratch based on an article dealing with a similar model, [2]. The Hopf-like bifurcation and connection parameters for the latter case were located and proven to be close to that of the smooth framework. From canard behaviour we were able to present other phenomena such as bursting after a slow passage through a bifurcation as well as antimonotonicity. To induce these phenomena, we added a periodic forcing term to one of the equations, which is characterized by a frequency and an amplitude. While variations in frequency related to different bursting configurations, changes in amplitude triggered the antimonotonicity. A key qualitative difference detected between the smooth and PWL models' behaviour was the presence of a delay during the slow passage through the Hopf bifurcation at sufficiently small frequencies. However, the PWL system was proven to also manifest the delay effect when properly modified, at the cost of losing the nullcline shape.

The final stage of the project consisted on observing the physical behaviour of the circuit. First, we implemented the schematic in SPICE software via *LTSpice*, following the parameter configurations dictated by the theoretical PWL model. The resulting nullclines differed slightly due to simplifications made during the modelling process; however, the subcritical or supercritical nature of the bifurcation diagrams matched the theoretical predictions. Nevertheless, the specific bifurcation and connection

6. CONCLUSIONS

parameters shifted due to these nullcline discrepancies. Both the bursting and anti-monotonicity phenomena were properly detected, and no delay was observed, just as the PWL model predicted.

Eventually, an analogue implementation of the circuit was developed in the laboratory. It has been shown that the theoretical parameters of the circuit did not have the expected implications on reality, likely due to component tolerances, connectivity errors, and parasitic interference. For this reason, parameters were adjusted directly on the physical circuit to ensure the nullcline resembled the shape of the theoretical models. In this way, we worked with a setup respecting theoretical parameters' configurations and with another optimized through laboratory experience. We successfully observed both subcritical and supercritical bifurcation diagrams for the circuit, even though not being able to obtain both characters at any circumstances, and even bifurcation parameters close to those of the SPICE simulations. The bursting and anti-monotonicity phenomena were also reported with their singularities. The theoretically established setup managed to obtain bursting and monotonicity scenarios qualitatively close to that of theory, and without delay in the passage through the bifurcation, though this occurred under significantly different forcing frequencies. Conversely, the empirically adjusted setup expressed the same phenomena within frequency ranges closer to the theoretical case, yet presenting a strong delay during the bursting and a period-doubling through the appearance of a plateau.

Future improvements to the analog circuit implementation may yield a substantially higher degree of alignment with the theoretical models, particularly matching the SPICE simulations more closely.

Overall, this project has required development across different disciplines and scenarios within science. On the one hand, closely tied to mathematics, precise theoretical material in the field of dynamical systems has been reviewed, studied, and developed. On the other, the study of the real behaviour of the circuit has required the use of different engineering techniques. First, specific software was used to model the circuit in a computer environment. Finally, several days were dedicated to laboratory tasks to carry out the study of an analogue implementation of the circuit using actual hardware and instruments like the oscilloscope, function generators and power supplies. Computational tools have also been used for numerical resolution of equations and for data extraction and visualisation. Moreover, the nature of said tasks has been quite distinct: while mathematics demanded building precise theoretical machinery, working in reality required the capacity for improvisation and adaptation to the circumstances that were developing, in order to be able to overcome real-world discrepancies with theoretical assumptions.

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